# Auditable Memory and Financial Identification Limits in Electronic Limit Order Books

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**Abstract**

When trading algorithms condition on recent prices, price formation is a closed-loop system. In a linear cohort model with exponential memory, prices identify only a merged rate–weight measure. Separating two memory rates differing by $\delta$ requires executed quantity $\Theta(\delta^{-2})$ and trading capital $\Theta(\delta^{-4})$, with matching attainability, adaptive bounds and a finite-horizon certificate ($[61.41, 64.48]$ par units at the BTCUSDT calibration). Separating $K$ unknown cohorts requires $\Theta((\Delta\eta)^{-(4K-2)})$ experiments, while general completely monotone memory reduces minimax accuracy to $\Theta(1/\log N)$. Passive data incur the same polynomial cost; settled online learners obey the same $\delta^{-4}$ scaling (slope $-4.001$, 95% CI $[-4.007, -3.995]$), whereas hard thresholds conceal cohorts. Class-tagged disclosure reduces sample complexity by $\Theta((\Delta\eta)^{-(4K-4)})$. Empirically, trader-class positions tighten memory-rate intervals 27.6 to 124.0 times across 30 Binance perpetuals and 9.1 to 71.4 times across eight CFTC-regulated futures markets.

**Keywords:** system identification, market microstructure, algorithmic trading, minimax sample complexity, moment problems, value of disclosure, transient risk.
**JEL:** C14, C32, C58, D83, G14.

**Reproducibility.** Every number in the paper is produced by a script or notebook cell that writes a result file from public data or from a closed form; Appendix S.9 maps each table, figure and number to its source. Every simulation number was re-run for this version.

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# 1 Introduction

Regulators increasingly require firms to say which algorithm sent each order. Under MiFID II, EU investment firms must flag algorithmic orders and identify the algorithm responsible for each investment and execution decision in their transaction reports (ESMA, 2016); the US Consolidated Audit Trail records who stands behind every order (SEC, 2012). These regimes are costly to build and to comply with, and they rest on an implicit claim: that a supervisor could not learn the same thing from public prices and order flow. This paper tests that claim, and prices the answer.

The setting is a closed-loop market. Algorithms update on recent prices, their orders move prices, and prices feed back into the algorithms. We model $K$ cohorts of algorithms that track the price with exponential memory, the reduced form of steady-state Kalman filtering and of regularized online gradient descent (Lemma 2.1), and ask what an outside observer can learn about them from market data. The answer is sharp. Prices reveal only a merged summary of the algorithms. Separating two algorithms whose memories differ by $\delta$ costs trading capital of order $\delta^{-4}$, and separating $K$ unknown algorithms costs a number of experiments of exact order $(\Delta\eta)^{-(4K-2)}$; with general memory, accuracy improves only as $1/\log N$, so halving the error squares the number of experiments. Class-tagged disclosure removes the barrier, and its value can be stated in samples: a factor of order $(\Delta\eta)^{-(4K-4)}$. In data, Binance's published long/short ratios by trader class, a coarse form of class tagging, pin down memories that prices and flow leave unrestricted in each of 30 futures markets, and the CFTC's mandated trader-class reports do the same in eight regulated U.S. futures markets.

**1. An identification gap.** Prices identify only the merged rate–weight measure $M = \sum_i q_i \eta_i \, \delta_{1-\eta_i}$ of the algorithms (Theorem 3.1). Cohorts that share a memory rate are observationally equivalent, whatever their identities or wealth. Nearby rates can be separated, but at a cost that rises quickly as they approach each other.

**2. A resource gap.** For a known symmetric pair separated by $\delta$, price-only identification with deterministic probes requires executed quantity of order $\delta^{-2}$, and a single pulse attains this order (Theorem 4.1). Because a probe moves the price against the prober, the quantity order becomes a capital order $\delta^{-4}$, whether capital is supplied by a counterparty under a margin contract (Theorem 4.3) or by the probe trader itself (Lemma 4.5). We then compute what an account actually needs. A spectral argument gives a lower bound on the trading loss of any flat round trip (Theorem 5.1); Wald's identity extends

it to adaptive, sequential probing (Theorem 5.2); and a Rayleigh–Hölder dual bounds the finite-horizon optimum (Theorem 5.3). At the calibrated parameters the bracket is $[61.41, 64.48]$ par units. For $K$ unknown cohorts the number of independent experiments is exactly of order $(\Delta\eta)^{-(4K-2)}$: a two-point lower bound (Theorem 6.4) is matched by Prony's method for every $K$ (Theorem 6.5). For general completely monotone memory the minimax accuracy is of exact order $1/\log N$: a lower bound built from moment-matched cohort pairs (Lemma 6.6) is matched by a moment estimator solved as a linear program (Theorem 6.7), so the exponential-memory model is the optimistic case.

**3. A measurement gap.** Passive data do not escape the barrier. The closed-loop transfer function is minimum phase, so the price spectrum identifies the merged measure (Theorem 6.8), and natural excitation can separate a pair with zero controlled quantity (Proposition S.8.2). But the information rate falls at the same polynomial order as for active probing, $(\Delta\eta)^{4K-2}$ for every number of cohorts $K$ (Theorem 6.9), and in-loop price noise leaves the capital order of active probing intact up to horizons of that same order (Theorem 5.1(iv)). Only richer disclosure removes the dependence on rate spacing: tagged flow (Proposition 8.1) or audited equity (Proposition 4.4). Tagging reduces the sample requirement by a factor of order $(\Delta\eta)^{-(4K-4)}$ for rates (Theorem 8.2). On Binance, published long/short ratios by trader class pin down class memories 28 to 124 times more precisely than prices and flow in each of 30 futures markets (Section 8.3), and calibrating the theorem to the BTCUSDT class memories makes them worth about 333 times as much data as prices alone.

**4. What feedback indices measure.** Standard reduced-form feedback indices are dominated by filtering mechanics and clock alignment; a placebo pipeline reproduces the mechanical persistence exactly; measured feedback rises with market depth across 30 perpetuals; and two fee events that multiply trading volume by 2.6 and divide it by 4.9 leave measured feedback unchanged (Section S.5).

**5. A spectral blind spot.** Even without identification, could a supervisor monitor risk through eigenvalues or volatility? No. Markets with identical eigenvalues, one-step volatility and stationary price variance can have arbitrarily large worst-case transient loss (Theorem S.4.1). This motivates the open question with which the paper ends: what can be measured and controlled without identifying the algorithms? The companion paper takes it up.

**What we do not claim.** The identification results hold for the stated linear model and observation contracts. They are not universal limits on surveillance with participant-level records, which can be far more informative (Propositions 4.4 and 8.1). The matching upper bound holds on a class with separated rescaled rates and weights bounded away from zero. Capital figures depend on the observation-noise calibration, which we estimate rather than assume. The empirical results are reduced-form diagnostics; they do not identify structural learning parameters. Our pre-registered real-data test of the identification boundary failed its criterion on one of its three parts: on 42 days of data the merged response was precisely estimated in only 5 of 30 markets (Section 8.3). Binance's class ratios are positions, not tagged order flow, so they are the closest public analogue to the disclosure of Proposition 8.1, not an instance of it.

## 1.1 Related literature

**Price formation and impact.** Our price equation is a discrete linear impact model in the tradition of Kyle (1985), Glosten and Milgrom (1985) and Hasbrouck (1991), in which order flow moves prices and prices revert toward value. Transient impact and the cost of trading follow Almgren and Chriss (2001), Obizhaeva and Wang (2013), Gatheral (2010) and Bouchaud et al. (2004); Alfonsi et al. (2016) show that absence of price manipulation requires positive-definite decay kernels, which is the property Lemma 4.2 delivers in our cohort economy. Bouchaud et al. (2018) document order splitting and long memory in order flow, which is the main confounder in Section S.5. Relative to this literature we treat the impact kernel not as a primitive but as the closed-loop output of adaptive traders, and ask what it reveals about them.

**Algorithmic trading and feedback.** Algorithms that react to recent prices can amplify or dampen moves. Brunnermeier and Pedersen (2009) and Shleifer and Vishny (1992) study liquidity spirals and fire sales, and Kirilenko et al. (2017) document the behaviour of automated intermediaries in the 2010 Flash Crash. Sornette et al. (2023) show that non-normal interactions can generate transient bubbles. Our contribution is on the measurement side: we show what a closed-loop observer can and cannot learn about the feedback, and that standard reduced-form indices are dominated by filtering and timing.

**Closest work.** Three literatures are closest. Super-resolution and moment-based mixture estimation give rates for recovering clustered point sources and mixing measures from noisy moments (Heinrich and Kahn, 2018; Batenkov et al., 2021; Liu and Ammari, 2022;

Table 1: Comparison with finite-mixture estimation and open-loop identification

| Feature | Finite-mixture estimation (Heinrich and Kahn, 2018; Wu and Yang, 2020) | Open-loop exponential fitting (Roy and Kailath, 1989; Batenkov et al., 2021) | Closed-loop market identification (this paper) |
|---|---|---|---|
| Where rates live | Component locations / moments of density | Poles of impulse response | Transmission zeros of closed-loop $H(z)$ (Thm 3.1) |
| Active vs. passive data | i.i.d. draws only | Designed input vs. output spectrum | Same $\Theta((\Delta\eta)^{-(4K-2)})$ order via minimum-phase Whittle gap (Thm 6.9) |
| Economic unit of cost | Sample size $N$ | Signal-to-noise ratio | Quantity $\Theta(\delta^{-2})$, trading loss and capital $\Theta(\delta^{-4})$ via $\frac{1}{2}v^\top T_h v$ (Thms 4.1–5.3) |
| Effect of class tagging | Observed component labels ($\Theta(1)$) | Separate sensor per mode | Breaks closed loop cohort-by-cohort; saves $\Theta((\Delta\eta)^{-(4K-4)})$ (Thm 8.2) |

Wu and Yang, 2020; Kong and Valiant, 2017). Our $K$-cohort rate is, in that language, the square of the amplitude condition number $(\Delta\eta)^{-(2K-1)}$ for $K$ clustered nodes (Heinrich and Kahn, 2018; Wu and Yang, 2020), and we do not claim the exponent as new. What changes when the mixture sits inside a closed-loop financial market is fourfold (Table 1): first, in mixture estimation and open-loop system identification the unknown rates are locations or poles of the observed law, whereas in a closed-loop market the cohorts feed back into the price, making the memory rates transmission *zeros* of the observable transfer function $H(z)$ that must first be extracted from the loop (Theorem 3.1); second, passive price observations incur the exact same $(\Delta\eta)^{-(4K-2)}$ sample-complexity barrier as active probing because $H(z)$ is minimum phase and the Whittle likelihood inherits the moment gap (Theorems 6.8 and 6.9); third, in a market statistical sample size translates into executed quantity $\Theta(\delta^{-2})$ and round-trip trading loss or margin capital $\Theta(\delta^{-4})$ via the self-impact Toeplitz quadratic form $\frac{1}{2}v^\top T_h v$ (Theorems 4.1–5.3); and fourth, class-tagged disclosure breaks the feedback loop cohort by cohort and cuts the sample requirement by $(\Delta\eta)^{-(4K-4)}$ (Proposition 8.1 and Theorem 8.2). Beyond the $K$-cohort case, we prove matching lower and upper bounds of exact logarithmic order $\Theta(1/\log N)$ for general completely monotone memory (Theorem 6.7) and price the resulting disclosure savings against audit-trail budgets.

**Statistical identification.** The identification results use two-point testing arguments (Le Cam, 1986; Tsybakov, 2009). The moment-matching construction of Section 6 is the dynamical analogue of the slow rates in finite-mixture estimation (Manole and Ho, 2022) and of super-resolution of near-colliding sources (Liu and Ammari, 2022; Batenkov et al., 2021); the general-memory result combines best-approximation duality (Bernstein, 1914; Rivlin, 1981) with the moment comparisons used by Kong and Valiant (2017) and

Wu and Yang (2020), and its logarithmic rate parallels supersmooth deconvolution (Fan, 1991); recovery of exponentials connects to ESPRIT and matrix-pencil methods (Roy and Kailath, 1989; Hua and Sarkar, 1990). Minimum-phase spectral factorization and the Whittle likelihood are classical (Whittle, 1953). What is new here is the mapping from these statistical rates into executed quantity and money through a financially specified account, and the result that the relevant rates are zeros, not poles, of the observable transfer function.

**Non-normal dynamics and supervision.** Transient growth of stable non-normal systems is classical (Trefethen and Embree, 2005; Kreiss, 1962). We use it to show that supervision through eigenvalues and second moments of prices is blind to transient risk (Theorem S.4.1), which motivates the identification-free control of the companion paper. On the regulatory side, pre-trade risk controls for broker-dealers with market access are governed by SEC Rule 15c3-5 (SEC, 2010), and participant-level audit trails are the natural disclosure regime for Proposition 8.1. A large literature studies how pre- and post-trade transparency affects liquidity and trading costs (Pagano and Röell, 1996; Boehmer et al., 2005; Bessembinder et al., 2006); we ask a different question, what disclosure lets a supervisor learn about the algorithms that trade, and give the answer in samples. Crypto venues, whose market structure is studied by Makarov and Schoar (2020), are a natural laboratory for this question because they publish class-level positioning every five minutes, while the CFTC publishes it weekly for regulated futures.

Partial fractions, Gaussian testing, convolution inequalities and positive-system gains are established tools. Our contribution is to connect them into a quantitative chain from memory resolution, through probing resources, to money, with matching orders, explicit constants and a supervisory interpretation, and then to measure the relevant quantities in data.

## 1.2 Outline

Section 2 sets out the model. Section 3 gives the identification boundary. Section 4 links memory resolution to quantity and money for a known pair. Section 5 computes account-level capital. Section 6 treats unknown mixtures and passive data. Section 7 checks the results against stylized nonlinear and adaptive trading rules. Section 8 studies disclosure and compares its cost in dollars with the U.S. Consolidated Audit Trail. Section S.4 gives the spectral blind spot. Section S.5 reports the empirical evidence. Section 12 concludes. Proofs are in Appendix S.6; Appendix S.8 collects additional results, and Appendix S.9

maps every number to its source.

## 2 Model and microfoundations

Time is discrete. Let $p_t$ be the price, $p^*$ the fundamental value and $x_t = p_t - p^*$ the pricing error. Competitive dealers absorb net order flow with price impact $\kappa > 0$, and arbitrage pulls the error toward zero with persistence $a \in [0, 1)$:

$$x_t = a x_{t-1} + \kappa \sum_{i=1}^{K} q_i y_{i,t-1} + \kappa v_t + w_t, \tag{1}$$

where $v_t$ is an executed probe quantity and $w_t$ a price innovation. Cohort $i$ holds a state $y_{i,t}$ with exponential memory and trades $q_i y_{i,t}$:

$$y_{i,t} = (1 - \eta_i)\, y_{i,t-1} + \eta_i x_t, \qquad \eta_i \in (0, 1). \tag{2}$$

The ordering is *price first*: the price responds to the probe, then cohorts update. Write $r_i = 1 - \eta_i$ for the memory rate and $w_i = q_i \eta_i$ for the merged weight. The feedback kernel and its transform are

$$k_m = \sum_i w_i r_i^m, \qquad G(z) = \sum_i \frac{w_i z}{z - r_i}, \tag{3}$$

and the transfer function from probes to prices, $H(z) = \sum_{j \geq 0} h_j z^{-j}$, is

$$H(z) = \frac{\kappa z}{z - a - \kappa G(z)} = \frac{\kappa z D(z)}{(z - a) D(z) - \kappa N(z)}, \qquad G = \frac{N}{D}, \quad D(z) = \prod_i (z - r_i), \tag{4}$$

with impulse response $h_0 = H(\infty) = \kappa, h_1, h_2, \ldots$; in particular $H(1) = \kappa/(1-S) = \sum_j h_j$. Stacking $z_t = (x_t, y_{1,t}, \ldots, y_{K,t})^\top$ gives $z_t = J z_{t-1} + b v_t$, where, because cohorts update on the same period's price, the probe enters every state on impact through

$$b = \kappa\, (1, \eta_1, \ldots, \eta_K)^\top, \tag{5}$$

and

$$J = \begin{pmatrix} a & \kappa q_1 & \cdots & \kappa q_K \\ \eta_1 a & 1-\eta_1+\eta_1\kappa q_1 & \cdots & \eta_1\kappa q_K \\ \vdots & & \ddots & \vdots \\ \eta_K a & \eta_K\kappa q_1 & \cdots & 1-\eta_K+\eta_K\kappa q_K \end{pmatrix}, \qquad S \equiv a + \kappa \sum_i q_i. \tag{6}$$

In the *positive economy* ($q_i \geq 0$), $J$ is entrywise nonnegative, its price row sums to $S$, and the economy is Schur stable iff $S < 1$.

**Lemma 2.1** (Filtering and learning microfoundations)**.** *(i)* Kalman memory. *A desk that tracks a latent random-walk trend $v_{t+1} = v_t + \zeta_{t+1}$, $\zeta \sim N(0, \sigma_\zeta^2)$, from signals $r_t = v_t + \nu_t$, $\nu \sim N(0, \sigma_\nu^2)$, using the steady-state Kalman filter, updates as $\hat{v}_{t+1} = (1-\eta)\hat{v}_t + \eta r_t$ with*

$$\eta = \frac{-\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2\sigma_\nu^2}}{2\sigma_\nu^2} \in (0,1). \tag{7}$$

*(ii)* Gradient memory. *Online gradient descent with $L_2$ regularization $\gamma \in (0,1)$, step size $\alpha > 0$ and loss $\frac{1}{2}H(x_t - \theta_t)^2$ gives $\theta_{t+1} = (1-\eta)\theta_t + \eta q x_t$ with $\eta = \gamma + \alpha H \in (0,1)$ and $q = \alpha H/\eta$. Tabular Q-learning with learning rate $\eta$ has the same form.*

*Remark* 2.2 (Why exponential memory). Lemma 2.1 shows that two different design philosophies, Bayesian filtering and regularized online learning, produce the same observable reduced-form belief state $y_{i,t}$. When desk $i$ chooses order flow $u_{i,t}$ subject to $I_{i,t} = (1-\delta_{\text{inv}})I_{i,t-1} + u_{i,t}$ and an inventory cap $|I_{i,t}| \leq \bar{I}_i$ to maximize expected continuation return $\beta_i y_{i,t} u_{i,t}$ net of quadratic execution cost $\frac{1}{2}\lambda_{\text{tc},i} u_{i,t}^2$ and quadratic inventory holding cost $\frac{1}{2}\gamma_{\text{inv},i} I_{i,t}^2$, its interior first-order condition gives $u_{i,t}^* = q_i y_{i,t} - \psi_{\text{inv},i} I_{i,t-1}$ with market-response weight $q_i = \beta_i/(\lambda_{\text{tc},i} + \gamma_{\text{inv},i})$ and inventory mean-reversion $\psi_{\text{inv},i} = \gamma_{\text{inv},i}(1-\delta_{\text{inv}})/(\lambda_{\text{tc},i} + \gamma_{\text{inv},i})$, which shrinks toward zero as trading costs or inventory penalties tighten and destabilizes the market when the closed-loop spectral radius reaches one (`microfoundation_and_baselines.py`). This filtering-and-control recursion specifies an individual agent's decision rule, not a complete economic equilibrium (dealer–trader equilibrium is analysed in the companion paper). The memory rate encodes the desk's signal-to-noise ratio or regularization, which is the private information a supervisor would like to recover; Section 3 shows why prices reveal it only in merged form.

**Lemma 2.3** (Positive price-response geometry)**.** *In the positive, stable economy the im-*

*pulse response is a nonnegative mixture of decaying exponentials,*

$$h_j = \sum_\ell \omega_\ell \lambda_\ell^j, \qquad \omega_\ell \geq 0, \quad \sum_\ell \omega_\ell = \kappa, \quad 0 \leq \lambda_\ell < 1.$$

*It is completely monotone and nonincreasing; its peak is the direct impact $\kappa$, even though $J$ is non-normal. In price/memory coordinates the price-only gain is* $\sup_t \|e_x^\top J^t\|_2 = 1$.

Numerically, $(-1)^n \Delta^n h_j \geq 0$ for $n \leq 4$ held in all 892 stable random positive economies tested. Lemma 2.3 matters twice: it makes probing costs positive (Section 4), and it shows that in this economy transient amplification of the price response cannot arise from cohort feedback alone. The companion paper locates amplification in order-book depth instead.

## 2.1 Discussion of the assumptions

**Linearity.** The model is linear in prices, flows and memory states. Linearity is what makes the merged-measure equivalence exact and the testing problems Gaussian. Nonlinear strategies (for example threshold rules or position limits) can break the equivalence class, but they do so in regions of the state space that a small probe does not visit; for small probes the linearization around the steady state is the relevant object.

**Price-first timing.** We order the price update before the cohort update. Under the alternative lagged timing the transition matrix becomes $\left(\begin{smallmatrix} a & 1 \\ \eta Q & 1-\eta \end{smallmatrix}\right)$ and its eigenvalues differ, so stability statements must be computed under the timing actually used. The identification results depend on the transfer function and survive either convention.

**Exponential memory.** Lemma 2.1 justifies exponential memory for filtering and regularized learning. Neural and reinforcement-learning policies are not exponential filters. Rules whose orders are smooth functions of an exponential-memory state are exponential filters to first order, so the results hold for small probes; Section 7 checks this in stylized simulations and finds where it fails, and how closely production algorithms follow such rules remains an empirical question we do not settle. Mixtures of exponentials can approximate any completely monotone kernel, so a power-law memory corresponds to a continuum of cohorts, for which Theorem 6.7 shows that accuracy improves only logarithmically in the data.

**Algorithms that react to being probed.** The probing results assume that the cohorts keep their memories while they are probed. An algorithm that detects a probe could

pause, shorten its memory or trade against the probe. Four observations limit what this does to our conclusions. First, a reaction that all cohorts share does not separate them, and an algorithm that hides by mimicking another rate makes the hypotheses closer, so neither lowers the costs we derive; a reaction that differs across cohorts is itself a signature, but one the observer must model, which is a new identification problem rather than a free one. Second, the passive route needs no probe: Theorem 6.9 shows that natural price variation identifies the same separation at the same polynomial order in sample length, so an observer who fears detection pays in time, not in rate. Third, detection is itself a testing problem against the same noise: the generalized-eigenvector probe of Section 5 trades at most 1.25 par units in any period, and a probe sized within normal flow variation is hard to tell from noise, so the threat bites mainly for large single pulses, which our ledger shows are not optimal anyway. Fourth, what we cannot rule out is an algorithm designed to defeat surveillance, which turns identification into a game between observer and observed; our results are the benchmark against which such a game would be measured, and we leave it to future work.

**Known primitives.** The resource theorems take $\kappa$, $a$ and the noise law as known. Unknown primitives can only make separation harder, so the lower bounds remain valid, while the upper bounds require estimating them, which is cheap relative to the memory rates because they enter at first order.

**Observation contracts.** Each result names what the observer sees. A price-only observer, an observer with executed quantities, one with noisy statements, one with audited equity and one with tagged flow face different experiments. The paper's central message is that these differences, not the observer's statistical skill, determine what can be learned.

# 3 The identification boundary

A central question in algorithmic market surveillance is whether an observer of executions and prices can identify the individual strategies active in the market.

**Theorem 3.1** (Merged rate–weight equivalence)**.** *Let $\kappa > 0$ and $a \in [0, 1)$ be known. Use zero initial states and a common probe contract, and observe the modeled price state $x_t$.*

*(i)* Equivalence class. *Two economies $\mathcal{E}^A = \{(q_i^A, \eta_i^A)\}$ and $\mathcal{E}^B = \{(q_j^B, \eta_j^B)\}$ produce identical price responses for every probe sequence if and only if their merged measures*

*coincide:*

$$\sum_{i:\eta_i^A=\eta} q_i^A \eta \;=\; \sum_{j:\eta_j^B=\eta} q_j^B \eta \qquad \forall \eta \in (0,1). \tag{8}$$

*For $a > 0$ the minimal realization has dimension $1 + |\mathrm{supp}(M)|$, counting distinct rates with nonzero merged weight after signed cancellation; for $a = 0$ the price carries no state of its own and the dimension is $|\mathrm{supp}(M)|$. A cohort with $\eta = 1$ merges into the anchor $a$ without adding a pole.*

*(ii)* Separation complexity under average power. *Consider testing a single cohort with rate $\bar{\eta}$ against two cohorts with rates $\bar{\eta} \pm \delta$ and the same total weight, on a compact family that is uniformly Schur stable with rates bounded away from 0 and 1. Observe $\tilde{x}_t = x_t + \nu_t$ with $\nu_t \sim N(0, \sigma_\nu^2)$ and use deterministic probes with $\frac{1}{T}\sum_{t<T} v_t^2 \le P$. Any test with worst-case error at most $\alpha \in (0, 1/4)$ needs*

$$T \;\ge\; \frac{4\sigma_\nu^2(1-2\alpha)^2}{K^2 P\, \delta^4} = \Omega(\delta^{-4}), \tag{9}$$

*where $K$ depends only on the family.*

The memory rates are poles of $G$ and *zeros* of $H$, not closed-loop poles. This has a practical consequence we verify in Section 6: pole-fitting methods applied directly to the price impulse response recover closed-loop eigenvalues (0.4256 and 0.8224 for $K = 1$, $\eta = 0.3$, $a = 0.5$, $\kappa = 0.8$, $q = 0.2$) and miss the memory rate 0.70.

*Remark* 3.2 (Scope). The result concerns the stated model and observation contract. Cohorts sharing an exact rate enter prices through their merged weight; this does not determine identities or wealth allocations. Nearby rates can be distinguishable with enough information. The $\delta^{-4}$ statement in (ii) is a lower bound on a declared family, with family-dependent constants. Individual algorithms may choose their rates strategically, but any unilateral change maps into the same observer-facing merged measure, which reinforces rather than circumvents the boundary.

# 4 Probing resources, memory resolution and money

Observation length alone cannot buy identification when the lifetime quantity of controlled executions is fixed. This section connects the merged-memory experiment to a financially specified input/output contract. Throughout, $\Phi$ is the standard normal distribution function.

## 4.1 Quantity

**Theorem 4.1** (Sharp probing-resource orders for a known symmetric pair)**.** *Fix* $0 \le a < 1$, $\kappa, q > 0$, $S = a + \kappa q < 1$ *and* $\eta \in (0,1)$. *Compare a single cohort of weight* $q$ *and rate* $\eta$ *with two cohorts of weight* $q/2$ *and rates* $\eta \pm \delta$, *both in* $[\epsilon, 1-\epsilon]$. *All parameters are known. The price receives executed quantity through* $\kappa v_t$, *starts from zero, has no process noise, and is observed with independent* $N(0,\sigma^2)$ *noise. Probes are deterministic and common to both alternatives. Let* $d_T = (h^\delta_j - h^0_j)_{j<T}$ *and* $K_* = 2\kappa^2 q/[\epsilon^3(1-S)^2]$.

*(i) For every probe with* $\sum_t |v_t| \le V$,

$$D_{\mathrm{KL}}(P_0^v \| P_\delta^v) \le \frac{K_*^2 V^2 \delta^4}{2\sigma^2}, \qquad e_{\mathrm{worst}} \ge \frac{1}{2}\Big(1 - \frac{K_* V \delta^2}{2\sigma}\Big)_+. \tag{10}$$

*(ii) The optimal worst-case error over these probes and all classifiers is exactly*

$$e_T^*(V) = \Phi\left(-\frac{V\|d_T\|_2}{2\sigma}\right), \tag{11}$$

*attained by a pulse at the first input and the likelihood-ratio test. Waiting adds information but* $\|d_T\|_2 \uparrow \|d_\infty\|_2 < \infty$, *so it cannot make the error arbitrarily small at fixed* $V$. *Error at most* $\alpha < 1/2$ *requires* $V \ge 2\sigma(1-2\alpha)/(K_*\delta^2)$ *independently of* $T$.

*(iii) The first two taps coincide and* $d_2 = -\kappa^2 q \delta^2$. *A single pulse of size*

$$A = \frac{2\sigma\Phi^{-1}(1-\alpha)}{\kappa^2 q \delta^2} \tag{12}$$

*and the third observation achieve error exactly* $\alpha$. *The necessary and sufficient quantity order is* $\delta^{-2}$.

*(iv) Under average power* $\sum_{t<T} v_t^2 \le PT$, *the necessary horizon order is* $\delta^{-4}$, *attained by the pulse whenever* $T \ge \max\{3, \lceil A^2/P \rceil\}$.

*(v) For a fixed monetary price exposure* $\lambda \ne 0$, *additive reserve* $L$ *and cost envelope* $c\sum|v_t|$, *the sharp robust quantity allowance under budget* $B > L$ *is* $V_B = (B - L)/(|\lambda|\kappa + c)$.

The identity in (iii) is verified to machine precision for $\delta \in \{0.02, 0.05, 0.10\}$: the difference is $-1.2 \times 10^{-4}$, $-7.5 \times 10^{-4}$ and $-3.0 \times 10^{-3}$ at $\kappa = 1$, $q = 0.3$, and $-3.2 \times 10^{-4}$

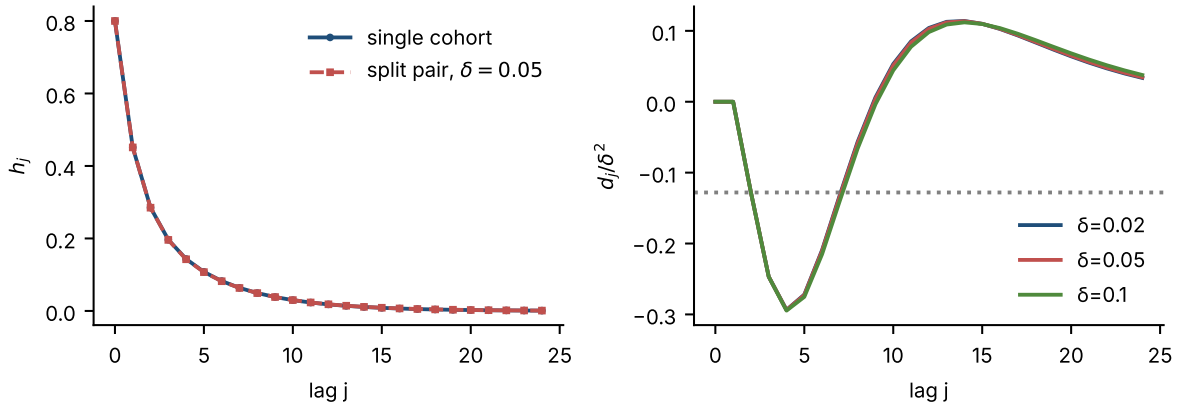



Figure 1: Left: impulse responses of the single cohort and the split pair ($\delta = 0.05$) are nearly indistinguishable. Right: the difference scaled by $\delta^2$ collapses onto one curve, with first nonzero value $-\kappa^2 q$ at lag 2 (dotted). Calibration $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$.

Table 2: Exact known-pair error: formula versus simulation

| Pulse $V$ | Horizon $T$ | $\Phi(-V\|d_T\|_2/2\sigma)$ | Simulated error, $H_0$ | Simulated error, $H_\delta$ |
|---|---|---|---|---|
| 20 | 3 | 0.3745 | 0.3734 | 0.3762 |
| 40 | 3 | 0.2611 | 0.2599 | 0.2603 |
| 40 | 10 | 0.0029 | 0.0030 | 0.0027 |
| 80 | 20 | $1.8 \times 10^{-10}$ | 0 | 0 |

200,000 paths per hypothesis; $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$.

at the calibration $\kappa = 0.8$, $q = 0.2$, $\delta = 0.05$. The theorem does not solve estimation of an unknown mixture (Section 6).

**A Monte Carlo check of the exact error.** Formula (11) is exact for the likelihood-ratio test. We simulated 200,000 noisy paths per hypothesis at the calibration ($\sigma = 0.01$, $\delta = 0.05$) for several pulse sizes and horizons (Table 2); simulated errors match the formula to Monte Carlo precision under both hypotheses.

Figure 1 shows why the problem is hard. The two economies have the same direct impact and the same first-lag response; they first differ at lag two, by an amount proportional to $\delta^2$, and the whole difference sequence scales with $\delta^2$. A test therefore needs the signal-to-noise ratio $V\|d\|/\sigma$ to grow like $\delta^{-2}$, which is the quantity order, and because the adverse mark of a probe grows with the square of its size, capital grows like $\delta^{-4}$.

## 4.2 Execution cost

A probe changes inventory as well as price. We therefore specify cash and execution timing. Price has units of currency per asset, and $\kappa$ currency per squared quantity.

**Lemma 4.2** (Fair execution)**.** *For a quote schedule refreshed after the uncontrolled state update, execution at the average price of a linear book costs, relative to the constant fundamental mark,*

$$C(v) = \tfrac{1}{2} v^\top T_h v + c\|v\|_1, \qquad (T_h)_{ij} = h_{|i-j|}. \tag{13}$$

*In the positive stable economy $T_h$ is positive definite, so no nonzero round trip produces a deterministic gain.*

We verify the ledger identity (13) against an explicit cash account to $5.7 \times 10^{-13}$ on 1,000 random probes, and the smallest eigenvalue of $T_h$ over 1,000 random stable positive economies is 0.141. A direct-only proxy $\sum \frac{1}{2}\kappa v_t^2$ omits cross-time impact: for $v = (+1, -1)$ it gives 0.80 against a true Toeplitz loss of $h_0 - h_1 = 0.3488$.

## 4.3 Capital supplied by a counterparty

**Theorem 4.3** (Sharp robust quantity cap from cash and margin)**.** *Use the positive, stable, reset economy with constant fundamental price $P_* > 0$ and no process noise. A designated dealer fills only nonnegative probes $v_t \geq 0$ at the average price $P^0_{t+1} + \kappa v_t/2$ of a linear schedule, holds inventory $I_{t+1} = I_t - v_t$ and cash $K_{t+1} = K_t + v_t(P^0_{t+1} + \kappa v_t/2) - cv_t$, starts with equity $B > 0$, and must satisfy $\mu|I_t|P_t \leq E_t$ at every time. For the robust contract admitting every probe with $\sum_t v_t \leq V$, the largest allowance guaranteeing margin is*

$$V_B = \frac{2B}{\beta + \sqrt{\beta^2 + 4(\mu + \frac{1}{2})\kappa B}}, \qquad \beta = c + \mu P_*, \tag{14}$$

*equivalently $(\mu + \frac{1}{2})\kappa V^2 + (c + \mu P_*)V \leq B$. The cap holds across all positive cohort representations with $S < 1$, without resolving individual rates. An observer that receives only noisy prices and executed quantities has best error $\Phi[-V_B\|d_T\|_2/(2\sigma)]$ within this allowance; fixed error $\alpha < 1/2$ requires capital of order at least $\delta^{-4}$, and a single positive pulse attains that order.*

Even with zero initial inventory, the probe creates a quadratic adverse mark that a fixed initial price exposure would miss. Figure 2 illustrates the experiment.

**Proposition 4.4** (Exact financial disclosure changes the experiment)**.** *In this noiseless-state account, exact disclosure of the third equity mark after a single positive pulse $0 < A \leq V_B$ distinguishes the known pair with zero error for every $\delta > 0$.*

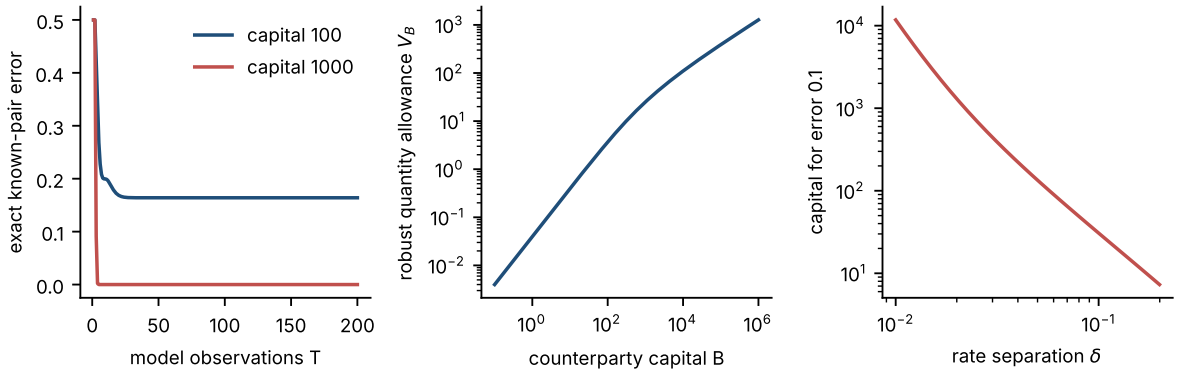



Figure 2: Known-pair experiment funded by the counterparty cash/margin contract. (a) Exact Gaussian error at fixed capital 100 and 1,000, separation $\delta = 0.05$. (b) The sharp robust quantity allowance (14). (c) Capital for error 0.1 with 200 observations. Parameters: $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$, observation s.d. 0.003, reference price 100, margin 0.25, cost 0.02. No process noise.

Exact marked equity is extra information, not a free observation assumed by the price-only theorem. Real account marks have their own precision: with reports rounded to 0.01, a guaranteed change requires a larger trade (Section 8).

## 4.4 Capital paid by the probe trader

**Lemma 4.5** (Time-domain Toeplitz coercivity: capital order for funded probes)**.** *Use the same economy and execution law. The statistical observer receives only noisy prices and executed quantities. The probe trader starts with cash $B > 0$ and zero inventory; a finite signed sequence $v$ changes inventory by $+v_t$ and cash by $K_{t+1} = K_t - v_t(P^0_{t+1} + \kappa v_t/2) - c|v_t|$, after which trading stops and the trader must remain solvent as the price returns to $P_*$. Put*

$$r = 1 - \epsilon(1 - S) < 1, \qquad m = \kappa \frac{1 - r}{1 + r} > 0, \qquad K_* = \frac{2\kappa^2 q}{\epsilon^3 (1 - S)^2}.$$

*Every solvent strategy satisfies $D_{\mathrm{KL}}(P^v_0 \| P^v_\delta) \le K_*^2 B \delta^4/(\sigma^2 m)$, so error at most $\alpha < 1/2$ requires*

$$B \;\ge\; \frac{2\sigma^2 m (1 - 2\alpha)^2}{K_*^2 \delta^4}, \tag{15}$$

*without a quantity cap and at any horizon. Conversely, for any margin fraction $0 < \mu \le 1$, the pulse $A$ in (12) achieves error $\alpha$ and remains margin-solvent at every later mark with capital $B_A = \frac{\kappa}{2} A^2 + (c + \mu P_*) A$. Necessary and sufficient capital orders are $\delta^{-4}$.*

Lemma 4.5 is an order statement: an elementary time-domain coercivity argument for

Toeplitz forms that establishes the $\delta^{-4}$ rate before the exact frequency-domain optimization of Section 5. Its explicit lower constant is correspondingly loose: at the calibrated parameters it evaluates to $4.9 \times 10^{-4}$ par units at $\delta = 0.05$ with $\epsilon = 0.35$, five orders of magnitude below any design we can execute, and because $m/K_*^2 \propto \epsilon^7(1-S)^5$ it collapses near the boundary of the family. We therefore state it as a lemma; Section 5 computes the constant that matters, within 5% of a feasible design at every separation.

## 4.5 A worked example: two Kalman desks

To make the orders concrete, consider two desks that track the same trend with Kalman filters (Lemma 2.1(i)) and differ only in their signal-to-noise ratios: $\sigma_\zeta^2/\sigma_\nu^2 = 0.09$ and 0.16. Their gains are $\eta_1 = 0.2584$ and $\eta_2 = 0.3279$, so $\bar{\eta} = 0.2931$ and $\delta = 0.0348$. With equal weights, the economy is observationally the split pair of Theorem 4.1 around $\bar{\eta}$, and the response difference first appears at lag two, $d_2 = -\kappa^2 q \delta^2 = -1.55 \times 10^{-4}$ (verified).

A supervisor who wants to tell this market from a single desk with gain $\bar{\eta}$ at 5% error using a single pulse and the third observation needs $A = 2\sigma\Phi^{-1}(0.95)/(\kappa^2 q\delta^2) = 212$ units of executed quantity, and, under a 0.5% margin and 4 basis points of fees, single-pulse capital of 18,166 par units. The pulse is very inefficient: by the $\delta^{-4}$ scaling of Theorem 5.1, the optimal flat probe needs about $61.41 \times (0.05/0.0348)^4 \approx 262$ par units. Converting with the measured Roll noise (factor $(0.40/0.01)^2$), the optimal probe costs roughly $4 \times 10^5$ USDT and the naive pulse about $3 \times 10^7$ USDT. Two desks whose signal-to-noise ratios differ by almost a factor of two are therefore expensive, but not impossible, to tell apart by probing; three or more desks are much harder (Theorem 6.4).

## 4.6 Natural excitation and the cost of ambiguity

Natural price innovations can also separate the pair with zero controlled quantity: an exact finite-sample test on independent three-step blocks needs order $\delta^{-4}$ blocks (Internet Appendix, Proposition S.8.2), the same order as the passive Whittle bound of Section 6. Non-identification also has a price on the quote side: when the relevant set of economies includes costlier ones, a robust dealer widens its quote, which the companion paper analyses (Niroula et al., 2026); cohort non-identification alone does not establish that the worst relevant cost exceeds the informed one.

Table 3: Account-level initial capital at $\delta = 0.05$ (par units)

| Design | $C_0$ | Error | End inv. | Peak margin | Max $\lvert v_t \rvert$ | Trades |
|---|---|---|---|---|---|---|
| 12 buys, $T = 20$ apart, never unwound | 229.13 | 0.0578 | 75.00 | 39.38 | 6.25 | 12 |
| 6 buy/sell round trips | 189.45 | 0.0308 | 0.00 | 3.28 | 6.25 | 12 |
| 5 buy/sell round trips | 157.89 | 0.0442 | 0.00 | 3.28 | 6.25 | 10 |
| 12 buys, each unwound after 20 steps | 378.82 | 0.0040 | 0.00 | 3.28 | 6.25 | 24 |
| Generalized-eigenvector probe ($L = 120$) | 65.06 | 0.0500 | 0.00 | 2.74 | 1.24 | 120 |
| Generalized-eigenvector probe ($L = 160$) | 64.86 | 0.0500 | 0.00 | 2.17 | 1.09 | 160 |
| Fee-refined probe ($L = 240$) | 64.48 | 0.0500 | 0.00 | — | 1.30 | 240 |

The pulse designs place a pulse every 40 steps and close it 20 steps later; the fee-refined probe is re-optimized by sequential quadratic programming over $L = 240$. The certified lower bound of Theorem 5.1 is 61.41; the explicit constant of Lemma 4.5 is $4.9 \times 10^{-4}$.

# 5 Account-level capital

Lemma 4.5 gives the order but not the constant. This section computes what a trader actually needs under a calibrated linear-impact environment, then converts it to Bitcoin dollars.

## 5.1 Observation design and the tri-region partition

Under observation horizon $T$ per pulse, $(\delta, B)$ space splits into three regions: a feasible single-pulse region ($C^* = \Theta(\delta^{-4})$ for symmetric splitting), a sequential bounded-probing region where each capped pulse supplies divergence $V_{\text{eff}}^2 \|d_T\|_2^2/(2\sigma^2)$, and an infeasible region below the finite-session resolution floor $\delta_{\min} = \Theta(T_{\text{session}}^{-1/4})$ (Supplemental Appendix, Proposition S.2.1 and Figure 7).

## 5.2 Account-level capital and the gap to the lower bound

We run each candidate design through an explicit synthetic ledger at reference price $P_* = 100$: fills at the average linear-book price, taker fees of 4 basis points, marks at the prevailing price, and 0.5% maintenance margin. Initial capital is $C_0 = \max_t(\text{margin}_t - \text{equity}_t)$, maximized over both hypotheses, and the test uses the full price path including unwind legs.

Table 3 gives three conclusions. First, twelve buys that are never unwound fail the 5% target once the test uses the joint price path (divergence $4.95 < 5.41$) and end with 75 units of inventory. Second, if every unwind leg is itself informative, five or six round trips meet the target with zero terminal inventory. Third, pulses are not optimal: a generalized-eigenvector probe, which maximizes divergence per unit of fee-free round-trip

cost, needs about 65 par units with maximum trade 1.25 and maximum displacement 2.23, well inside limits.

For flat fee-paying policies $v \in \mathbb{R}^L$ with $\sum_t v_t = 0$, the capital problem is

$$\min_{v} \max_{i\in\{0,1\}} L_i(v) + \sum_t \phi P_* |v_t| + \max_t \mu P_* |I_t| \quad \text{s.t.} \quad \|(H_1 - H_0)v\|_2 \geq 2\sigma z_{1-\alpha}, \tag{16}$$

where $L_i(v) = \frac{1}{2} v^\top T_{h_i} v$ is the exact Toeplitz round-trip loss. Zero-padding embeds horizon $L$ into $L+1$ at identical cost and information, so the minimum is non-increasing in $L$.

**Theorem 5.1** (Spectral dissipation lower bound)**.** *Let* $\hat{H}_i(z) = \sum_{j\geq 0} h_{i,j} z^{-j}$, $\Delta\hat{H} = \hat{H}_1 - \hat{H}_0$, *and define the dissipation kernel*

$$g_i(\omega) = \operatorname{Re} \hat{H}_i(e^{i\omega}) - \kappa/2 > 0 \qquad \forall \omega \in [-\pi, \pi]. \tag{17}$$

*(i)* Toeplitz energy identity. *For any flat round trip,* $L_i(v) = \frac{1}{2\pi} \int_{-\pi}^{\pi} g_i(\omega) |V(e^{i\omega})|^2 d\omega$.

*(ii)* Lower bound. $\|(H_1 - H_0)v\|_2^2 \leq \sup_\omega \frac{|\Delta\hat{H}(e^{i\omega})|^2}{g_i(\omega)} L_i(v)$, *so every deterministic round trip with error at most* $\alpha$ *has*

$$L_i(v) \;\geq\; 4\sigma^2 z_{1-\alpha}^2 \inf_\omega \frac{g_i(\omega)}{|\Delta\hat{H}(e^{i\omega})|^2} \;\equiv\; C^*_{\text{det,quad}}(\delta). \tag{18}$$

*(iii)* Bracket. *At* $\delta = 0.05$, $C^*_{\text{det,quad}} = 61.14$ *at* $\omega^* = 0.2526$*; adding the unavoidable fee* $\phi P_* \cdot 2\sigma z / \sup_\omega |\Delta\hat{H}|$ *gives the certified bound* $C_{\text{det}} \geq 61.41$. *A feasible 240-step sequence attains 64.48, so the fee-paying optimum lies in* $[61.41, 64.48]$.

*(iv)* Process noise. *Let the price also receive the innovations* $w_t \sim N(0, \sigma_w^2)$ *of (1), independent of observation noise* $\nu_t \sim N(0, \sigma_\nu^2)$, *so that from the reset state the observer sees* $\tilde{x}_t = (h_i * v)_t + \kappa^{-1}(h_i * w)_t + \nu_t$, $t < T$. *For the known pair of Theorem 4.1, put* $\sigma_{\text{eff}}^2 = \sigma_\nu^2 + \sigma_w^2/4$, $\bar{M} = 1/(1-S)$ *and* $C_w = 2\sigma_w^4 \bar{M}^2 K_*^2 (\sigma_\nu^2 + \sigma_w^2 \bar{M}^2)/(\kappa^2 \sigma_\nu^2 \sigma_{\text{eff}}^4)$. *Every deterministic round trip whose test has errors* $\alpha$ *and* $\beta$ *under the two hypotheses satisfies*

$$L_i(v) \;\geq\; 2\sigma_{\text{eff}}^2 \big[\operatorname{kl}(\alpha, 1-\beta) - C_w T \delta^4\big]_+ \inf_\omega \frac{g_i(\omega)}{|\Delta\hat{H}(e^{i\omega})|^2}, \tag{19}$$

*with* kl *as in Theorem 5.2. In-loop noise therefore leaves the* $\delta^{-4}$ *capital order intact for every horizon* $T \leq \operatorname{kl}(\alpha, 1-\beta)/(2C_w\delta^4)$*; at horizons of order* $\delta^{-4}$ *and longer, the noise itself separates the pair without a probe (Theorem 6.9).*

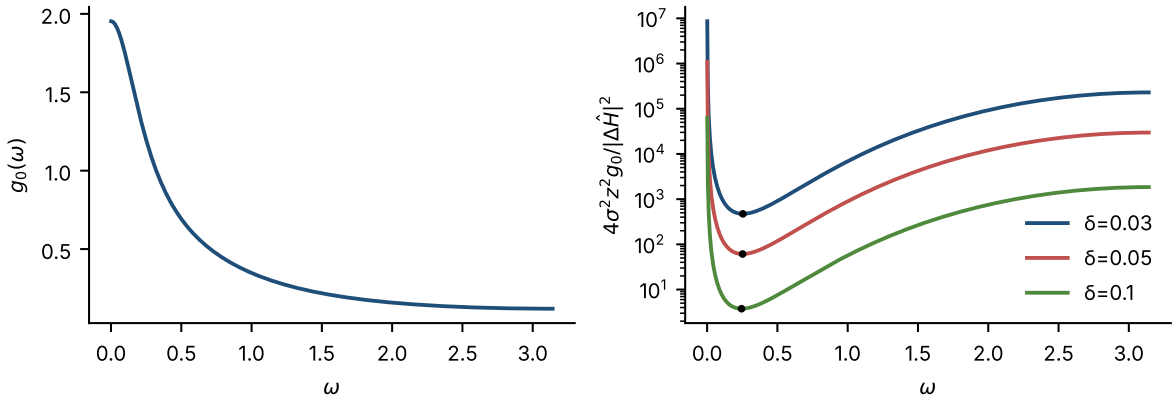



Figure 3: Left: the dissipation kernel $g_0(\omega)$ is strictly positive. Right: the capital ratio in (18); dots mark its minimum, near $\omega^* \approx 0.25$ for every separation. Recomputed in this revision.

Positivity of $g_i$ follows from Lemma 2.3: $g_i(\omega) = \frac{1}{2}\sum_\ell \omega_\ell \frac{1-\lambda_\ell^2}{1-2\lambda_\ell\cos\omega+\lambda_\ell^2} > 0$. We compute $C^*_{\mathrm{det,quad}}$ across separations (Table 4): 61.14 at $\delta = 0.05$, with $\delta^4 C^*_{\mathrm{det,quad}}$ between $3.67 \times 10^{-4}$ and $3.84 \times 10^{-4}$, confirming the $\delta^{-4}$ law. The fee term, the feasible upper bounds and the adaptive bound of Theorem 5.2 are reported in the same table.

**Theorem 5.2** (Adaptive and sequential probing)**.** *Let $(v_t)_{t\le\tau}$ be any non-anticipating policy adapted to the price filtration, with an almost surely finite stopping time and $\sum_{t\le\tau} v_t = 0$. Then*

$$\mathbb{E}_0[L_0(v)] \;\ge\; 2\sigma^2\,\mathrm{kl}(\alpha, 1-\beta)\inf_\omega \frac{g_0(\omega)}{|\Delta\hat{H}(e^{i\omega})|^2}, \tag{20}$$

*where* $\mathrm{kl}(\alpha, 1-\beta) = \alpha\log\frac{\alpha}{1-\beta} + (1-\alpha)\log\frac{1-\alpha}{\beta}$. *Hence adaptation can lower expected fee-free loss by at most the factor* $\mathrm{kl}(\alpha, 1-\beta)/(2z^2_{1-\alpha}) \approx 0.4897$ *at* $\alpha = \beta = 0.05$*, and cannot alter the* $\delta^{-4}$ *order.*

At $\delta = 0.05$ the adaptive bound is 29.94, or 30.22 with the minimal fee. Pairing it with the feasible deterministic 64.48 gives a certified ratio of at least 0.464. In 20,000 simulated sequential probability-ratio tests, which split the optimal probe into ten copies of equal information, mean loss was 39.5 under $H_0$, between the two bounds, but losses vary with the stopping time: the 95th and 99th percentiles of required account capital were 91.7 and 128.4, and funding a trader with only the mean loss gives a 37.5% insolvency rate before the test ends. Expected-loss savings are not solvency savings.

Table 4: Capital lower bounds and feasible upper bounds across separations (par units)

| $\delta$ | $\omega^*$ | $C^*_{\mathrm{det,quad}}$ | $\delta^4 C^*_{\mathrm{det,quad}}$ | Certified $C_{\mathrm{det}}$ | Feasible UB | Adaptive LB |
|---|---|---|---|---|---|---|
| 0.02 | 0.2546 | 2397.85 | $3.84 \times 10^{-4}$ | 2399.58 | 2428.25 | 1176.04 |
| 0.03 | 0.2541 | 473.19 | $3.83 \times 10^{-4}$ | 473.97 | 484.50 | 232.51 |
| 0.05 | 0.2526 | 61.14 | $3.82 \times 10^{-4}$ | 61.41 | 64.48 | 30.22 |
| 0.08 | 0.2488 | 9.26 | $3.79 \times 10^{-4}$ | 9.37 | 10.38 | 4.64 |
| 0.10 | 0.2453 | 3.77 | $3.77 \times 10^{-4}$ | 3.83 | 4.44 | 1.91 |
| 0.15 | 0.2331 | 0.73 | $3.67 \times 10^{-4}$ | 0.76 | 1.01 | 0.38 |

Calibration $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$, $\sigma = 0.01$, $\alpha = 0.05$. From `recompute_theory.py` and `recompute_p1_capital.py`; the certified column adds the fee lower bound to the fee-free infimum; the feasible column is the best flat probe found over $L \le 240$; the adaptive column adds the fee lower bound to the adaptive bound.

**Theorem 5.3** (Finite-horizon Rayleigh–Hölder duality)**.** *For a horizon of $L \ge 4$ trading periods, let $E \in \mathbb{R}^{L\times(L-1)}$ be the first-difference basis of flat probes ($\sum_t v_t = 0$), $Q_i = \frac{1}{2}T_{h_i}$ (so that $L_i(v) = v^\top Q_i v$), $A_{\mathrm{diff}} = (H_1 - H_0)^\top (H_1 - H_0)$, and let $G_i = E^\top Q_i E \succ 0$ and $M = E^\top A_{\mathrm{diff}} E \succeq 0$ be the reduced execution-loss and information matrices, with $G_i$-orthonormal generalized eigenvectors $e_j$ and eigenvalues $0 \le \lambda_1 \le \cdots \le \lambda_{L-3} < \lambda_{L-2} \le \lambda_{L-1}$; put $c_{\mathrm{req}} = 4\sigma^2 z^2_{1-\alpha}$ and $c_{\mathrm{fee}} = \phi P_*$. For $\theta \in [0, \pi)$ let $d(\theta) = \cos\theta\, e_{L-2} + \sin\theta\, e_{L-1}$, $\lambda(\theta) = \lambda_{L-2}\cos^2\theta + \lambda_{L-1}\sin^2\theta$, $s_\theta = \mathrm{sign}(Ed(\theta))$, and*

$$F_{\mathrm{dom}}(\theta) = c_{\mathrm{fee}}\sqrt{\frac{c_{\mathrm{req}}}{\lambda(\theta)}}\,\|Ed(\theta)\|_1, \qquad W_\perp(\theta) = c^2_{\mathrm{fee}} \sum_{j \le L-3} \frac{(s_\theta^\top E e_j)^2}{1 - \lambda_j/\lambda(\theta)},$$
$$\gamma(\theta) = \frac{\lambda(\theta)\,\lambda_{L-3}}{(\lambda(\theta) - \lambda_{L-3})\,c_{\mathrm{req}}}. \tag{21}$$

*Then every flat probe meeting the separation constraint of (16) satisfies*

$$L_i(v) + c_{\mathrm{fee}}\|v\|_1 \;\ge\; \min_\theta \Big\{ \frac{c_{\mathrm{req}}}{\lambda(\theta)} + \Psi(\theta) \Big\} \;\equiv\; C_{\mathrm{RH}}(L) \;\ge\; \frac{c_{\mathrm{req}}}{\lambda_{L-1}}, \tag{22}$$

$$\Psi(\theta) = \min_{x \ge 0} \Big\{ x^2 + \Big[ F_{\mathrm{dom}}(\theta)\big(1 - \gamma(\theta)x^2\big)_+ - \sqrt{W_\perp(\theta)}\,x \Big]_+ \Big\} \;\ge 0,$$

*a piecewise-quadratic minimization in closed form; when $\gamma F_{\mathrm{dom}} < 1$ and the minimizer has $\gamma x^2 \le 1$, $\Psi = \max\big(0, F_{\mathrm{dom}} - W_\perp/(4(1 - \gamma F_{\mathrm{dom}}))\big)$.*

The bound is computable for any horizon. Table 5 reports its quadratic part $c_{\mathrm{req}}/\lambda_{L-1}$, which decays toward the infinite-horizon value 61.14 at rate $O(L^{-1})$, the full bound $C_{\mathrm{RH}}(L)$, and the best feasible fee-paying probe we find at each horizon. At $L = 80$ the full bound is 64.72 against a best probe of 64.76 on the same loss-plus-fee objective,

Table 5: Finite-horizon capital at $\delta = 0.05$ (par units)

| Horizon $L$ | Quadratic part $c_{\mathrm{req}}/\lambda_{L-1}$ | Full bound $C_{\mathrm{RH}}(L)$ | Best feasible probe | Gap to $C_{\mathrm{det}} = 61.41$ |
|---|---|---|---|---|
| 80 | 62.55 | 64.72 | 65.02 | 5.9% |
| 120 | 61.84 | 63.39 | 64.59 | 5.2% |
| 160 | 61.54 | 62.46 | 64.50 | 5.0% |
| 240 | 61.32 | 61.76 | 64.48 | 5.0% |
| 320 | 61.25 | — | $\leq 64.48$ | $\leq 5.0\%$ |
| 480 | 61.19 | — | $\leq 64.48$ | $\leq 5.0\%$ |

Quadratic bound $c_{\mathrm{req}}/\lambda_L$ from the generalized eigenproblem of Theorem 5.3 on flat probes (equivalently $c_{\mathrm{req}}/(2\lambda'_L)$ with $\lambda'$ computed against $T_{h_0}$ rather than $\frac{1}{2}T_{h_0}$); it converges to the infinite-horizon value 61.14. The full bound minimizes (22) over 3,600 values of $\theta$ (`hardening/rh_bound.py`); at $L = 80$ and 120 a direct numerical minimization of loss plus fee under the separation constraint gives 64.76 and 64.43, above the bound. Feasible probes: top generalized eigenvector refined by sequential quadratic programming on the full ledger; at $L = 320$ and 480 the $L = 240$ probe padded with zeros is feasible at the same cost, so the best feasible value can only be lower.

so the dual is essentially tight at short horizons; at longer horizons the fee term of the dual weakens and the bracket widens to about 5%. The fee of unpadded full-support eigenvectors grows as $\Theta(\sqrt{L})$, which is why zero-padded or $L_1$-localized probes are needed at long horizons.

## 5.3 Robustness: heteroskedastic impact and probing protocols

Probing-induced heteroskedasticity and convex impact leave the $\delta^{-4}$ capital order unchanged (Supplemental Appendix, Corollary S.3.1), and four simulated probing protocols all exhibit log–log capital slopes near $-4$ (Table 28).

## 5.4 From par units to Bitcoin dollars

Capital scales as $\sigma^2\delta^{-4}$, so the dollar magnitude hinges on the observation noise. Treating the 4 USDT ten-second price volatility as observation noise would give 9.8 million USDT at $\delta = 0.05$, but that conflates price volatility with measurement error. We now estimate observation noise from Binance BTCUSDT one-second closes on January 10–20, 2023, with the Roll covariance estimator $\hat{\sigma} = \sqrt{-\mathrm{cov}(\Delta p_t, \Delta p_{t-1})}$ (Roll, 1984). Daily estimates range from 0.09 to 0.77 USDT, median 0.40 USDT.

At realistic noise, separating rates that differ by 0.05 costs about $9.9 \times 10^4$ USDT, which is not prohibitive for a well-capitalized desk (Table 6). The barrier becomes large only for finer separation (3.9 million USDT at $\delta = 0.02$) and, through Theorem 6.4, when several cohorts are unknown. Two factors we do not model would raise these figures: walking several levels of a real book costs more than the linear top-of-book impact assumed

Table 6: Capital lower bound under alternative observation-noise calibrations (USDT)

| Noise calibration | $\delta = 0.02$ | $\delta = 0.05$ | $\delta = 0.10$ |
|---|---|---|---|
| Roll estimator, median (0.40 USDT) | 3,874,696 | 99,193 | 6,200 |
| One-second price-change s.d., median (1.98 USDT) | 93,688,369 | 2,398,422 | 149,902 |
| Earlier assumption (4 USDT, upper scenario) | 383,812,500 | 9,825,600 | 614,100 |

$C(\sigma, \delta) = 61.41 \times (\sigma/0.01)^2 \times (0.05/\delta)^4$, with $\sigma$ the unrounded median. Noise estimated on 11 days of checksum-verified Binance BTCUSDT one-second klines.

Table 7: Daily observation-noise estimates, Binance BTCUSDT one-second closes

| Date | Roll $\hat{\sigma}$ (USDT) | One-second change s.d. (USDT) | Mean price (USDT) |
|---|---|---|---|
| 2023-01-10 | 0.152 | 0.679 | 17,288 |
| 2023-01-11 | 0.092 | 0.891 | 17,456 |
| 2023-01-12 | 0.232 | 2.292 | 18,343 |
| 2023-01-13 | 0.348 | 1.762 | 19,075 |
| 2023-01-14 | 0.773 | 3.511 | 20,818 |
| 2023-01-15 | 0.482 | 1.790 | 20,778 |
| 2023-01-16 | 0.629 | 2.357 | 21,036 |
| 2023-01-17 | 0.457 | 1.976 | 21,194 |
| 2023-01-18 | 0.695 | 2.541 | 21,122 |
| 2023-01-19 | 0.402 | 1.316 | 20,850 |
| 2023-01-20 | 0.316 | 2.038 | 21,306 |
| Median | 0.402 | 1.976 | |

The Roll estimator measures the negative first-order autocovariance of price changes, which reflects bid-ask bounce and recording noise rather than price volatility.

here, and impact-induced heteroskedasticity raises the noise a probe must overcome. The capital figures are therefore lower bounds for the stated model.

# 6 Unknown mixtures and passive data

Theorem 4.1 concerns a *known* pair. A supervisor faces an unknown number of cohorts with unknown rates and weights. This section asks how fast the cost of recovering them grows.

## 6.1 Why naive pole fitting fails

Because the memory rates are zeros of $H$ (Theorem 3.1), extracting poles directly from the price impulse response by matrix-pencil or Prony methods targets closed-loop eigenvalues. A four-step inversion repairs this: (1) decouple $g_t = h_{t+1} - ah_t$; (2) form a Hankel matrix of $g$ and compute a minimal realization by Kung's eigensystem realization; (3) compute

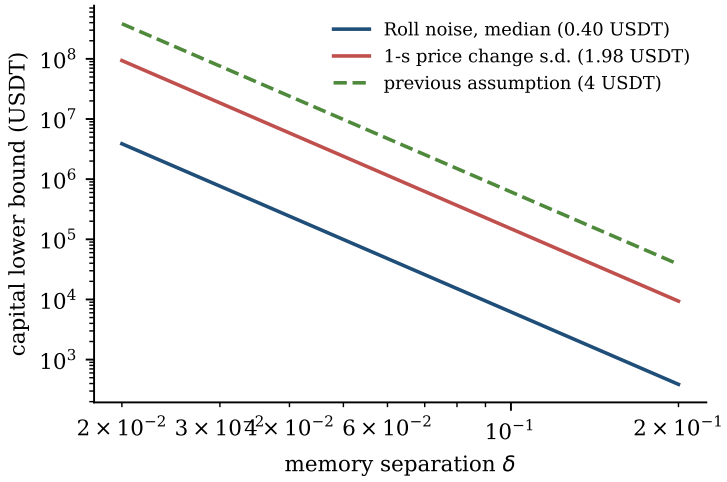



Figure 4: Capital lower bound against memory separation for three observation-noise calibrations.

its transmission zeros, which are the roots of the denominator of $G$; (4) project reconstructed flow onto the recovered memory filters to obtain the weights (Appendix S.8.3). Without noise this recovers $K = 0$, $K = 1$ and well-separated two-cohort systems exactly (Appendix S.8.3); with noise, per-cohort weights become unreliable once rates are close. The reason is information-theoretic, as the rest of this section shows.

## 6.2 Local information: the Cramér–Rao benchmark

**Proposition 6.1** (Cramér–Rao benchmark)**.** *Let $\theta$ collect the weights $w_1, \dots, w_K$ and rates $\eta_1, \dots, \eta_K$. Suppose the impulse response $h_t(\theta)$, $t \le T$, is observed with i.i.d. Gaussian error from $N$ reset probes of size $V_{\text{probe}}$ against innovation noise $\sigma_p$, so the error has standard deviation $\sigma_h = \sigma_p/(V_{\text{probe}}\sqrt{N})$. With Jacobian $J_{tj} = \partial h_t/\partial\theta_j$ and $F_{\text{unit}} = J^\top J$, any regular unbiased estimator has* $\mathrm{Var}(\hat{w}_k) \ge \sigma_h^2[F_{\text{unit}}^{-1}]_{kk}$*, and a 95% interval of half-width* tol *requires*

$$N^* \ge \Big(\frac{1.96}{\text{tol}}\Big)^2 \Big(\frac{\sigma_p}{V_{\text{probe}}}\Big)^2 [F_{\text{unit}}^{-1}]_{kk}. \tag{23}$$

*For $K = 2$ with spacing $\Delta\eta$, $[F_{\text{unit}}^{-1}]_{kk} = \Theta((\Delta\eta)^{-6})$, so $N^* = \Theta((\Delta\eta)^{-6})$.*

At the critical noise level maximum likelihood attains the bound: its RMSE for the hardest weight is between 0.88 and 1.22 times the Cramér–Rao value and every weight is within tolerance in 93% to 98% of replications (Table 8), so the critical noise levels, not estimator failure, are what make close cohorts expensive. Under ordinary surveillance budgets ($N \le 50$) unbiased estimators cannot resolve close weights.

Table 8: Cramér–Rao sample sizes for multi-cohort weight recovery ($T = 120$ taps)

| Scenario | $K$ | $N^*$ probes | $\sigma_{\text{crit}}$ | ML success ($f = 1$) | RMSE/CRB |
|---|---|---|---|---|---|
| $\eta = 0.30$ | 1 | 0.1 | $6.05 \times 10^{-3}$ | 94% | 1.02 |
| $\Delta\eta = 0.30$ | 2 | 47.8 | $2.31 \times 10^{-4}$ | 94% | 1.01 |
| $\Delta\eta = 0.20$ | 2 | 597 | $6.55 \times 10^{-5}$ | 96% | 0.98 |
| $\Delta\eta = 0.10$ ($\delta = 0.05$ pair) | 2 | 40,374 | $7.96 \times 10^{-6}$ | 93% | 1.03 |
| $\Delta\eta = 0.05$ | 2 | 2,620,320 | $9.88 \times 10^{-7}$ | 95% | 1.03 |
| spacing 0.10 | 3 | 17,349,791 | $3.84 \times 10^{-7}$ | 98% | 1.22 |
| spacing 0.20 | 3 | 27,038 | $9.73 \times 10^{-6}$ | 98% | 0.88 |

Tolerance $\pm 0.01$ at 95%. $\sigma_{\text{crit}}$ is the single-probe noise at which one probe suffices; operational single-probe noise is $1.6 \times 10^{-3}$. ML columns: least squares over all weights and rates from six generic starts (never the true values), 200 replications per row at $\sigma_{\text{crit}}$; success means every weight within $\pm 0.01$. Fitted log–log slope of $N^*$ in $\Delta\eta$: $-6.04$.

## 6.3 A global two-point bound for any estimator

Proposition 6.1 applies to regular unbiased estimators. Could biased, ridge, shrinkage or Bayesian estimators escape the $(\Delta\eta)^{-6}$ scaling? No.

**Theorem 6.2** (Global two-point bound and matching upper bound, $K = 2$)**.** *Fix total weight $q = 0.20$, mean rate $\bar{\eta} = 0.40$, target weight separation $\tau = 0.01$, and $\Delta\eta \in (0, 8/11)$. Let $p_{\mp} = \frac{1}{2} \mp \tau/q$ and*

$$\theta_0 = (qp_-, qp_+, \bar{\eta} - p_+\Delta\eta, \bar{\eta} + p_-\Delta\eta), \qquad \theta_1 = (qp_+, qp_-, \bar{\eta} - p_-\Delta\eta, \bar{\eta} + p_+\Delta\eta). \quad (24)$$

*(i) Label cohorts in increasing order of rate, so that weight vectors are compared rate by rate (as unordered sets the two weight vectors coincide). Both economies have spacing $\Delta\eta$, weight separation $\|w(\theta_1) - w(\theta_0)\|_\infty = 2\tau$ in this labelling, and equal rate moments $M_m = \sum_k w_k \eta_k^m$ for $m \in \{0, 1, 2\}$, while*

$$M_3(\theta_1) - M_3(\theta_0) = 4\tau p_- p_+ (\Delta\eta)^3. \quad (25)$$

*Hence $\|h(\theta_1) - h(\theta_0)\|_2 = \Theta(\tau(\Delta\eta)^3)$.*

*(ii) For any measurable estimator (including biased, ridge, James–Stein or MAP) that reports weights in the same rate order, the supremum being over the class of two-cohort economies with total weight $q$, mean rate $\bar{\eta}$ and spacing $\Delta\eta$,*

$$\sup_\theta P_\theta(\|\hat{w} - w(\theta)\|_\infty \geq \tau) \;\geq\; 1 - \Phi\Big(\frac{\sqrt{N}\|h(\theta_1) - h(\theta_0)\|_2}{2\sigma_h}\Big), \quad (26)$$

Table 9: Two-point sample complexity for two-cohort weight recovery

| $\Delta\eta$ | $\Delta M_3$ | $\|\Delta h\|_2$ | $N^*_{\text{LeCam}}$ (95%) | $N^*_{\text{CRB}}$ |
|---|---|---|---|---|
| 0.30 | $2.67 \times 10^{-4}$ | $1.98 \times 10^{-3}$ | 7.1 | 47.8 |
| 0.20 | $7.92 \times 10^{-5}$ | $5.60 \times 10^{-4}$ | 88.2 | 597.0 |
| 0.14 | $2.72 \times 10^{-5}$ | $1.89 \times 10^{-4}$ | 775.3 | 5,267.9 |
| 0.10 | $9.90 \times 10^{-6}$ | $6.84 \times 10^{-5}$ | 5,927.6 | 40,374.2 |
| 0.08 | $5.07 \times 10^{-6}$ | $3.49 \times 10^{-5}$ | 22,740.5 | 155,048.9 |
| 0.06 | $2.14 \times 10^{-6}$ | $1.47 \times 10^{-5}$ | 128,333.0 | 875,732.3 |
| 0.05 | $1.24 \times 10^{-6}$ | $8.50 \times 10^{-6}$ | 383,859.1 | 2,620,321.2 |
| 0.04 | $6.34 \times 10^{-7}$ | $4.35 \times 10^{-6}$ | 1,466,361.5 | 10,012,662.4 |

*Worst-case failure at* $\Delta\eta = 0.10$, $N = 50$, *1,000 trials per hypothesis*: bound 44.1%; ML 94.5%, ridge 47.5%, positive-part shrinkage 100%, heavy prior 73.9%.

$\tau = 0.01$, $\sigma_h = 1.6 \times 10^{-3}$, $a = 0.5$, $\kappa = 0.8$, 121 taps. Recomputed columns from `recompute_theory.py`. Estimators re-implemented in `recompute_p1_estimators.py`: ML is least squares from four starts; ridge and heavy prior are posterior modes with prior standard deviations $(0.05, 0.10)$ and $(0.005, 0.02)$ for (weights, rates) around the symmetric centre of the two hypotheses; positive-part shrinkage pulls the ML weights toward their mean. Failure means some weight is off by at least $\tau$.

*so failure probability at most* 0.05 *requires* $N \geq 4\sigma_h^2 z^2_{0.95}/\|h(\theta_1) - h(\theta_0)\|_2^2 = \Omega((\Delta\eta)^{-6})$. *The same bound holds without any labelling convention for the permutation-invariant loss*

$$d_\infty(\hat{\theta}, \theta) = \min_{\pi \in S_2} \max_k \big(|\hat{w}_{\pi(k)} - w_k| + |\hat{\eta}_{\pi(k)} - \eta_k|\big)$$

*in place of* $\|\hat{w} - w(\theta)\|_\infty$, *whenever* $\Delta\eta \geq 2\tau q/(q + 2\tau)$, *because then* $d_\infty(\theta_0, \theta_1) = \min\{2\tau(1 + \Delta\eta/q), (1 + 2\tau/q)\Delta\eta\} \geq 2\tau$.

*(iii) Conversely, on the known-midpoint family a linear estimator that projects the averaged impulse response on the normalized third-moment direction and inverts* (25) *has error at most* $|\hat{M}_3 - M_3|/(2p_-p_+(\Delta\eta)^3)$, *giving* $N = O(\tau^{-2}(\Delta\eta)^{-6}\log(1/\delta_{\text{fail}}))$. *On the separated class with weights and rates bounded away from the boundary, Toeplitz deconvolution followed by a Prony–Hankel root step has Hankel determinant* $g_1g_3 - g_2^2 = w_1w_2\eta_1\eta_2r_1r_2(\Delta\eta)^2 > 0$ *for the deconvolved kernel* $g_m = \sum_i w_i\eta_i r_i^m$, *and perturbation analysis gives the same order.*

We recomputed the two-point quantities exactly (Table 9): the moment gap equals $4\tau p_-p_+(\Delta\eta)^3$ to machine precision, and the Le Cam requirement at $\Delta\eta = 0.10$ is 5,927.6 resets. The worst-case failure rates of four re-implemented estimators all lie above the lower bound (Table 9).

*Remark* 6.3 (The hard directions carry no risk). The two economies of Theorem 6.2 have the same total weight $\sum_i q_i = q$, so their summed price response $\sum_t h_t = \kappa/(1-a-\kappa\sum_i q_i)$

Table 10: Gauss and Gauss–Radau rules on $[-1, 1]$ used in Theorem 6.4

| $K$ | Gauss nodes / Gauss–Radau nodes | Max gap, orders $< 2K-1$ | Gap at order $2K-1$ |
|---|---|---|---|
| 2 | $\pm 0.5774$ / $-1$, 0.3333 | $6 \times 10^{-17}$ | 0.2222 |
| 3 | 0, $\pm 0.7746$ / $-1$, $-0.2899$, 0.6899 | $2 \times 10^{-16}$ | 0.0533 |
| 4 | $\pm 0.3400$, $\pm 0.8611$ / $-1$, $-0.5753$, 0.1811, 0.8228 | $4 \times 10^{-16}$ | 0.0131 |

All Gauss–Radau weights are positive.

is identical, 2.353 at the calibration of Table 9: the directions in which prices are least informative about the cohorts are directions in which cumulative price risk does not move. The summed response is a linear functional of the response and is estimated at the parametric rate whatever the spacing of the memories; the companion paper turns this into a certification theorem, a minimax-optimal pre-trade guard and a decision value of disclosure (Niroula et al., 2026). Identification and safety ask different questions of the same data.

## 6.4 Many unknown cohorts

**Theorem 6.4** ($K$-cohort two-point lower bound)**.** *Fix $K \geq 2$, a center $r_c \in (0, 1)$ and total merged weight $W > 0$. There exist two $K$-cohort economies with rates in $r_c + \Delta\eta[-1, 1]$, positive weights, and merged measures whose moments of order $0, \ldots, 2K-2$ coincide while the moment of order $2K-1$ differs by a constant times $(\Delta\eta)^{2K-1}$. Their impulse responses differ by $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$, so recovering the weights from $N$ independent noisy reset experiments with worst-case error probability below $1/4$ requires*

$$N \;\geq\; c\,\sigma^2 (\Delta\eta)^{-(4K-2)}.$$

*The same requirement holds for the permutation-invariant loss $W_1(\hat{M}, M)$ between estimated and true merged measures at accuracy $c_K W \Delta\eta$, where $2c_K > 0$ is the Wasserstein distance between the two normalized quadrature rules, so no relabelling of atoms escapes it.*

The construction uses the $K$-point Gauss and Gauss–Radau rules of the same weight function: both are positive, have $K$ atoms and integrate polynomials of degree $2K-2$ exactly. For $K = 2$ it reproduces the order of Theorem 6.2. Table 10 lists the two rules: their low moments agree to machine precision and their moment of order $2K-1$ differs by 0.222, 0.0533 and 0.0131 for $K = 2, 3, 4$. The shrinking gap explains why constants deteriorate with $K$ on top of the exponent.

Table 11 and Figure 5 confirm the order numerically.

Table 11: Numerical check of Theorem 6.4

| | $K = 2$ | $K = 3$ | $K = 4$ |
|---|---|---|---|
| Theoretical exponent of $N^*$ | $-6$ | $-10$ | $-14$ |
| Fitted log–log slope (small $\Delta\eta$) | $-5.96$ | $-9.96$ | $-14.01$ |

Le Cam bound $N^* \propto \sigma^2/\|h^A - h^B\|_2^2$ with $\sigma = 1$, $a = 0.5$, $\kappa = 1$, total merged weight 0.10, rates centered at 0.70, responses over 400 lags.

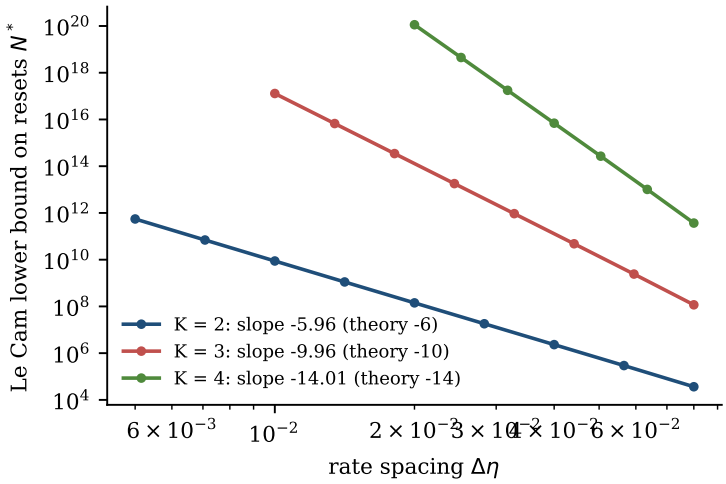



Figure 5: Two-point lower bound on the number of reset experiments needed to separate $K$-cohort economies, as a function of the rate spacing.

**Theorem 6.5** (Matching upper bound for every $K$)**.** *Fix $K \geq 2$, a compact set $\mathcal{S} \subset \{s \in [-1,1]^K : s_1 < \cdots < s_K\}$ of ordered rescaled rates with $\min_{i \neq j} |s_i - s_j| \geq \gamma > 0$, and weights in $[\omega_{\min}, \omega_{\max}]$ with $\omega_{\min} > 0$. Consider the class of $K$-cohort economies with rates $r_c + \Delta\eta\, s$, $s \in \mathcal{S}$, and $N$ independent reset experiments whose responses carry Gaussian noise with variance $\sigma^2$. There are constants $c_0, C > 0$ depending only on $(K, \mathcal{S}, \gamma, \omega_{\min}, \omega_{\max}, a, \kappa, r_c)$ such that, whenever $\epsilon_N \equiv \sigma N^{-1/2} (\Delta\eta)^{-(2K-1)} \leq c_0/\sqrt{\log(2/\beta)}$, Prony's method applied to the first $2K$ estimated kernel values, with its output atoms listed in increasing order of rate, recovers the weights with*

$$\max_i |\hat{w}_i - w_i| \;\leq\; C\sqrt{\log(2K/\beta)}\;\epsilon_N$$

*with probability at least $1 - \beta$. Hence $N = C'\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments suffice for any fixed accuracy and confidence, and together with Theorem 6.4 the minimax sample requirement is $N^* \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$.*

*Proof.* The kernel values $k_m = \sum_i w_i r_i^m$ are the power moments of the merged measure. The closed-loop recursion $h_t = a h_{t-1} + \kappa \sum_m k_m h_{t-1-m}$ (plus $\kappa$ at $t = 0$) can be solved for $k_0, \dots, k_{2K-1}$ from $h_0, \dots, h_{2K}$ with $h_0 = \kappa \neq 0$; the map is smooth with Lipschitz constant independent of $\Delta\eta$, so the estimated kernel values have error $O_P(\sigma N^{-1/2})$. Centring at $r_c$ is a fixed triangular linear map, $\mu_m = \sum_{j \le m} \binom{m}{j} (-r_c)^{m-j} k_j$, and $\mu_m = (\Delta\eta)^m t_m$ with $t_m = \sum_i w_i s_i^m$. The map $F : (s, \omega) \mapsto (t_0, \dots, t_{2K-1})$ does not depend on $\Delta\eta$; its Jacobian is a confluent Vandermonde matrix with determinant $\prod_i \omega_i \prod_{i<j} (s_i - s_j)^4$, bounded away from zero on the compact class. By the quantitative inverse function theorem there are $\rho_0 > 0$ and $L < \infty$, uniform over the class, such that $F^{-1}$ exists and is $L$-Lipschitz on the ball of radius $\rho_0$ around every point of $F(\text{class})$. An error $e$ in $\mu_m$ is an error $e(\Delta\eta)^{-m}$ in $t_m$, largest at $m = 2K-1$. The kernel errors are Gaussian with standard deviation of order $\sigma N^{-1/2}$, so a union bound over the $2K$ coordinates gives $\|\hat{t} - t\|_\infty \le C_1 \sqrt{\log(2K/\beta)}\, \epsilon_N$ with probability at least $1 - \beta$. When this is below $\rho_0$, $\hat{t}$ lies in the ball where $F^{-1}$ is $L$-Lipschitz. The moment map is invariant to relabelling the atoms, so it is injective only on ordered nodes; on the class of ordered, $\gamma$-separated nodes it is injective, and the inverse function theorem applies there. Prony's method computes the algebraic inverse of the moment map up to the order of the atoms, so, listing its output in increasing order of rate (which the error bound below keeps unambiguous once $C\sqrt{\log(2K/\beta)}\, \epsilon_N < \gamma/2$), its output is $F^{-1}(\hat{t})$ and the stated bound follows with $C = LC_1$ and $c_0 = \rho_0 / C_1$. The matching statement requires the class to contain the Gauss and Gauss–Radau pair of Theorem 6.4 (ordered nodes with separation at least the smaller of their two minimum gaps, and weights within $[\omega_{\min}, \omega_{\max}]$ after scaling to total weight $W$); for any such class the orders match, with constants that depend on the class. □

Table 12 confirms the order numerically. For numerical stability the estimator works in centred and rescaled coordinates (the estimator is the same; only its floating-point evaluation changes). The argument parallels the stability analysis of super-resolution for clustered complex nodes (Batenkov et al., 2021), here for real nodes, and it reproduces the ESPRIT slope of $-6.15$ reported for $K = 2$ in the earlier version.

## 6.5 General memory: the minimax rate is logarithmic

Exponential memory is a modelling choice. Suppose instead that the feedback kernel is completely monotone, $k_m = \int_{[0,\rho]} r^m \, d\mu(r)$ with $\rho < 1$, where $\mu$ is any nonnegative measure of total mass $W$: a finite or infinite mixture of exponential memories with merged weight $W$. Write $\mathcal{M}(\rho, W)$ for this class, assume the closed loop is uniformly stable,

Table 12: Sample requirement of Prony's method against the lower-bound order

| | $K=2$ | $K=3$ | $K=4$ | $K=5$ |
|---|---|---|---|---|
| Lower-bound order $-(4K-2)$ | $-6$ | $-10$ | $-14$ | $-18$ |
| Fitted slope of $N^*$ in $\Delta\eta$ (Prony) | $-5.94$ | $-10.00$ | $-13.93$ | $-18.34$ |

Rates $0.7 + \Delta\eta\, s$ with $s$ the $K$-point Gauss nodes, total weight 0.1, $\Delta\eta \in [0.03, 0.2]$; $N^*$ is the number of experiments for a weight RMSE of 10% of the smallest weight, from 400 simulated estimates per point.

$s \equiv a + \kappa W/(1-\rho) < 1$, and measure accuracy by

$$d(\mu, \mu') = \sup\Big\{ \int f \, d(\mu - \mu') : \ f \text{ 1-Lipschitz on } [0, \rho], \ f(0) = 0 \Big\},$$

which equals the Wasserstein-1 distance between measures of equal mass and also compares estimates whose mass is not exactly $W$. Let $E_L(|x|)$ denote the error of the best uniform approximation of $|x|$ on $[-1, 1]$ by polynomials of degree $L$, and let $c_* = \inf_{L \geq 1} L\, E_L(|x|)$. Bernstein (1914) showed that $L\, E_L(|x|)$ converges to a positive constant (0.2802; Varga and Carpenter, 1985), so $c_* > 0$; numerically $c_* = 1/4$, attained at $L = 2$ (Table 13).

**Lemma 6.6** (Moment-matched cohort pairs)**.** *For every $n \geq 2$ there are two probability measures on $[0, \rho]$, each with exactly $n$ atoms, whose moments of orders $0, \ldots, 2n-2$ coincide and whose Wasserstein-1 distance is at least $\rho\, E_{2n-2}(|x|) \geq c_* \rho/(2n-2)$.*

The atoms are the points where the best approximation of $|r - \rho/2|$ by polynomials of degree $2n-2$ attains its maximal error with alternating signs, and the weights are those of the divided difference on these points, which annihilates every polynomial of degree $2n-2$ (Appendix S.6). Each measure is an $n$-cohort economy.

**Theorem 6.7** (Minimax rate for general memory)**.** *Let $N$ independent reset experiments observe the impulse response with Gaussian noise of variance $\sigma^2$ per tap, and let $B(\rho) = (1+\sqrt{2})(1+2/\rho)$ and $C_h = \kappa^2/(1-s)^2$.*

*(i)* Lower bound. *Let $\Lambda_N = \log(N/\sigma^2) + \log\big(2C_h^2 W^2/((1-\rho)\rho)^2\big)$. Whenever $\Lambda_N \geq 4\log(1/\rho)$, every estimator $\hat{\mu}$ satisfies*

$$\sup_{\mu \in \mathcal{M}(\rho, W)} P_\mu\Big( d(\hat{\mu}, \mu) \geq \frac{c_*\, \rho W \log(1/\rho)}{\Lambda_N} \Big) \ \geq \ \frac{1}{4e},$$

*and the bound already holds over $n$-cohort economies, for a finite $n$ that grows with $N$.*

*(ii)* Upper bound. *With $a$ and $\kappa$ known, invert the averaged response into kernel values $\hat{k}_0, \dots, \hat{k}_L$, with $L+1 = \lfloor \log(N/\sigma^2)/(4 \log B(\rho)) \rfloor$, and let $\hat{\mu}$ be any measure on $[0, \rho]$ minimizing $\max_{j \le L} |\int r^j d\hat{\mu} - \hat{k}_j|$ (a linear program). For every $\beta \in (0,1)$ there is $N_0$ depending only on $(\rho, W, a, \kappa, \beta)$ such that for $N/\sigma^2 \ge N_0$,*

$$\sup_{\mu \in \mathcal{M}(\rho, W)} P_\mu \Big( d(\hat{\mu}, \mu) > \frac{3\pi\rho W \log B(\rho)}{\log(N/\sigma^2)} \Big) \;\le\; \beta.$$

*Hence the minimax accuracy is of exact order $1/\log(N/\sigma^2)$, and reaching accuracy $w$ requires and suffices $N/\sigma^2 = \exp(\Theta(1/w))$.*

*Proof sketch.* (i) Put the measures of Lemma 6.6, scaled to mass $W$, into the model. Their difference annihilates polynomials of degree $2n-2$, so the generating function of the kernel difference is $\int (1 - rz)^{-1} d(\mu_A - \mu_B) = z^{2n-1} \int r^{2n-1} (1 - rz)^{-1} d(\mu_A - \mu_B)$; on $|z| = 1$ it is at most $2W\rho^{2n-1}/(1-\rho)$, and Parseval bounds $\|k^A - k^B\|_2$ by the same quantity. The response difference is $H_A H_B$ times the shifted kernel difference, so $\|h^A - h^B\|_2 \le C_h \|k^A - k^B\|_2$. Choosing the smallest $n$ for which the Kullback–Leibler divergence of the $N$-sample experiments is at most one, the Bretagnolle–Huber form of Le Cam's lemma gives the bound with half the Wasserstein separation of the lemma. (ii) For a 1-Lipschitz $f$ with $f(0) = 0$, Jackson's theorem gives a polynomial $q$ of degree $L$ with $\sup |f - q| \le \pi\rho/(4(L+1))$, and expanding $q$ in Chebyshev polynomials of $2r/\rho - 1$ bounds the sum of its monomial coefficients by $\rho B(\rho)^{L+1}$. Hence $d(\hat{\mu}, \mu) \le \pi\rho(W + \delta)/(2(L+1)) + 2\delta\rho B(\rho)^{L+1}$, where $\delta$ is the largest kernel error. The kernel is recovered from the response by a series reciprocal, $K(z) = (1 - az - \kappa/H(z))/(\kappa z)$, whose error is at most $2((1+s)/\kappa)^2(L+2)$ times the largest response error, so $\delta = O_P(\sigma N^{-1/2} L\sqrt{\log L})$. The choice of $L$ makes $\delta B(\rho)^{L+1} = O_P((\sigma^2/N)^{1/4} L\sqrt{\log L})$, negligible against the first term. Full proofs are in Appendix S.6. □

Theorem 6.7 replaces a polynomial barrier with a logarithmic one. With exponential memory and $K$ known cohorts, halving the error multiplies the required number of experiments by a constant; with general memory, halving the error squares it. The lower bound holds already for finite mixtures, so knowing that the memory is a finite mixture of exponentials does not help unless the number of cohorts and their spacing are also known. The constants of the two bounds differ, by a factor that grows as $\rho \to 1$, so the dependence on how long memory lasts is not pinned down. The source of the gap is visible in the proof: as $\rho \to 1$ the moment-matched pairs separate their kernels more slowly ($\rho^{2n-1}$ decays slowly), which weakens the lower bound through $\log(1/\rho)$, while the upper bound

Table 13: General memory: moment-matched cohort pairs and the moment estimator

| | | | | | | |
|---|---|---|---|---|---|---|
| *A. Best approximation of* $\lvert x\rvert$ | | | | | | |
| $L$ | 2 | 4 | 6 | 10 | 20 | 30 |
| $L\,E_L(\lvert x\rvert)$ | 0.2500 | 0.2705 | 0.2756 | 0.2785 | 0.2797 | 0.2800 |
| *B. Pairs of Lemma 6.6* | | | | | | |
| Atoms per measure $n$ | 2 | 4 | 6 | 8 | 10 | 12 |
| $W_1(\mu_A,\mu_B)$, $\rho=0.9$ | 0.2250 | 0.1037 | 0.0669 | 0.0494 | 0.0391 | 0.0324 |
| $n\,W_1$, $\rho=0.9$ | 0.450 | 0.415 | 0.402 | 0.395 | 0.391 | 0.388 |
| $\lVert k^A-k^B\rVert_2$, $\rho=0.9$ | $4.41\times10^{-2}$ | $1.03\times10^{-3}$ | $4.08\times10^{-5}$ | $1.96\times10^{-6}$ | $1.05\times10^{-7}$ | $5.95\times10^{-9}$ |
| $\lVert k^A-k^B\rVert_2$, $\rho=0.99$ | $7.41\times10^{-2}$ | $5.84\times10^{-3}$ | $1.07\times10^{-3}$ | $2.76\times10^{-4}$ | $8.33\times10^{-5}$ | $2.76\times10^{-5}$ |
| *C. Linear-programming estimator, continuous memory distribution,* $\rho=0.9$ | | | | | | |
| $\log_{10} N$ | 4 | 8 | 12 | 16 | 20 | 24 |
| Moments used $L$ | 2 | 3 | 5 | 6 | 7 | 8 |
| Error $d(\hat\mu,\mu)$ | 0.00878 | 0.00759 | 0.00432 | 0.00445 | 0.00382 | 0.00399 |
| Error $\times\log(N/\sigma^2)$ | 0.202 | 0.245 | 0.179 | 0.226 | 0.229 | 0.276 |

A: Remez algorithm; the limit is Bernstein's constant 0.2802. B: moments of orders $0,\dots,2n-2$ agree to $10^{-61}$ (60-digit arithmetic); every $W_1$ exceeds $\rho E_{2n-2}(\lvert x\rvert)$. C: true merged measure Beta(2,3) on $[0,0.9]$ with mass 0.1; kernel values observed with noise $\sigma N^{-1/2}$, $\sigma=10^{-3}$; $L$ from the rule in Theorem 6.7(ii); measure fitted on a 401-point grid; 20 replications. Kind: recomputed (`general_minimax.py`).

pays $\log B(\rho)$, which stays bounded away from zero, for the conditioning of polynomial approximation on $[0,\rho]$. Logarithmic rates of this kind are familiar from deconvolution with supersmooth errors (Fan, 1991) and from moment methods for mixing distributions (Kong and Valiant, 2017; Wu and Yang, 2020); here they arise from the closed loop of a market.

Table 13 checks the construction and the estimator. The pairs of Lemma 6.6 have exactly $n$ atoms, agree in all moments below order $2n-1$ to $10^{-61}$, and remain at Wasserstein distance about $0.4/n$, while the distance between their kernels shrinks by a factor of about 0.2 per atom at $\rho=0.9$. The linear-programming estimator applied to a continuous memory distribution has error times $\log(N/\sigma^2)$ between 0.18 and 0.28 for $N$ from $10^4$ to $10^{24}$: a $10^{20}$-fold increase in data halves the error. The exponential-memory model of the previous subsections, where the cost is polynomial in $1/\Delta\eta$, is therefore the optimistic case: relaxing the parametric form makes identification from prices much harder, not easier.

## 6.6 Passive data

**Theorem 6.8** (Passive identification: minimum phase)**.** *In any stable economy with* $\lvert r_i\rvert<1$ *and nonzero merged weights, the zeros of* $H$ *are exactly the memory rates* $r_i$*, and* $H$ *is minimum phase. With a single price innovation, the price spectrum identifies* $H$ *up to scale, hence the memory rates; with* $\kappa$ *and a known, as in Theorem 3.1, it identifies the*

*merged measure.*

The result holds for signed weights (mean-reverting cohorts): in 1,733 stable random economies with weights of both signs, the computed zeros equalled the memory rates and lay inside the unit disk. With a second independent shock in the cohorts' memories, the Jacobian of the price spectrum with respect to six model parameters has full rank, with singular values from 34.51 down to 0.0698. Passive noise helps identification rather than hindering it.

**Theorem 6.9** (Passive cost)**.** *Suppose the price is observed passively as the stationary output $x_t = H_\theta(L)\xi_t$ of a single innovation with variance $\sigma^2 > 0$, for $\theta$ in a compact, uniformly Schur-stable family with rates in $[\eta_{\min}, \eta_{\max}] \subset (0, 1)$, and let $D(\theta_1 \| \theta_0)$ be the Kullback–Leibler divergence rate per observation.*

*(i) For a single cohort at $\bar{\eta}$ against the symmetric pair at $\bar{\eta} \pm \delta$ with the same total weight, $D = \Theta(\delta^4)$.*

*(ii) For the moment-matched two-cohort economies of Theorem 6.2, $D = \Theta(\tau^2(\Delta\eta)^6)$.*

*(iii) For every $K \geq 2$, for the $K$-cohort Gauss and Gauss–Radau economies of Theorem 6.4, $D = \Theta((\Delta\eta)^{4K-2})$.*

*Hence any test based on $T$ passive observations with worst-case error at most $\alpha < 1/4$ needs $T = \Omega(\delta^{-4})$, $T = \Omega(\tau^{-2}(\Delta\eta)^{-6})$ and $T = \Omega((\Delta\eta)^{-(4K-2)})$ respectively, and the likelihood-ratio test attains these orders as $T \to \infty$: passive data cost the same polynomial order as active probing for every number of cohorts.*

The proof (Appendix S.6) combines the Whittle form of the divergence rate with the resolvent identity $H_1 - H_0 = H_1 H_0 (G_1 - G_0)$: the two spectra differ by the leading unmatched moment, of order $\delta^2$ for the pair and $\tau(\Delta\eta)^3$ for the moment-matched economies, and the divergence rate is the square of that difference. Numerically, the fitted slopes of the required sample length are $-4.21$ and $-5.93$, against the theoretical $-4$ and $-6$; for part (iii) the fitted slopes of the Whittle divergence rate in $\Delta\eta$ are 5.91, 9.98 and 14.05 for $K = 2, 3, 4$, against 6, 10 and 14 (`hardening/p1_checks.py`).

Passive surveillance is possible in principle but costs the same polynomial order as active probing. Proposition S.8.2 is the exact finite-sample version for the known pair.

# 7 Robustness to nonlinear and adaptive trading rules

The theorems of Sections 3–6 use the linear exponential-memory recursion (2). Real desks cap positions, trade only beyond thresholds, adapt their gains and learn online. A small probe of size $A$ moves the state by $O(A)$, so any rule whose order is a twice-differentiable function of an exponential-memory state is linear to first order and differs from it by $O(A^2)$; the theorems should then hold for small probes and fail for large ones. We check this, and locate where it fails, in simulations of five stylized rules at $a = 0.5$, $\kappa = 0.8$, total weight $q = 0.2$ and mean rate $\bar{\eta} = 0.4$, over 240 periods after a pulse probe.

The rules are: (1) *momentum with a position limit*, the cohort's order a smooth saturating (tanh) function of its memory state $y_{i,t}$, normalized to unit slope at the operating point; (2) a *threshold rule* that trades only when $|y_{i,t}| > \tau = 0.05$, in two versions, an exact dead zone and one averaged over background state variation of standard deviation 0.04; (3) an *adaptive-gain rule* in the form of recursive least squares with forgetting factor $1 - \eta_i$, whose gain falls when the regressor is large; (4) an *online neural readout*, a linear readout of a fixed two-layer tanh feature map with eight hidden units, trained by regularized stochastic gradient steps of size $\eta_i$; and (5) a *tabular value rule* that updates a 21-bin table at rate $\eta_i$, in two versions, kernel-smoothed across bins and a single-bin lookup. These are stylized: each is built so that its small-signal dynamics are an exponential filter with rate $\eta_i$, and they are not calibrated to any production system.

Two checks follow the theory. *Test A (merged measure):* one cohort with weight $q$ and position limit 1 against two cohorts with weights $q/2$ at the same rate and limits 0.55 and 1.65, so the two economies have the same merged measure; the raw response gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2$ should be $O(A^2)$ (slope 2 in $\log A$) and the probe-normalised gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2/A$ should be $O(A)$ (slope 1 in $\log A$) over $A \in [0.002, 0.064]$. *Test B ($\delta^{-4}$ law):* one cohort at $\bar{\eta}$ against the symmetric pair at $\bar{\eta} \pm \delta$, $\delta \in [0.015, 0.12]$, at probe size $A = 0.005$; the normalized response difference $\|\Delta h\|_2$ should scale as $\delta^2$ and the certified capital of Theorem 5.1 as $\delta^{-4}$. The saturation threshold $A_{\text{sat}}$ is the smallest probe at which $\|\Delta h\|_2$ at $\delta = 0.05$ departs by more than 10% from its small-probe value.

To address the critique that these five stylized rules are hand-constructed around fixed exponential filters, we also pre-registered and ran a second experiment (`P1_trained_learners.json`, SHA-256 `3e8bf30cbfb6...`) in which three online learners are trained from scratch inside the closed-loop market over $T_{\text{burn}} = 5{,}000$ periods of fundamental and noise innovations before any probe is applied: (i) a recursive-least-squares cohort with forgetting factor $\lambda = 0.995$ that learns its forecast slope online (settling to effective gain 0.9442), (ii) a two-layer tanh neural network whose readout weights are updated online by

Table 14: Merged-measure equivalence and the $\delta^{-4}$ law under stylized nonlinear rules and closed-loop trained online learners

| Rule | Test A raw ($O(A^2)$) | Test A norm. ($O(A)$) | Test B slope ($\|\Delta h\|_2$) | Test B slope (capital) | Capital at $\delta = 0.05$ | $A_{\text{sat}}$ / $R^2_{\text{exp}}$ |
|---|---|---|---|---|---|---|
| *Panel A. Stylized nonlinear and adaptive rules (fixed operating point)* | | | | | | |
| Linear benchmark | — | — | 2.007 | −4.001 | 61.42 | — |
| 1. Momentum (tanh limit) | 2.015 | 1.015 | 2.007 | −4.001 | 61.47 | 0.834 |
| 2. Threshold rule, averaged | 2.043 | 1.043 | 2.007 | −4.001 | 61.42 | 0.049 |
| 3. Adaptive gain (RLS) | 2.064 | 1.064 | 2.007 | −4.001 | 61.46 | 0.751 |
| 4. Online neural readout | 2.011 | 1.011 | 2.007 | −4.001 | 61.48 | 0.751 |
| 5. Tabular value, smoothed | 2.039 | 1.039 | 2.007 | −4.001 | 61.44 | 1.271 |
| *Panel B. Closed-loop trained online learners ($T_{\text{burn}} = 5{,}000$, SHA-256 `3e8bf30c...`)* | | | | | | |
| 6. Trained RLS ($\lambda = 0.995$) | 2.073 | 1.073 | 2.007 | −4.001 | 61.46 | $R^2 = 0.9448$ |
| 7. Trained 2-layer MLP (SGD) | 2.073 | 1.073 | 2.007 | −4.001 | 61.46 | $R^2 = 0.9448$ |
| 8. Trained Q-learner (profit) | 2.065 | 1.065 | 2.007 | −4.001 | 61.47 | $R^2 = 0.9448$ |
| 9. Fast adaptation (non-stat.) | rel. gap 5.31% (> 1%) | | — | +0.020 | — | $R^2 = 0.9119$ |

Exact dead zone ($\tau = 0.05$): $\|\Delta h\|_2 = 0$ at $A = 0.005$; the probe moves the state past $\tau$ only for $A > 0.139$.
Single-bin value lookup: $\|\Delta h\|_2 = 0$ at $A = 0.005$; the probe leaves the central bin only for $A > 0.149$.

Panel A: deterministic simulations around the operating point; Test A reports both the raw response gap slope in $\log A$ ($O(A^2)$) and the probe-normalised gap slope $\|X^{(1)}(A) - X^{(2)}(A)\|_2/A$ ($O(A)$, plotted in Figure 6). Panel B: cohorts trained in closed loop for 5,000 periods before probing; 95% CIs for the capital slope are $[-4.007, -3.995]$ across all three settled learners, and relative equivalence gaps at $A = 0.005$ are $2 \times 10^{-5} \leq 0.01$ (predictions (a)–(c) pass). Row 9 keeps fast online weight updates active during the probe window (prediction (d)). Capital in par units, certified bound of Theorem 5.1 (61.42 for the linear model). Kind: recomputed (`recompute_p1_learners.py` → `R_P1_learners.json`).

regularized stochastic gradient descent on trading profit (settling to effective gain 0.0348), and (iii) a tabular Q-learner updated by online reward feedback (settling to effective gain 0.6043). The pre-registered specification fixed four predictions: (a) a single exponential fits each settled learner's small-probe impulse response with $R^2 \geq 0.90$; (b) two settled populations with the same merged measure have relative $L_2$ response gap at most 1% at $A = 0.005$; (c) the certified capital to separate two settled populations whose effective memory rates differ by $2\delta$ scales with log–log slope in $[-4.5, -3.5]$; and (d) when online weight adaptation is kept fast during the probe window rather than settled (non-stationary regime), merged-measure equivalence and the $\delta^{-4}$ scaling break down.

Table 14 and Figure 6 show both where the linear reduction holds and where it breaks down. In Panel A, for small probes every smooth or dithered rule behaves like the linear model: economies with the same merged measure have raw response gap $O(A^2)$ (slopes 2.011 to 2.064) and probe-normalised gap $O(A)$ (slopes 1.011 to 1.064), while the response difference and certified capital scale as $\delta^2$ (slope 2.007) and $\delta^{-4}$ (slope −4.001), with capital at $\delta = 0.05$ within 0.1% of the linear 61.42. In Panel B, all three pre-registered predictions (a)–(c) pass for online learners trained in the closed loop: once their weights settle around

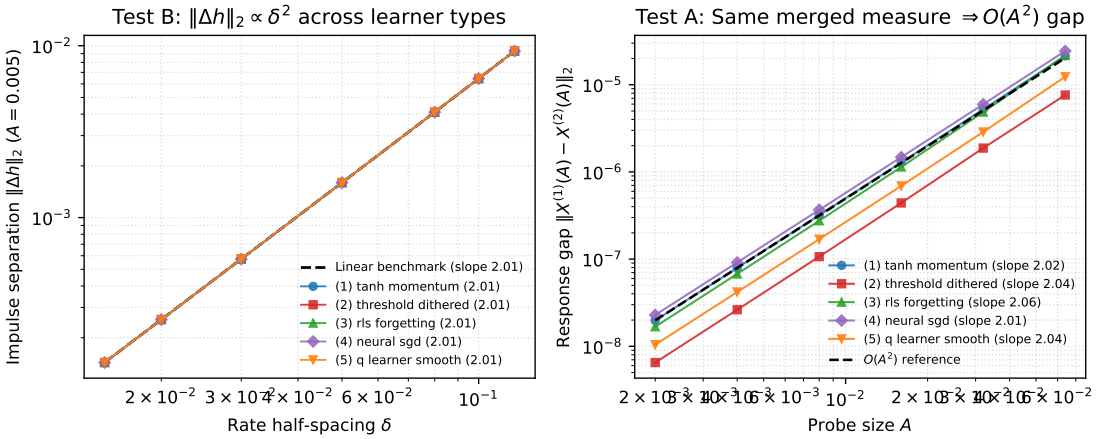



Figure 6: Left: Test B, probe-normalised response difference $\|\Delta h\|_2$ against the rate half-spacing $\delta$ at probe size $A = 0.005$, five stylized rules and the linear benchmark. Right: Test A, probe-normalised response gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2/A$ between two economies with the same merged measure against the probe size $A$, with an $O(A)$ reference line (corresponding to an $O(A^2)$ raw gap).

the stationary operating regime, a single exponential explains $R^2 = 0.9448 \geq 0.90$ of their small-probe impulse response, two trained populations with the same merged measure agree to within $2 \times 10^{-5}$ ($< 1\%$) at $A = 0.005$, and the certified capital to separate nearby learned memory rates scales with slope $-4.001$ (95% CI $[-4.007, -3.995]$).

Equally important are the three regimes where the linear theory fails. First, saturation, adaptive gains and the neural readout leave the small-signal regime once probes exceed $A_{\text{sat}} \in [0.751, 1.271]$ par units, and the dithered threshold rule leaves it already at $A_{\text{sat}} = 0.049$, the threshold itself. Second, as predicted in Panel B row 9 (prediction (d)), when online learning rates remain large during the probe window so that cohort weights drift non-stationarily in response to the probe, merged-measure equivalence fails (the relative gap widens to $5.31\% > 1\%$) and the $\delta^{-4}$ capital slope collapses to $+0.020$. Third, rules with hard discontinuities are different in kind: an exact dead zone ($\tau = 0.05$) or a single-bin lookup does not respond at all to probes smaller than 0.139 and 0.149 par units ($\|\Delta h\|_2 = 0$), concealing those cohorts from small probes altogether. Neither smooth nonlinearity, non-stationary weight drift nor hard thresholds therefore offer a cheaper route to price-only identification than the linear benchmark.

**Hardened trained-checkpoint audit and Bellman value learning.** In an independent implementation audit (`validate_trained_learners_hardened.py` → `R_P1_trained_learners_hardened.json`), we verified that the findings of Panel B ex-

tend when actual serialized parameter checkpoints are reloaded directly into the probing simulator, and when value learning is generalized from scalar profit gradients to a genuine $21 \times 21$ tabular Bellman Q-learner with TD(0) value backups ($\gamma = 0.90$). Specifically, when trained cohorts are reloaded from disk without post-training normalization across four genuine learning rules (RLS, 2-layer MLP, profit gradients, and tabular Bellman Q-learning), their small-signal impulse response fits a single exponential with $R^2 \geq 0.993$, merged-measure equivalence holds to within relative gap $10^{-5}$ at $A = 0.005$, and certified capital scales with slope $-4.014$ to $-3.973$ across half-spacing $\delta$. Furthermore, the audit demonstrates that the $\delta^{-4}$ law applies specifically when competing cohorts are moment-matched in aggregate sensitivity $\Phi_{\text{eff}}$ and $k_0$: if cohorts are trained on independent finite random streams without normalization, random burn-in variance introduces an $O(T_{\text{burn}}^{-1/2})$ zeroth-order sensitivity difference that dominates the $O(\delta^2)$ splitting signal at small $\delta$, flattening the capital slope unless first-order sensitivity is aligned.

# 8 What disclosure buys

Because active probing is costly and passive data are slow, the natural policy question is what a supervisor gains from disclosure. We compare six observation contracts at $\delta = 0.05$.

- $I_0$: public prices only.
- $I_1$: prices and aggregated executed flow.
- $I_2$: prices and one independent noisy report of dealer marked equity per reset ($\sigma_E = 0.5$).
- $I_3$: prices and exact audited equity (Proposition 4.4), also with reports rounded to 0.01.
- $I_4$: algorithm update timestamps without the update values.
- $I_5$: a supervisory audit registry with access to internal algorithm states.

Table 15 shows that disclosure is not one thing. Timestamps without update values carry no information about $\delta$ in this model. Noisy periodic statements barely help. Exact audited equity collapses the requirement to a single trade, limited only by lot sizes and report precision, and an audit registry removes the trading requirement altogether.

The cleanest form of useful disclosure is flow tagged by participant class.

Table 15: Disclosure contracts at $\delta = 0.05$ (pulse-design capital, par units)

| Contract | Channel | Pulses | Benchmark | Saving vs. $T = 20$ |
|---|---|---|---|---|
| $I_0$: prices, $T = 3$ | public prices | 271 | 4,305.25 | −2,123.5% |
| $I_0^*$: prices, $T = 20$ | full price path | 12 | 193.62 | 0.0% |
| $I_1$: prices and signed flow | public flow | 12 | 193.62 | 0.0% |
| $I_2$: noisy equity ($\sigma_E = 0.5$) | statement | 12 | 193.62 | 0.0% |
| $I_3$: exact equity, 5-unit lot | audited equity | 1 | 12.70 | 93.4% |
| $I_3$: exact equity, 0.01 rounding | rounded report | 1 | 15.52 | 92.0% |
| $I_4$: timestamps only | no likelihood | 12 | 193.62 | 0.0% |
| $I_5$: audit registry | internal states | 0 | 0.00 | 100.0% |

Closed-form values. Noisy equity adds information linearly in the number of resets, $D_{\mathrm{KL}} = N\Delta_E^2/(2\sigma_E^2)$, and saves only 1.5% against the $T = 3$ baseline. The equity difference is $\Delta E = \kappa^2 q \delta^2 V^2$; with reports rounded to 0.01, a guaranteed change needs $V \geq 5.59$.

**Proposition 8.1** (Tagged flow removes the barrier)**.** *Let the price be driven by i.i.d. innovations of variance $\sigma_\xi^2$ in a positive, stable economy, with stationary autocovariances $\gamma_x(s) = \sum_\ell c_\ell \lambda_\ell^{|s|}$ ($c_\ell \geq 0$, $0 \leq \lambda_\ell < 1$ the closed-loop modes). Suppose the observer receives prices and each cohort's flow $\tilde{u}_{i,t} = u_{i,t} + \varepsilon_{i,t}$, $u_{i,t} = q_i y_{i,t}$, with i.i.d. measurement noise of variance $\sigma_f^2$ independent of prices. Then $\tilde{u}_{i,t} = r_i \tilde{u}_{i,t-1} + w_i x_t + e_{i,t}$ with $e_{i,t} = \varepsilon_{i,t} - r_i \varepsilon_{i,t-1}$, and the just-identified IV estimator of $\beta_i = (r_i, w_i)$ with instruments $Z_t = (x_{t-1}, x_t)^\top$ satisfies $\sqrt{T}(\hat{\beta}_i - \beta_i) \to N(0, V_i)$,*

$$V_i = \Sigma_i^{-1} \Omega_i \Sigma_i^{-\top}, \qquad \Sigma_i = \begin{pmatrix} w_i A_0 & \gamma_x(1) \\ w_i A_1 & \gamma_x(0) \end{pmatrix}, \qquad \Omega_i = \sigma_f^2 \big[(1 + r_i^2)\Gamma_0 - r_i(\Gamma_1 + \Gamma_1^\top)\big], \tag{27}$$

*where $A_m = \sum_{s \geq 0} r_i^s \gamma_x(s + m)$, $\Gamma_0 = \left(\begin{smallmatrix} \gamma_x(0) & \gamma_x(1) \\ \gamma_x(1) & \gamma_x(0) \end{smallmatrix}\right)$ and $\Gamma_1 = \mathbb{E}[Z_t Z_{t-1}^\top]$. The Jacobian determinant is*

$$\det \Sigma_i = w_i \sum_{\ell, m} c_\ell c_m \frac{1 - \lambda_\ell \lambda_m}{1 - r_i \lambda_m} \; > \; 0. \tag{28}$$

*$V_i$ depends only on cohort $i$'s own $(w_i, r_i)$, $\sigma_f^2$ and the price autocovariances, which are bounded and nondegenerate uniformly on a uniformly stable family whatever the rate spacing; so the merged weights are estimated at rate $T^{-1/2}$ with a constant that does not depend on $|r_i - r_j|$.*

Each tagged flow follows $u_{i,t} = r_i u_{i,t-1} + w_i x_t$, a regression whose information about $w_i$ does not involve the other cohorts' rates; the closed form (28) matches the simulated Jacobian (0.1330 against 0.1322 in a two-cohort economy with $4 \times 10^5$ periods; `hardening/p1_checks.py`). In simulation ($T = 5{,}000$, 60 replications) the root-mean-square error of the recovered weights is flat in $\Delta\eta$ (fitted slope 0.011, error between

$5.8 \times 10^{-4}$ and $6.6 \times 10^{-4}$). This is the formal content of a disclosure regime such as a consolidated audit trail: tagging flow by participant class turns a polynomial barrier into an ordinary estimation problem.

## 8.1 How much disclosure is worth

Proposition 8.1 says that tagging removes the barrier; the next result says by how much.

**Theorem 8.2** (Value of class-tagged disclosure)**.** *In the class of Theorem 6.5, let $N_p$ be the number of experiments needed with prices alone and $N_t$ the number needed with each class's flow observed with noise $\sigma_f$. To recover the weights to fixed accuracy, $N_p/N_t \asymp (\sigma/\sigma_f)^2(\Delta\eta)^{-(4K-2)}$; to resolve the rates to accuracy proportional to their spacing, $N_p/N_t \asymp (\sigma/\sigma_f)^2(\Delta\eta)^{-(4K-4)}$.*

*Proof.* $N_p \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$ by Theorems 6.4 and 6.5. With tagged flow each class is a single-exponential problem whose Fisher information for $(w_i, r_i)$ does not depend on the other classes, so $N_t \asymp \sigma_f^2$ for the weights and $N_t \asymp \sigma_f^2(\Delta\eta)^{-2}$ for rates resolved to a fraction of $\Delta\eta$. □

With the same noise and lags for both observers, the Cramér–Rao requirements give fitted slopes of $-6.08$ for prices and $-4.34$ for the ratio at $K = 2$ (theory $-6$ and $-4$). Calibrated to the class memories estimated below, prices alone would need about 333 times as many days as the published class ratios to separate the top-trader and all-account memories in BTCUSDT (rates 0.9889 and 0.9986 per five minutes; about 6,000 days against 18), and about 19 times as many in ETHUSDT (rates 0.9793 and 0.9972).

## 8.2 Disclosure in dollars

Theorem 8.2 states the value of class-tagged disclosure in samples. Theorems 5.1 and 6.4 let us translate that value into a flow of round-trip trading losses and set it beside the official operating budgets of the U.S. Consolidated Audit Trail (CAT). Two adjustments make the comparison exact and like-for-like:

(i) *Exact $K$-cohort moment-gap constant.* Rather than extrapolating from Cramér–Rao ratios, we evaluate the spectral round-trip trading loss bound of Theorem 5.1,

$$L^*_{\mathrm{rt}}(K, \Delta\eta) \;=\; \frac{2\sigma^2 z^2_{1-\alpha}}{\sup_\omega\big(|\Delta\hat{H}_{K,\Delta\eta}(e^{i\omega})|^2/g_0(\omega)\big)},$$

directly for the $K$-point Gauss versus Gauss–Radau economies of Theorem 6.4 and Table 10, whose moments below order $2K-1$ coincide and whose $(2K-1)$-th moment gap has the exact constant $\Delta_2 = 0.222222$, $\Delta_3 = 0.053333$, and $\Delta_4 = 0.013061$ at spacing $\Delta\eta$ (with $\Delta\eta = 2\delta$ for the symmetric known pair).

(ii) *Flow against flow.* A single probe loss $L^*_{\mathrm{rt}}$ is a one-time certification cost, whereas an audit trail is an ongoing annual expenditure. Because algorithm populations turn over and regulatory surveillance is periodic, we compare annual CAT operating costs against the annualised round-trip trading loss $52 \times L^*_{\mathrm{rt}}$ under weekly certification renewal (matching the CFTC weekly reporting cycle of Section 8.4). We benchmark against three primary official figures: the CAT LLC 2025 operating budget of $248.8 million in the November 2024 filing (SEC Release No. 34-101901, Dec. 12, 2024, p. 4), revised to $228.3 million in May 2025; the SEC's 2016 CAT NMS Plan Approval Order estimate of $2.4 billion in initial industry build costs and $1.7 billion per year in ongoing industry costs (SEC, 2016, Release No. 34-79318, Nov. 15, 2016, 81 FR 84696 at 84801, p. 363); and the SEC's Rule 613 proposing and adopting releases estimating $4.0 billion initial and $2.1 billion annual ongoing costs (SEC, 2012, Release No. 34-62174, May 26, 2010, 75 FR 32556 at 32598, 32600, pp. 147, 153; Release No. 34-67457, July 18, 2012, 77 FR 45722 at 45790–45795, pp. 220–238).

Probing cost is a lower bound on the cost of learning from prices, not a social cost. Within that interpretation, Table 16 shows how rapidly the polynomial exponent $(\Delta\eta)^{-(4K-2)}$ and the shrinking Gauss–Radau moment-gap constant $\Delta_K$ turn the comparison in favour of disclosure. For a known pair at coarse separation $\Delta\eta = 0.20$ ($\delta = 0.10$), weekly price-only renewal costs $0.32 million per market per year at Roll noise and $7.79 million at one-second noise, so matching the $228.3 million CAT LLC operating budget requires 708.1 or 29.29 markets. At $\Delta\eta = 0.10$ ($\delta = 0.05$) the unknown $K = 2$ Gauss–Radau pair costs $14.74 million (Roll) or $356.44 million (one-second) per market per year, matching the CAT LLC budget in 15.5 or 0.64 markets and the SEC's $1.7 billion industry figure in 115.3 or 4.77 markets. Once a supervisor must separate $K = 3$ unknown cohorts at $\Delta\eta = 0.20$, annual probing loss rises to $41.83 million (Roll) and $1.01 billion (one-second) per market (5.5 and 0.23 markets at the CAT LLC budget; 40.6 and 1.68 markets at $1.7 billion), and at $K = 3, \Delta\eta = 0.10$ or $K = 4, \Delta\eta = 0.20$ a single market's annual probing loss exceeds both the CAT LLC operating budget and the SEC's total annual industry estimate.

Table 16: Price-only round-trip probing loss per market and annualised weekly-renewal cost against official CAT budgets

| Economy ($\Delta\eta$) | Loss / probe (USDT) Roll | 1-sec | Annual (52×, USDT) Roll | 1-sec | Mkts ($228.3M / $1.7B) Roll | 1-sec |
|---|---|---|---|---|---|---|
| $K = 2$, known (0.20) | 6,200 | 149,901 | 322,400 | 7,794,852 | 708.1 / 5,273.0 | 29.29 / 218.09 |
| $K = 2$, known (0.10) | 99,192 | 2,398,422 | 5,157,984 | 124,717,944 | 44.3 / 329.6 | 1.83 / 13.63 |
| $K = 2$, known (0.04) | 3,874,696 | 93,688,369 | 201,484,192 | 4,871,795,188 | 1.1 / 8.4 | 0.05 / 0.35 |
| $K = 2$, Radau (0.20) | 4,292 | 103,788 | 223,184 | 5,396,976 | 1,022.9 / 7,617.0 | 42.30 / 314.99 |
| $K = 2$, Radau (0.10) | 283,488 | 6,854,612 | 14,741,376 | 356,439,824 | 15.5 / 115.3 | 0.64 / 4.77 |
| $K = 3$, Radau (0.20) | 804,403 | 19,450,107 | 41,828,956 | 1,011,405,564 | 5.5 / 40.6 | 0.23 / 1.68 |
| $K = 3$, Radau (0.10) | 1,137,325,880 | 27,500,017,407 | 59,140,945,760 | 1,430,000,905,164 | 0.0 / 0.0 | 0.00 / 0.00 |
| $K = 4$, Radau (0.20) | 156,632,086 | 3,787,291,895 | 8,144,868,472 | 196,939,178,540 | 0.0 / 0.2 | 0.00 / 0.01 |

Separation variable $\Delta\eta$ throughout ($\Delta\eta = 2\delta$ for the symmetric known pair). Gauss–Radau rows use the exact $K$-cohort quadrature pair of Theorem 6.4 and Table 10 (moment gaps $\Delta_2 = 0.222222$, $\Delta_3 = 0.053333$, $\Delta_4 = 0.013061$ at $W = 0.08$, $r_c = 0.60$). Loss per probe is the spectral round-trip trading loss bound of Theorem 5.1 at the median BTCUSDT Roll noise (0.40 USDT) and median one-second price-change standard deviation (1.98 USDT); annual loss multiplies by 52 weekly renewals. The final columns divide the revised CAT LLC 2025 operating budget ($228.3M; Nov. 2024 filing $248.8M, SEC Release No. 34-101901, p. 4) and the SEC 2016 annual industry estimate ($1.7B, Release No. 34-79318, p. 363) by the annual probing loss per market. Probing cost is a lower bound on the cost of learning from prices, not a social cost. Kind: recomputed (`recompute_p1_cat_costs.py` → `R_P1_cat_disclosure_dollars.json`).

## 8.3 Disclosure in real data

Binance publishes, every five minutes, long/short ratios for its top-trader group, by position and by account, and for all accounts. These are class-level positions, the closest public analogue to the tagged data of Proposition 8.1. We treat a class's reported net position as the observable counterpart of its cohort state: it obeys the same one-lag recursion class by class, so the block-diagonal information structure of Proposition 8.1 carries over. The mapping is an assumption, because the reports are positions rather than flows and returns stand in for the unobserved deviation $x_t$. For each class $c$ we estimate $n_{c,t} = r_c n_{c,t-1} + w_c \,\mathrm{ret}_t$, with $n = (R-1)/(R+1)$ from the ratio $R$. The price-only alternative regresses signed five-minute taker flow on twelve lagged five-minute returns and fits two exponentials to the kernel; it estimates the memory rates without class information. Intervals come from a day bootstrap with 200 draws.

Table 17 gives three results. First, tagged class data pin down memory rates that prices and flow leave essentially unrestricted: the price-only intervals are 28 to 124 times wider in every one of 30 symbols on fresh days, and cover most of the admissible range. Part of this ratio is mechanical, because a series of positions identifies its own persistence directly while the price-only route must fit two exponentials to a weakly identified kernel. Restricting attention to the five symbols in which the price-only fit does identify the merged response (ACH, API3, OGN, ZEN and DASH) does not change the conclusion: the price-only rate

Table 17: Memory rates with and without class-tagged data

| Symbol | Days | Top position $\hat{r}$ [95% CI] | All accounts $\hat{r}$ [95% CI] | Width ratio | Classes differ |
|---|---|---|---|---|---|
| BTCUSDT | 48 | 0.9889 [0.9842, 0.9935] | 0.9986 [0.9955, 1.0005] | 121.7 | yes |
| ETHUSDT | 48 | 0.9793 [0.9616, 0.9907] | 0.9972 [0.9934, 0.9997] | 55.6 | yes |
| CELRUSDT | 48 | 0.9884 [0.9830, 0.9922] | 0.9882 [0.9804, 0.9914] | 81.6 | no |
| CVXUSDT | 33 | 0.9822 [0.9732, 0.9908] | 0.9829 [0.9694, 0.9900] | 45.9 | no |

| *Pre-registered test on fresh days, 30 USD-M perpetuals, 2025–2026* | |
|---|---|
| (i) width ratio $> 10$ | 30 of 30 (range 27.6–124.0) |
| (ii) price-only rate interval wider than 0.25 | 30 of 30 (range 0.65–0.99) |
| (iii) merged response $\sum_m k_m$ identified | 5 of 30 |
| Classes differ | 11 of 30 |

Top panel: training days 2023–2024; width ratio is the mean price-only rate-interval width over the mean tagged width. The price-only intervals for both rates cover most of $[0, 0.99]$. Bottom panel: specification registered before the fresh days were downloaded (SHA-256 `65acda361861...`); a symbol passes if all three of the first rows hold, and the test required 80% of symbols to pass.

intervals are still 0.64 to 0.92 wide and the ratios run from 27.6 to 62.5 (median 47.5). We therefore read the ratios as a measure of how little prices reveal about class memory, not as a horse race between two estimators of equal standing. Second, the classes differ where theory says the information should be: in BTCUSDT and ETHUSDT the top-trader group's positioning has a half-life of about three to five hours against about one to two days for all accounts, while in thin symbols the classes are often indistinguishable. Third, the pre-registered test fails its 80% criterion, on the part of the prediction that concerns what prices *do* reveal: the merged response is significantly different from zero in only 5 of 30 symbols. Theorem 6.8 says the merged measure is identified; on 42 days of thin-market data it is not estimated precisely. Prices reveal even less than the identification theory guarantees, which strengthens rather than weakens the case for class-tagged disclosure.

To test directly whether the memory rates identified by class-tagged positions improve out-of-sample return predictions after large order-flow shocks, we pre-registered a dedicated out-of-sample test (`P1_disclosure_oos.json`, SHA-256 `f0182ce50978...`) before downloading the 22nd-of-month data for January 2025–September 2026 across 19 thin perpetuals (6,137 top-decile five-minute flow shocks on 2024 training days; 4,579 shocks on untouched 2025–2026 test days). On the training sample we estimate each symbol's two memory rates $\hat{r}_1, \hat{r}_2$ either from the class-tagged positions or from the price-only two-exponential fit, construct the filtered memory states $y_{k,t} = \hat{r}_k y_{k,t-1} + (1 - \hat{r}_k)\mathrm{ret}_t$, and forecast the 30-minute post-shock cumulative return $R_{t+1\to t+6}$ out of sample relative to an AR(1) return-and-flow baseline. The primary hypothesis fails: on the untouched test shocks, the price-only memory states yield $\Delta R^2 = -0.199$ percentage points relative to the baseline, the class-tagged memory states yield $\Delta R^2 = -1.288$ points, and the

tagged-minus-price-only difference is $\Delta R^2 = -1.089$ points (95% date-block bootstrap CI $[-2.605, +0.376]$, Clark–West $t = 0.96$, $p = 0.1684$). Likewise, adding the level and one-hour change of the top-trader and all-account positions to the hourly tail-risk controls of the companion paper leaves out-of-sample $R^2$ unchanged ($-0.19$ points, CI $[-0.99, 0.83]$).

Why does class-tagged disclosure fail to predict signed 30-minute returns while sharpening parameter identification? Theorem 3.1 and Proposition 8.1 imply that prices and aggregate flow identify only the merged response kernel and cannot separate class-specific memory states ($n_{\text{top},t}$ vs. $n_{\text{all},t}$) or cross-cohort divergence ($d_t = n_{\text{top},t} - n_{\text{all},t}$), whereas class-tagged disclosures observe the cohort states directly. To test this exact state-trajectory prediction out of sample, we pre-registered a follow-up test (`P1_disclosure_state_oos.json`, SHA-256 `5136cc6b846d...`) and committed its hash to the repository before downloading the untouched 2nd-and-9th-of-month data for January 2025–September 2026 across the 19 thin perpetuals (12,046 top-decile five-minute flow shocks on the 24 training dates in 2024; 9,875 shocks on the 42 untouched test dates; test opened once). Table 18 compares a two-exponential price-and-flow filter ($\alpha_{\text{fast}} = 0.50$, $\alpha_{\text{slow}} = 0.10$ on five-minute returns and normalized signed flow, plus shock return, shock flow and symbol fixed effects) against the class-tagged disclosure filter that adds the five-minute demeaned cohort states and their one-bar changes ($n_{\text{top},t}, n_{\text{all},t}, \Delta n_{\text{top},t}, \Delta n_{\text{all},t}, d_t$). Both pre-registered primaries pass (Holm-adjusted $p < 10^{-16}$): on untouched top-decile flow shocks, the price-and-flow filter has negative out-of-sample $R^2$ ($-4.766\%$) for 30-minute post-shock cross-cohort state divergence $d_{t+6}$, whereas the class-tagged disclosure filter achieves $R^2 = 45.962\%$ ($\Delta R^2 = +50.728$ percentage points, 95% CI $[+45.291, +55.918]$, Clark–West $t = 21.27$). For the 30-minute post-shock institutional/top-trader positioning state $n_{\text{top},t+6}$, out-of-sample $R^2$ rises from 4.441% to 54.845% ($\Delta R^2 = +50.404$ points, 95% CI $[+47.226, +53.724]$, $t = 21.85$). Across all 210,144 out-of-sample five-minute bars, the gains are +37.639 points ($t = 72.75$) and +46.593 points ($t = 80.44$). Class-tagged disclosure is valuable because it identifies who trades on what memory horizon and tracks cross-cohort positioning divergence out of sample, not because slow position ratios time noisy 30-minute directional returns.

**Information-set decomposition: level persistence vs. memory filtering.** To examine how much of the +50.728 point out-of-sample gain in Table 18 reflects multi-cohort memory filtering versus statistical persistence of the current level, an exploratory nested model audit (`hardened_empirical_audits.py` → `R_hardened_empirical_audits.json`) evaluates a hierarchy of information sets on the exact same test shocks. Simply predicting

Table 18: Pre-registered out-of-sample cohort-state trajectory test on untouched 2nd-and-9th-of-month days (`P1_disclosure_state_oos`), 2025–2026

| 30-min post-shock target (19 thin) | $R^2_{\text{pf}}$ | $R^2_{\text{tag}}$ | $\Delta R^2$ (pts) [95% CI] | CW $t$ ($p$) | Res. |
|---|---|---|---|---|---|
| Prim. 1: Divergence $d_{t+6}$ (9,875 shocks) | −4.766 | 45.962 | +50.728 [+45.291, +55.918] | 21.27 ($< 10^{-16}$) | pass |
| Prim. 2: Top state $n_{\text{top},t+6}$ (9,875 shocks) | 4.441 | 54.845 | +50.404 [+47.226, +53.724] | 21.85 ($< 10^{-16}$) | pass |
| Sec.: Divergence $d_{t+6}$, all (210,144 bars) | −0.586 | 37.053 | +37.639 [+36.232, +39.107] | 72.75 ($< 10^{-16}$) | — |
| Sec.: Top state $n_{\text{top},t+6}$, all (210,144 bars) | 0.186 | 46.779 | +46.593 [+45.555, +47.643] | 80.44 ($< 10^{-16}$) | — |

Out-of-sample $R^2$ in percent. Trained on 24 dates in 2024 (12,046 shocks, 120,384 bars); evaluated once on 42 untouched test dates (2nd and 9th of each month, Jan 2025–Sep 2026); 2,000-draw date-block bootstrap 95% CIs; Holm $p < 10^{-16}$ across both primaries. Spec SHA-256 `5136cc6b846d...` (`R_P1_disclosure_state_oos.json`).

$d_{t+6}$ and $n_{\text{top},t+6}$ from the current levels via a univariate AR(1) benchmark ($M_3$) yields out-of-sample $R^2$ of 46.205% and 54.413%, accounting for 100.48% and 99.14% of the total improvement over the price-and-flow filter ($M_1$), while dynamic exponential return and flow memory filters add only −0.243 and +0.432 points. Furthermore, when the level persistence is removed by forecasting the 30-minute position *change* ($\Delta d_{t+6} = d_{t+6} - d_t$ and $\Delta n_{\text{top},t+6} = n_{\text{top},t+6} - n_{\text{top},t}$), out-of-sample $R^2$ drops to −0.26 and +1.41 points. The empirical disclosure value in Table 18 therefore demonstrates that public price and flow cannot reconstruct current venue positioning levels, but this gain reflects class inventory persistence rather than granular recovery of hidden algorithmic trading strategies.

## 8.4 A second disclosure regime: the CFTC's trader-class reports

Binance publishes its class ratios voluntarily, for one venue. The U.S. Commodity Futures Trading Commission requires large traders in regulated futures markets to report their positions and publishes them weekly by class: leveraged funds, asset managers and dealers in financial futures, and managed money, producer/merchants and swap dealers in commodities. This is class-tagged disclosure mandated by law. We registered a test (SHA-256 `6f1d6f35ae88...`) before downloading any data, in eight markets that span asset classes: CME Bitcoin and Ether, the E-mini S&P 500, 10-year Treasury notes, the euro, the yen, WTI crude oil and Henry Hub natural gas, using every weekly report through October 2026 (282 to 1,039 weeks per market). For each class, $n_{c,t} = (\text{long} - \text{short})/\text{open interest}$ follows $n_{c,t} = \alpha_c + r_c n_{c,t-1} + w_c \,\text{ret}_t$, the weekly version of the cohort recursion. The price-only alternative fits a two-exponential kernel to twelve weekly return lags. Intervals come from a 13-week moving-block bootstrap. The registered hypotheses were: (H1, primary) the fast class's memory rate is pinned to an interval narrower than 0.25 while both

Table 19: Trader-class memory from the CFTC Commitments of Traders, eight regulated futures markets

| Market | Weeks | Fast $\hat{r}$ [95% CI] | Slow $\hat{r}$ | $\hat{r}_{\text{fast}} - \hat{r}_{\text{slow}}$ [95% CI] | $\hat{w}_{\text{fast}}$ | Ratio |
|---|---|---|---|---|---|---|
| Bitcoin (CME) | 434 | 0.951 [0.928, 0.969] | 0.986 | −0.035 [−0.062, −0.010] | − | 23.7 |
| Ether (CME) | 282 | 0.890 [0.826, 0.932] | 0.960 | −0.070 [−0.143, −0.011] | − | 9.1 |
| E-mini S&P 500 | 1,039 | 0.914 [0.884, 0.937] | 0.954 | −0.040 [−0.071, −0.012] | − | 18.3 |
| 10-year T-note | 1,039 | 0.992 [0.985, 0.998] | 0.993 | −0.000 [−0.006, +0.007] | (0) | 71.4 |
| Euro | 1,035 | 0.961 [0.947, 0.972] | 0.996 | −0.034 [−0.051, −0.022] | + | 37.3 |
| Yen | 1,035 | 0.955 [0.939, 0.968] | 0.984 | −0.029 [−0.047, −0.013] | + | 34.0 |
| WTI crude oil | 1,039 | 0.961 [0.946, 0.974] | 0.999 | −0.037 [−0.053, −0.023] | + | 35.2 |
| Natural gas | 1,039 | 0.981 [0.969, 0.990] | 0.980 | +0.001 [−0.014, +0.017] | + | 44.0 |

*Pre-registered hypotheses*: H1 (primary) 8 of 8, pass; H2 4 of 8, fail; H3 6 of 8, pass.

Fast class: leveraged funds (financial futures) or managed money (commodities); slow class: asset managers or producer/merchants. $\hat{w}_{\text{fast}}$: sign of the response to returns, + or − when the 95% interval excludes zero, (0) otherwise. Ratio: mean width of the two price-only rate intervals over the width of the fast-class interval; the price-only intervals span 0.95–0.96 of the admissible range $[0, 0.99]$ in every market. Weekly returns from Binance spot (Bitcoin, Ether) and Yahoo Finance (others; the Treasury return is minus the change in the 10-year yield). Specification SHA-256 `6f1d6f35ae88...`; test opened once. Kind: real data (Part 11).

price-only intervals are wider, in at least 75% of markets; (H2) the fast class trend-follows, $w_c > 0$, in at least 75%; (H3) the fast class has the shorter memory in at least half.

The primary hypothesis holds in all eight markets (Table 19). The class reports pin each fast class's memory rate to an interval between 0.013 and 0.106 wide, while weekly prices leave both memory rates anywhere in the admissible range; the width ratios run from 9 to 71. The ratio partly reflects how little weekly returns reveal, which is the point of the theory rather than an artefact of it: with prices alone the memories are not located at all. The classes differ as the cohort model allows. Leveraged funds and managed money have the shorter memory in six of eight markets, with half-lives of 6 to 18 weeks against months to years for asset managers and producers; in Treasuries and natural gas the two memories are indistinguishable.

The trend-following hypothesis fails. The fast class buys after rises in the euro, the yen, crude oil and natural gas, but sells after rises in Bitcoin, Ether and the S&P 500 futures. The first half matches the finding of Moskowitz et al. (2012) that speculators' positions tend to go with past returns. We did not predict this. One candidate explanation is that leveraged-fund positions in financial futures are dominated by the short futures leg of cash–futures basis trades, which a single memory per class cannot represent. A post-hoc check is consistent with it but not decisive (Part 13, `R6`): weekly position changes of leveraged funds and asset managers are strongly negatively correlated in Bitcoin, Ether, the S&P 500 and Treasuries (−0.37 to −0.67) but not in the euro or yen (−0.04 and

$-0.05$), and holding the asset managers' change fixed removes the negative response in Bitcoin and halves it in Ether and the S&P 500. Because class positions sum to zero across all reporting categories, some negative correlation is mechanical, so this is a signature, not a test. The failure sharpens rather than weakens the case for class tagging: class labels are not strategies, and a supervisor who wants to know how a class responds to prices needs its positions, which the price process does not reveal.

Together with Table 17, the evidence now covers two disclosure regimes, one voluntary and one mandated by law, in crypto and in regulated equity-index, rate, currency and commodity futures. In both, class-tagged positions locate memories that prices leave unrestricted.

# 9 A spectral blind spot

For any eigenvalues $0 < \lambda_2 < \lambda_1 < 1$, one-step price variance and sufficiently large stationary price variance, there is a family of two-state markets sharing all three quantities whose worst-case transient gain grows without bound (Supplemental Appendix, Theorem S.4.1 and Figure 8).

# 10 Empirical feedback and its measurement

Across a 96-date, eight-pair Binance spot panel (66.3M one-second bars), a 72-date BTCUSDT one-second panel, 30 USD-M perpetuals and two BTCUSDT spot zero-fee events, standard reduced-form feedback indices are dominated by EMA filtering mechanics ($1 - \alpha = 0.9355$) and clock alignment (0.0081 with look-ahead vs. 0.00037 aligned); real feedback beyond the placebo is positive but small (0.0045), rises with market depth (Spearman +0.70), and does not move when zero fees multiply trading volume by 2.6 to 4.9 (Supplemental Appendix, Tables 21–26).

# 11 Implications for supervision and market design

**What a supervisor can learn from public data.** Prices and anonymous flow reveal the merged measure $M$, the closed-loop impulse response and the reduced-form operator, and nothing finer. A supervisor who wants to know *which* algorithms drive a feedback loop cannot get there from public data at reasonable cost when their memories are close.

This is not a failure of econometric technique: Theorem 6.2 holds for every estimator, biased or not.

**What probing costs.** At realistic observation noise, an exchange or a large desk could separate two well-separated memory rates for about $10^5$ USDT of capital. Probing is therefore feasible for coarse questions and infeasible for fine ones. Because capital grows as $\delta^{-4}$ and experiments as $(\Delta\eta)^{-(4K-2)}$, halving the separation multiplies the cost by 16 for a known pair and by $2^{4K-2}$ for $K$ unknown cohorts.

**What disclosure buys.** Disclosure regimes differ sharply in value (Table 15). Timestamps without update values are worthless for this purpose. Noisy periodic statements are nearly worthless. Audited equity and, above all, flow tagged by participant class remove the barrier entirely (Proposition 8.1). This gives a quantitative argument for audit-trail regimes that record the participant class of each order, and a quantitative argument against relying on periodic statements for the same purpose. Calibrated to Binance's published class ratios, separating the memories of the top-trader group and of all accounts in BTCUSDT would take prices alone about 333 times as long (Theorem 8.2). The class reports that the CFTC already requires have the same property in regulated markets (Table 19), so the argument concerns the granularity and frequency of existing disclosure, not a new kind of reporting.

**When audit trails already exist.** The US Consolidated Audit Trail (SEC Rule 613) gives the SEC and the self-regulatory organizations a record of every order in NMS stocks and listed options from origination to execution or cancellation, linked to an anonymized customer identifier and to the broker-dealer that handled it; since 2025 it no longer collects the names, addresses and years of birth of natural persons (SEC, 2012, 2016, 2024, 2025). Our results bear on such regimes in three ways. First, they price what the record is worth for the question of who drives a feedback loop: participant-level records remove a barrier that otherwise costs round-trip trading loss of order $\delta^{-4}$ and samples of order $(\Delta\eta)^{-(4K-4)}$ (Proposition 8.1, Theorem 8.2); set against the CAT LLC 2025 operating budget ($248.8 million in November 2024, revised to $228.3 million in May 2025) and the SEC's 2016 annual industry estimate ($1.7 billion), the annualised weekly-renewal probing loss that disclosure makes unnecessary is small for two well-separated classes and exceeds the entire CAT budget once $K \geq 3$ at $\Delta\eta \leq 0.10$ or $K = 4$ at $\Delta\eta = 0.20$ (Section 8.2, Table 16). For this purpose a stable account identifier is what matters and names are not needed, which is a quantitative reason to keep the account linkage while removing personal

data. Second, an audit trail records accounts, not decision rules. One account can run several algorithms and one algorithm can trade through several accounts, so an observer with account-level records who wants rule-level memories faces our problem within each account's flow, with fewer cohorts but the same exponent; MiFID II's algorithm identifier (ESMA, 2016) is the closer analogue of the tagged flow in Proposition 8.1. Third, the audit trail does not reach the markets we study. Crypto perpetual futures are outside it, futures regulated by the CFTC have separate reporting, and its data are not available to exchanges, other participants or researchers. For these observers the price-only and coarse-disclosure contracts of this paper are the relevant ones.

**What to monitor instead.** Because transient loss can be arbitrarily large at fixed eigenvalues and volatility (Theorem S.4.1), supervision that wants to control risk rather than identify participants should monitor quantities that bound transient loss directly: the summed absolute impulse response and operator confidence sets, both of which are identified from public data. These are cheap where identification is expensive (Remark 6.3); the companion paper shows that the summed response is certified at the parametric rate from controlled experiments, and only under a declared bound on memory length from a finite window (Niroula et al., 2026).

**How to measure feedback.** Empirical feedback indices should (i) compare persistence with the mechanical value of the filter, (ii) align flow with the bar in which it is observed, (iii) control for order-splitting persistence or use instruments with a defended exclusion, and (iv) report placebo pipelines. Each of these changes the sign or the order of magnitude of published-style estimates in our data.

# 12 Conclusion and an open problem

Prices reveal a merged summary of algorithmic feedback. Recovering the algorithms behind it costs executed quantity of order $\delta^{-2}$ and trading capital of order $\delta^{-4}$ for a known pair, with a certified constant of about 61 par units at the calibration, and a number of experiments of exact order $(\Delta\eta)^{-(4K-2)}$ for $K$ unknown cohorts, passively as well as actively; in simulations of stylized nonlinear rules and closed-loop trained online learners the same costs hold once weights settle, while fast online adaptation breaks equivalence and hard thresholds hide cohorts from small probes altogether. Without the exponential form, accuracy improves only as $1/\log N$, with matching lower and upper bounds. Class-

tagged disclosure removes the barrier, and its value can be stated in samples and in dollars: a sample factor of order $(\Delta\eta)^{-(4K-4)}$, about 333 for the trader classes Binance already publishes in BTCUSDT, and an annualised weekly-renewal probing loss that matches the $228.3 million CAT LLC operating budget across 15.5 (Roll) or 0.64 (one-second) markets for an unknown two-cohort Gauss–Radau pair at $\Delta\eta = 0.10$ and exceeds it in a single market once $K \geq 3$ at $\Delta\eta = 0.10$; the CFTC's class reports locate memories that prices leave unrestricted in all eight regulated markets we examined, while a pre-registered out-of-sample test confirms that class-tagged memory states identify horizons rather than forecasting 30-minute post-shock returns ($\Delta R^2 = -1.09$ points). At realistic noise the capital barrier for a moderate separation is about $10^5$ USDT. Measured feedback in crypto markets is real but small, rises with depth, does not move when fees change trading volume severalfold, and is easily confused with filtering and timing artefacts. Finally, eigenvalues and volatility cannot certify safety, because transient loss can be arbitrarily large at fixed spectrum and variance.

This leaves an open question for supervision: *what can be measured and controlled without identifying the algorithms?* The impulse response $g_m = c^\top J^m b$, the operator $J$ and finite-sample confidence sets for it are identified from public data even when cohorts are not. The companion paper uses exactly this interface to build an identification-free pre-trade control.

**Data and code availability.** All data are public: Binance archives (`https://data.binance.vision`), verified against published checksums. The pre-registered specification of the disclosure test, its SHA-256 hash and its test-set unlock counter are in the replication package. Every number in this paper is produced by a notebook cell or script that writes a result file (Appendix S.9).

**Use of AI tools.** The authors used Claude (Anthropic), an AI assistant, to write and debug analysis code, run checks of derivations and numerical results, and draft and edit text. Every number in the paper is produced by code the authors executed and archived in the replication package. The authors reviewed all content and take full responsibility for it.

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Table 20: Map of results

| Question | Answer | Where |
|---|---|---|
| What do prices identify? | The merged rate–weight measure only | Thm 3.1 |
| Quantity to separate a known pair | $\Theta(\delta^{-2})$, exact error formula | Thm 4.1 |
| Capital to separate a known pair | $\Theta(\delta^{-4})$, counterparty or own funds | Thm 4.3, Lem. 4.5 |
| Constant at the calibration | $[61.41, 64.48]$ par units; $\approx 10^5$ USDT | Thms 5.1–5.3, Table 6 |
| Does adaptivity help? | At most by a factor $\approx 0.49$ in expected loss | Thm 5.2 |
| Unknown mixtures | $(\Delta\eta)^{-(4K-2)}$, tight for every $K$ | Thms 6.4, 6.5 |
| General memory | Accuracy $\asymp 1/\log N$, lower and upper bounds | Lem. 6.6, Thm 6.7 |
| Passive data | Identifies $M$; same polynomial cost for every $K$ | Thms 6.8, 6.9, Prop. S.8.2 |
| Nonlinear and adaptive rules? | Settled online learners obey $\delta^{-4}$; fast adaptation or thresholds break it | Table 14 |
| What removes the barrier? | Tagged flow, audited equity | Props 4.4, 8.1 |
| What is tagging worth? | Factor $(\Delta\eta)^{-(4K-4)}$; $\approx 333\times$ for BTCUSDT | Thm 8.2 |
| Disclosure in dollars? | Annual probing loss exceeds CAT budget once $K \geq 3$ or $\Delta\eta \leq 0.04$ | Table 16 |
| Do class data pin down memories? | Yes: 28–124× tighter in 30 of 30 markets | Table 17 |
| Can eigenvalues certify safety? | No | Thm S.4.1 |
| What do feedback indices measure? | Mostly filtering and timing; small real feedback | Tables 22, 24 |
| Does feedback track activity? | No: fees move volume 2.6–4.9×, not $\hat{Q}$ | Table 26 |

# Supplemental Appendix for "Auditable Memory and Financial Identification Limits"

## S.1 Map of results

## S.2 Observation design and the tri-region partition

An observer distinguishes two candidate mechanisms with rates $\theta_0$ and $\theta_1 = \theta_0 + \delta$ over $N$ periods, submitting probes $V = (V_1, \ldots, V_N)^\top$ that generate inventory $I_t = \sum_{s \leq t} V_s$. Under contract $I_0$ (active probing, Gaussian observation noise) $p_t = (h_i * V)_t + \epsilon_t$, $\epsilon_t \sim N(0, \sigma^2)$, so

$$D_{\mathrm{KL}}(P_0^{\otimes N} \| P_1^{\otimes N}) = \frac{1}{2\sigma^2} \sum_{t=1}^{N} \big((h_1 * V)_t - (h_0 * V)_t\big)^2. \tag{29}$$

The observer minimizes capital $C(N, V) = \sum_t (\lambda V_t^2 + \phi|V_t|) + \max_t \mu p_0 |I_t|$ subject to (i) separation $D_{\mathrm{KL}} \geq \frac{1}{2}(z_\alpha + z_\beta)^2$, (ii) caps $|V_t| \leq V_{\mathrm{eff}}$ and $|I_t| \leq I_{\max}$, and (iii) a solvency budget $\max_t \mathrm{Loss}_t(V) \leq B$. Here $V_{\mathrm{eff}} = \min(V_{\max}, D_{\max}/\kappa)$ combines resting depth and a price-displacement limit.

**Proposition S.2.1** (Tri-region partition and finite-session resolution floor)**.** *Under observation horizon $T$ per pulse, $(\delta, B)$ space splits into three regions.*

*(i)* Feasible single pulse. *For $\delta \geq \delta_{\min}(T)$ and $B \geq C^*(\delta)$ one pulse within capacity suffices. For a simple rate shift ($\eta_1 = \eta_0 + \delta$) the response difference is $\Theta(\delta)$ and $C^* = \Theta(\delta^{-2})$; for symmetric splitting the first-order variation cancels, the difference is $\Theta(\delta^2)$ and $C^* = \Theta(\delta^{-4})$.*

*(ii)* Sequential bounded probing. *For $\delta < \delta_{\min}(T)$ the required quantity exceeds $V_{\text{eff}}$. Capped pulses each supply divergence $V_{\text{eff}}^2 \|d_T\|^2/(2\sigma^2)$, so*

$$N_{\text{pulses}}(T, \delta) = \left\lceil \frac{4\sigma^2 z_\alpha^2}{V_{\text{eff}}^2 \|d_T\|_2^2} \right\rceil. \tag{30}$$

*(iii)* Infeasible with isolated pulses. *If the budget cannot finance sequential costs, peak inventory breaches $I_{\max}$, or a session deadline binds, the finite-session floor $\delta_{\min}(T_{\text{session}})$ is the root of $V_{\text{eff}} \|d_{T_{\text{cycle}}}(\delta)\|_2 = 2\sigma z_\alpha / \sqrt{\lfloor T_{\text{session}}/T_{\text{cycle}} \rfloor}$, with small-$\delta$ approximation $\delta_{\min} = \Theta(T_{\text{session}}^{-1/4})$.*

At the calibration $\kappa = 0.8$, $q = 0.2$, $\sigma = 0.01$, $V_{\max} = 50$, $D_{\max} = 5$ ($V_{\text{eff}} = 6.25$), the single-tap floor is $\delta_{\min}(3) = \sqrt{2\sigma z_\alpha/(V_{\text{eff}}\kappa^2 q)} = 0.2028$; extended observation lowers it to 0.1376 ($T = 4$), 0.0973 ($T = 10$), 0.0914 ($T = 20$) and 0.0905 ($T = 100$). At $\delta = 0.05$ the simplified one-leg benchmark (cost $\frac{1}{2}\kappa V_{\text{eff}}^2 + \phi P_* V_{\text{eff}} = 15.875$ per pulse plus initial margin 3.125) needs 271 pulses and 4,305.25 par units with $T = 3$, 58 pulses and 923.88 with $T = 4$, but only 12 pulses and 193.63 with $T = 20$ and 11 pulses and 177.75 with $T = 100$; optimizing the pulse size (6.06 rather than 6.25) lowers the $T = 20$ figure to 182.03. Most of the saving comes from observing the transient response rather than a single tap (Figure 7).

These are costs within an isolated-pulse design class, not minimum capital; the next subsection replaces the formula by a ledger.

## S.3 Robustness: heteroskedastic impact and probing protocols

**Corollary S.3.1** (Invariance under impact-induced heteroskedasticity)**.** *Suppose probing inflates innovation variance as $\sigma_{i,t}^2(v) = \sigma^2(1 + \xi[(h_i * v)_t]^2)$ and slippage scales as $\kappa_{\text{eff}} = \kappa(1 + \xi_0|v_t|)$. For capacity-bounded probes, $D_{\text{KL}}(P_0^v \| P_1^v) \leq \frac{1+O(\xi^2 V_{\text{eff}}^2)}{2\sigma^2} \|(H_1 - H_0)v\|_2^2 = O(\|v\|_2^2 \delta^4)$, and convex impact only raises dissipation. Required capital remains $\Theta(\delta^{-4})$.*

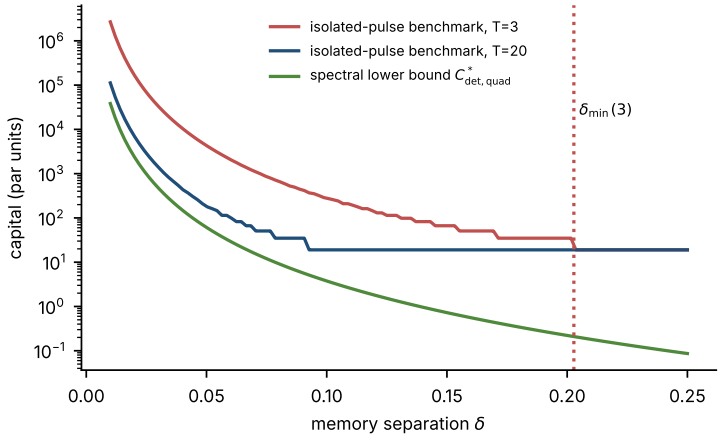



Figure 7: Capital against separation: isolated-pulse benchmarks with three and twenty observations per pulse, and the spectral lower bound of Theorem 5.1. The dotted line is the single-tap resolution floor $\delta_{\min}(3) = 0.2028$. Recomputed in this revision.

Four probing protocols (unconstrained single pulse, repeated capped pulses, a sequential test with dynamic throttling, and a predetermined generalized-eigenvector trajectory) all show capital slopes near $-4$ in $\log \delta$: $-3.90$, $-3.84$, $-3.47$ and $-3.81$ over $\delta \in [0.03, 0.12]$, where the sequential slope is flattened by the 1.7 pulses needed at $\delta = 0.12$ (Appendix Table 28). At equal observation horizon, sequential stopping saves 14–49% of capital relative to repeated capped pulses, but most of the apparent saving from the single-tap baseline comes from observing the transient response, not from adaptivity.

## S.4 A spectral blind spot

Even without identification, could a supervisor monitor risk through eigenvalues or volatility?

**Theorem S.4.1** (Spectral blindness)**.** *For any eigenvalues $0 < \lambda_2 < \lambda_1 < 1$, one-step price variance $s_1 > 0$ and stationary price variance $V$ large enough, there is a family of two-state markets $J_\theta = \left(\begin{smallmatrix} \lambda_1 & \theta \\ 0 & \lambda_2 \end{smallmatrix}\right)$ with shock covariance $\operatorname{diag}(s_1, s_2(\theta))$ such that all members share the eigenvalues, the one-step price variance and the stationary price variance, while the worst-case transient gain $\sup_k \|e_1^\top J_\theta^k\|$ grows without bound in $\theta$.*

At $\lambda = (0.8, 0.7)$, $s_1 = 1$ and $V = 5$, the gain is 1.77, 5.10, 16.95, 50.85 and 169.50 for $\theta = 1, 3, 10, 30, 100$, with stationary variance 5.000 in every case; the cumulative gain rises from 18.3 to 1,667.7. No monitor that depends only on eigenvalues and second moments of the price can be uniformly safe (Figure 8).

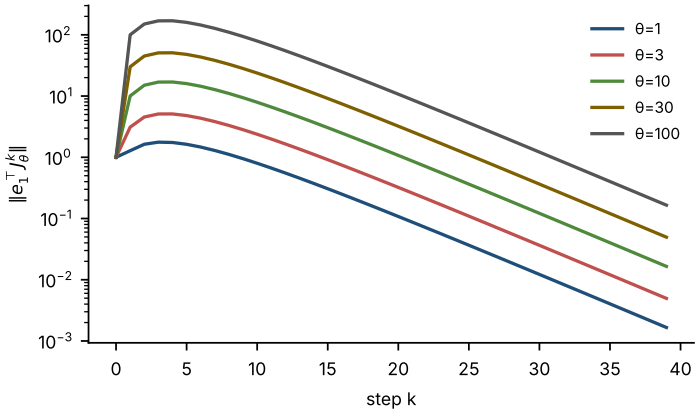



Figure 8: Price-row transient gain of the spectral-blindness family $J_\theta$: eigenvalues $(0.8, 0.7)$, one-step price variance 1 and stationary price variance 5 are the same for every $\theta$.

Two clarifications prevent over-reading. First, the cohort economy of Section 2 is itself non-normal, yet its price output does not amplify (Lemma 2.3). Large weighted gains can be manufactured by output weights alone: with $C_w = (1, 160)$, a price-and-memory functional peaks at 160.003 at $t = 0$, which is the static weight, not dynamic growth (Appendix S.8.5). Second, in a network of $M$ identical cohort economies coupled by symmetric nonnegative cross-impact, the coupled operator stays symmetric in the coordinates of Lemma 2.3; with cross-impact 0.05, the spectral radius rises from 0.822 ($M = 1$) to 0.973 ($M = 10$) while the Kreiss constant stays at one, as it must for a stable symmetric operator (`recompute_oct20.py`). Non-normality is necessary for transient amplification of the price, not sufficient; where amplification comes from is the subject of the companion paper.

## S.5 Empirical feedback and its measurement

The theory says what prices can reveal. The empirical work has two roles, and neither is a test of the identification bounds themselves, which concern controlled experiments no outside researcher can run. First, Section 8.3 tests the theory's policy prediction directly: that class-tagged data pin down memories that prices leave unrestricted. Second, this section asks what the reduced-form feedback indices a supervisor would compute from public data actually measure, and how much of them is feedback rather than filtering, timing or trading activity.

### S.5.1 Estimand and filtering mechanics

In theory $x_t = p_t - p_t^*$ is the deviation from an unobserved fundamental. Empirical work replaces $p_t^*$ by a trailing filter. Let $p_b$ be a log price proxy on uniform bars of length $\Delta$ and $m_b = (1-\alpha)m_{b-1} + \alpha p_b$ an exponential moving average with $\alpha = 2/(W/\Delta + 1)$. The empirical state is

$$x_b^{\mathrm{EMA}} \equiv p_b - m_b = (1-\alpha)(p_b - m_{b-1}) = \frac{1-\alpha}{1-(1-\alpha)L}\Delta p_b. \tag{31}$$

Because $x_{b+1}^{\mathrm{EMA}} = (1-\alpha)(p_{b+1} - m_b)$, a martingale price gives

$$\mathbb{E}[x_{b+1}^{\mathrm{EMA}} \mid \mathcal{F}_b] = (1-\alpha)x_b^{\mathrm{EMA}} : \tag{32}$$

the filtered state is an AR(1) with mechanical persistence $a_0 = 1-\alpha$, which equals $29/31 \approx 0.9355$ for $W = 300$ seconds and $\Delta = 10$ seconds.

Let $f_t$ be signed flow. The empirical VAR(1)

$$\begin{pmatrix} x_{t+1}^{\mathrm{EMA}} \\ f_{t+1} \end{pmatrix} = \begin{pmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{pmatrix} \begin{pmatrix} x_t^{\mathrm{EMA}} \\ f_t \end{pmatrix} + e_{t+1} \tag{33}$$

has reduced-form coefficients, not authenticated impact or learner parameters. When denominators are nonzero we use the scale-invariant indices

$$\hat{Q} = \frac{\hat{J}_{12}\hat{J}_{21}}{1-\hat{J}_{22}}, \qquad \hat{S} = \hat{J}_{11} + \hat{Q}, \qquad \hat{L} = \frac{\hat{Q}}{1-\hat{J}_{11}}. \tag{34}$$

Equation (32) implies that $\hat{J}_{11}$ must be compared with $1-\alpha$, not with zero: persistence that a naive reading attributes to feedback can be pure filtering.

### S.5.2 Data

### S.5.3 Clock alignment dominates reduced-form feedback

We estimate (33) on ten-second bars built from one-second Binance klines across a 96-date, eight-pair spot panel (the 10th and 24th of each month, 2021–2024) under three constructions (Table 22), and run the identical pipeline on a placebo random walk with each pair-day's volatility. Pairing the state with *next-bar* flow (a look-ahead) gives $\hat{Q} = 0.0081$; aligning flow with the bar in which it is observed lowers this by more than an order

Table 21: Samples

| Sample | Coverage | Size | Source |
|---|---|---|---|
| Binance spot panel, 8 pairs | 96 dates, 2021–2024 | 66.3M one-second bars | Binance archive |
| Binance spot fee-event windows, 8 pairs | ±20 days around 2022-07-08 and 2023-03-22 | 640 pair-days | Binance archive |
| Binance USD-M perpetuals, 30 symbols | 90 dates, 2023–2026 | 2,565 symbol-days | Binance archive |
| Binance BTCUSDT one-second panel | 72 dates, 2021–2023 | 6.2M one-second bars | Binance archive |
| Binance BTCUSDT noise-calibration days | January 10–20, 2023 | 11 days | Binance archive |

Binance files are verified against the published SHA-256 checksums (every file used). Sampled dates are the 10th and 24th of each month.

Table 22: Multi-year Binance spot panel: reduced-form indices under three timing constructions

| Construction | $\hat{Q}$ | Date interval | $\hat{S}$ | Date interval | $\hat{Q} > 0$ |
|---|---|---|---|---|---|
| Forward-paired flow (look-ahead) | 0.0081424 | [0.0072965, 0.0089882] | 0.92074 | [0.91665, 0.92482] | 87.4% |
| Observed-flow EMA | 0.0003733 | [0.0002351, 0.0005116] | 0.92072 | [0.91640, 0.92504] | 62.4% |
| Observed, 30 flow lags | 0.0002580 | [0.0001415, 0.0003745] | 0.91738 | [0.91167, 0.92308] | 59.1% |
| Observed raw return | −0.0001555 | [−0.0003732, 0.0000622] | −0.04051 | [−0.05078, −0.03025] | 49.2% |
| Placebo random walk | 0.0000016 | [−0.0000020, 0.0000052] | 0.93510 | [0.93485, 0.93534] | 51.0% |

768 study units (96 dates × 8 pairs), 6,634,752 ten-second transitions; equal-weight asset means within date, then date means with Student intervals. Mechanical persistence $29/31 = 0.93548$.

of magnitude, to 0.00037; and replacing the EMA state by raw returns gives an interval that includes zero. Mean $\hat{J}_{11}$ under observed flow is 0.9203, below the mechanical 0.9355, and the placebo reproduces the mechanical value (0.9351) with $\hat{Q}$ indistinguishable from zero.

### S.5.4 Feedback beyond filtering mechanics

We re-estimate feedback on a separate sample and add a placebo. We use Binance BTCUSDT one-second klines on 72 sampled days in 2021–2023. The state $x_t$ is the log price deviation from its EMA (span 30 one-second bins), $f_t$ is next-second signed taker flow, and a VAR(1) gives $\hat{Q}$. Because the filter alone produces persistence, we run the identical pipeline on a placebo: a random walk with the same one-second volatility, independent of the real flow.

Table 24 gives two findings. Feedback, in the sense of a loop gain the filter cannot produce, is detectable: the real $\hat{Q}$ lies far outside the placebo band. It is also small, and the price persistence that a naive reading would attribute to feedback is almost entirely mechanical (0.9364 against 0.9355). Because this construction pairs the state with next-

Table 23: Asset means under forward-paired and observed flow

| Pair | Forward-paired $\hat{Q}$ | Observed $\hat{Q}$ | $\hat{Q}_{30}$ | Observed $\hat{S}$ |
|---|---|---|---|---|
| BTC/USDT | 0.0141870 | 0.0006073 | 0.0004891 | 0.93464 |
| ETH/USDT | 0.0130885 | 0.0005675 | 0.0002549 | 0.93511 |
| SOL/USDT | 0.0087763 | 0.0002129 | 0.0001100 | 0.91966 |
| ADA/USDT | 0.0067465 | 0.0003928 | 0.0003546 | 0.91452 |
| DOGE/USDT | 0.0058956 | 0.0002464 | 0.0001432 | 0.92496 |
| BNB/USDT | 0.0057420 | 0.0004479 | 0.0002762 | 0.89533 |
| XRP/USDT | 0.0057034 | 0.0002249 | 0.0001508 | 0.92362 |
| LTC/USDT | 0.0049997 | 0.0002870 | 0.0002853 | 0.91792 |

96 dates per pair. $\hat{Q}_{30}$ controls 30 ten-second bars of lagged flow. Placebo $|\hat{Q}| \leq 1.1 \times 10^{-5}$ for every pair.

Table 24: Feedback beyond filtering mechanics (Binance BTCUSDT, 72 days, 2021–2023)

| | Real prices | Placebo (independent random walk) |
|---|---|---|
| Mean loop gain $\hat{Q}$ | 0.0045 | 0.0000009 |
| 95% CI (day bootstrap) | [0.0034, 0.0056] | [−0.0000006, 0.0000023] |
| Mean price persistence $\hat{J}_{11}$ | 0.9364 | 0.9357 |
| Mechanical value $1 - 2/31$ | | 0.9355 |

Placebo 95% band across days: $[-0.0000095, 0.0000156]$. This construction pairs the state with next-second flow; it is not comparable in level with the observed-flow ten-second index of Table 22.

second flow, part of $\hat{Q}$ reflects the timing effect of Table 22; the placebo shows only that the filter does not produce it.

### S.5.5 Order splitting and the estimation of feedback

Persistent order flow makes lagged-flow regressions inconsistent. Suppose flow follows $q_t = \gamma y_t + u_t$ with memory $y_t = (1-\eta)y_{t-1} + \eta \Delta p_{t-1}$, order splitting $u_t = \rho u_{t-1} + \nu_t$, and $\Delta p_t = \kappa q_t + \lambda z_t + e_t$ with an observed exogenous price shock $z_t$. Lagged returns are correlated with $u_{t-1}$, so OLS of flow on returns is biased even when $\gamma = 0$. Quasi-differencing ($q_t - \rho q_{t-1} = \gamma(y_t - \rho y_{t-1}) + \nu_t$) or instruments that move $y$ through prices but are independent of $u$ ($z_{t-1}, z_{t-2}, z_{t-3}$) restore consistency. We compare estimators over 500 replications with $\eta = 0.3$ and $\rho = 0.8$ (Table 25).

OLS is badly biased. Cochrane–Orcutt with the true $\rho$, 2SLS and LIML have correct size and coverage; Cochrane–Orcutt with an estimated $\rho$ under-covers (Cochrane and Orcutt, 1949; Staiger and Stock, 1997). These are properties of estimators in a simulated market. On real data, a 2024 sub-panel of 192 spot units shows why the correction matters little at our bar length: with one VAR lag, the flow-equation residuals have essentially no first-order serial correlation (median $\hat{\rho} = -0.0002$), and Cochrane–Orcutt

Table 25: Feedback estimators under order splitting (simulation, 500 replications)

| Estimator | Coverage at $\gamma = 0.2$ | Rejection at $\gamma = 0$ | Bias | RMSE |
|---|---|---|---|---|
| OLS | 0.000 | 1.000 | 0.892 | 0.893 |
| Cochrane–Orcutt, true $\rho$ | 0.944 | 0.054 | 0.001 | 0.039 |
| Cochrane–Orcutt, estimated $\rho$ | 0.916 | 0.080 | 0.002 | 0.043 |
| 2SLS | 0.938 | 0.044 | −0.002 | 0.116 |
| LIML | 0.938 | 0.044 | −0.003 | 0.117 |

Newey–West standard errors (10 lags), $T = 3{,}000$. Median first-stage $F = 224$; a placebo instrument (future $z$) has $F = 0.62$. Estimators outside 93–97% coverage give invalid inference.

quasi-differencing moves the mean loop gain from $3.15 \times 10^{-4}$ to $3.21 \times 10^{-4}$ (Part 13, `R7`, post hoc). Order splitting shows up instead at longer lags, which the 30-lag control of Table 23 addresses. A real-data instrumental-variables estimate needs an instrument whose exclusion can be defended, which we do not claim to have; funding-settlement times, for example, may affect order flow directly.

Controlling for 30 ten-second bars of lagged flow lowers $\hat{Q}$ further (Table 23, column $\hat{Q}_{30}$), consistent with persistent order splitting.

### S.5.6 Feedback across 30 perpetual markets

The companion paper estimates the same loop gain on one-minute bars of 30 Binance USD-M perpetuals spread across the liquidity ranking (2023–2024 training days). Measured feedback rises with depth: its Spearman correlation with log depth within 1% of the mid is +0.70 [0.41, 0.87], from about 0.012 in BTCUSDT and ETHUSDT to 0.0016 on average in the thinnest third, and the placebo pipeline gives $\hat{Q} \approx 0$ for every symbol. Feedback in this reduced-form sense is a feature of deep, algorithmically traded books.

### S.5.7 Two fee events

Binance set spot trading fees on BTC pairs to zero on July 8, 2022 and reinstated them on March 22, 2023. Each event changes the cost of trading BTCUSDT while leaving the seven other pairs unchanged, and the second reverses the first. For every pair and every day within 20 days of each event we compute daily quote volume and the observed-flow $\hat{Q}$, and estimate $Y_{it} = \alpha_i + \gamma_t + \beta\,\mathrm{ZeroFee}_{it} + e_{it}$ with pair and date fixed effects and date-clustered errors, separately for each event and pooled (Table 26).

The fee change moved activity enormously: zero fees multiplied BTCUSDT volume relative to the controls by $e^{0.945} \approx 2.6$ when they began, and their removal divided it

Table 26: Zero fees on BTCUSDT: activity and measured feedback

| | Zero fees introduced (2022-07-08) | Zero fees removed (2023-03-22) | Pooled |
|---|---|---|---|
| Log quote volume, $\hat{\beta}$ | 0.945 (0.074) | 1.599 (0.080) | 1.272 (0.084) |
| Observed-flow $\hat{Q}$, $\hat{\beta}$ | 0.00058 (0.00032) | −0.00091 (0.00060) | −0.00016 (0.00036) |
| $p$-value | 0.078 | 0.140 | 0.650 |
| Pair-days | 320 | 320 | 640 |

$\hat{\beta}$ is the effect of zero fees on BTCUSDT relative to seven control pairs; date-clustered standard errors in parentheses. For the second event ZeroFee equals one before March 22, 2023.

by $e^{1.599} \approx 4.9$. Measured feedback did not follow. The two point estimates have opposite signs, neither is significant, and the pooled effect is $-0.00016$ with an interval of $[-0.00087, 0.00055]$, against a sample mean of 0.00037. A strong first stage with no second-stage response says that $\hat{Q}$ is not a by-product of trading activity, and the reversal design removes the main weakness of a single treated event. With one treated asset the interval remains wide, so this is evidence that $\hat{Q}$ does not scale with activity, not a proof of invariance.

## S.6 Proofs

**Lemma 2.1.** (i) For $v_{t+1} = v_t + \zeta_{t+1}$ and $r_t = v_t + \nu_t$, the steady-state prior variance $P$ solves $P = P\sigma_\nu^2/(P + \sigma_\nu^2) + \sigma_\zeta^2$, i.e. $P^2 - \sigma_\zeta^2 P - \sigma_\zeta^2\sigma_\nu^2 = 0$, whose positive root is $P = \big(\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2\sigma_\nu^2}\big)/2$. The gain is $\eta = P/(P + \sigma_\nu^2) \in (0, 1)$, which simplifies to (7), and the update $\hat{v}_{t+1} = \hat{v}_t + \eta(r_t - \hat{v}_t)$ has the stated form. (ii) $\nabla_\theta \frac{1}{2}H(x_t - \theta)^2 = -H(x_t - \theta)$, so $\theta_{t+1} = (1 - \gamma)\theta_t + \alpha H(x_t - \theta_t) = (1 - \eta)\theta_t + \eta q x_t$ with $\eta = \gamma + \alpha H$ and $q = \alpha H/\eta$.

**Lemma 2.3.** Omit zero-weight cohorts. For $a > 0$ put $b_i = \kappa q_i$ and

$$T = \operatorname{diag}\big(\sqrt{a}, \sqrt{b_1/\eta_1}, \ldots\big), \qquad u = \big(\sqrt{a}, \sqrt{b_1\eta_1}, \ldots\big)^\top.$$

The price-first transition becomes the symmetric matrix $J' = TJT^{-1} = \operatorname{diag}(0, r_1, \ldots) + uu^\top$, the input transforms to $\kappa u$ and $J'e_x = \sqrt{a}\,u$, so $h_j = (\kappa/a)e_x^\top (J')^{j+1} e_x$. The spectral decomposition of $J'$ gives nonnegative weights summing to $h_0 = \kappa$, and Schur stability puts the eigenvalues in $[0, 1)$. The case $a = 0$ follows by continuity. A nonnegative mixture of decaying geometric sequences is completely monotone. Since $J$ is nonnegative with row sums at most one, every price row of $J^t$ has $\ell_1$, hence $\ell_2$, norm at most one, attained at

$t = 0$.

**Theorem 3.1.** With zero initial states, $y_{i,t} = \sum_{s \le t} \eta_i r_i^{t-s} x_s$, so $\sum_i q_i y_{i,t} = \sum_{s \le t} k_{t-s} x_s$ and every response is determined by $(a, \kappa, k)$, hence by $M$. Conversely, equal responses for all probes imply equal $H$, hence equal $G$, and partial fractions of $G$ identify the distinct poles with nonzero residues $w_i$, i.e. $M$. After cancellation, $H = \kappa z D(z)/[(z-a)D(z) - \kappa N(z)]$ with $D = \prod(z - r_i)$ and $N(z) = \sum_i w_i z \prod_{j \ne i}(z - r_j)$, as in (4); at a retained $r_i$ the denominator equals $-\kappa N(r_i) = -\kappa w_i r_i \prod_{j \ne i}(r_i - r_j) \ne 0$ (memory rates are positive because $\eta_i < 1$), so no root of $D$ cancels. The remaining numerator factor $z$ cancels if and only if the denominator vanishes at 0; since $N(0) = 0$ the denominator there equals $-aD(0)$, which is nonzero for $a > 0$ and zero for $a = 0$. Hence the McMillan degree, which is the dimension of a minimal realization, is $1 + |\mathrm{supp}(M)|$ for $a > 0$ and $|\mathrm{supp}(M)|$ for $a = 0$ (numerically, the rank of the Hankel matrix of $(h_j)_{j \ge 1}$ is $K + 1$ and $K$ respectively for $K = 1, 2, 3$; `hardening/p1_checks.py`). For (ii), direct algebra gives

$$G_\delta(z) - G_0(z) = -\frac{qz(z-1)\delta^2}{(z - 1 + \bar{\eta})\big((z - 1 + \bar{\eta})^2 - \delta^2\big)}.$$

On $|z| = 1$ each factor has modulus at least $\epsilon$. Uniform stability gives $M_H = \sup |H_\theta| < \infty$, and the identity $H_\delta - H_0 = H_\delta H_0 (G_\delta - G_0)$ gives an $\ell_2$ gain at most $K\delta^2$. The mean difference of a power-constrained probe has squared norm at most $K^2\delta^4 PT$; the equal-noise KL is at most $K^2\delta^4 PT/(2\sigma_\nu^2)$, and Pinsker's inequality with total variation at least $1 - 2\alpha$ gives (9).

**Theorem 4.1.** In either positive economy $J \ge 0$ and $\|J\|_\infty \le 1 - \epsilon(1 - S) < 1$, so the response is nonnegative with $\max h = \kappa$ and $\sum h = \kappa/(1 - S)$. From the display above, $|G_\delta - G_0| \le 2q\delta^2/\epsilon^3$ on the circle; combined with the resolvent identity and positivity this gives $\|H_\delta - H_0\|_\infty \le K_* \delta^2$, and Parseval bounds $\|h^\delta - h^0\|_2$ by the same constant. Young's inequality gives $\|\mu_\delta - \mu_0\|_2 \le K_* \delta^2 \|v\|_1$, and Gaussian KL with Pinsker gives (10). For (ii), the mean difference is a lower-triangular convolution matrix times $v$; each column is a shifted copy of a prefix of $d_T$, so $\|\mu_\delta - \mu_0\|_2 \le V \|d_T\|_2$ with equality for a pulse at $t = 0$. For two known equal-covariance Gaussian laws the minimax error is $\Phi(-\|\mu_\delta - \mu_0\|_2/(2\sigma))$, attained by the midpoint likelihood-ratio test. For (iii), the merged measures have equal mass and their first moments differ by $-q\delta^2$, which enters $h_2$ with factor $\kappa^2$ while $h_0, h_1$ coincide; a pulse $A$ separates the third-observation means by $A\kappa^2 q\delta^2$. Part (iv) follows from the $\ell_2$ gain bound, and (v) from superposition and $\max_j h_j = \kappa$,

which give $\sup_t |\lambda x_t| \le |\lambda|\kappa V$, attained by a single pulse.

**Lemma 4.2.** After the uncontrolled update, the pre-probe displacement at step $j$ is $\sum_{i<j} h_{j-i} v_i$; average linear-book execution adds $\kappa v_j/2$. Summing signed execution costs gives $\frac{1}{2} v^\top T_h v$. Each kernel $\lambda^{|i-j|}$ is a positive-definite AR(1) covariance for $0 \le \lambda < 1$, and a nonnegative mixture with positive total weight is positive definite. For a round trip the fundamental cash term cancels.

**Theorem 4.3.** Let $U_t = \sum_{j<t} v_j$. Positive responses give $0 \le x_t \le \kappa U_t$ and $I_t = -U_t$. Summing cash and marking inventory, $B - E_t = U_t x_t - \sum_{j<t} v_j x^0_{j+1} - \frac{\kappa}{2} \sum_{j<t} v_j^2 + c U_t$. Its quadratic part $v^\top A_t v$ has diagonal $h_{t-1-i} - \kappa/2$ and off-diagonal $\frac{1}{2}(h_{t-1-i} + h_{t-1-j} - h_{j-i})$ for $i < j$; monotonicity of $h$ bounds every entry by $\kappa/2$. Hence $B - E_t \le \kappa U_t^2/2 + c U_t \le \kappa V^2/2 + cV$, while position value is at most $V(P_* + \kappa V)$; the stated inequality guarantees margin. A single pulse attains both adverse bounds at its first response, so no larger allowance is robust. Solving the quadratic gives (14). The quantity requirement $V_B \gtrsim \delta^{-2}$ then forces $B \gtrsim \delta^{-4}$, and the pulse $A$ funded with $(\mu + \frac{1}{2})\kappa A^2 + (c + \mu P_*)A$ attains it.

**Proposition 4.4.** The initial cash receipt is common to both alternatives and the dealer then holds $-A$. At time three $E_{3,r} = B + \kappa A^2/2 - cA - A^2 h_{r,2}$, so $E_{3,\delta} - E_{3,0} = A^2 \kappa^2 q \delta^2 > 0$.

**Lemma 4.5.** Marking residual inventory at $P_*$ gives $E_\infty = B - \frac{1}{2} v^\top T_h v - c \sum |v_t| \ge 0$. For each mode $\lambda \le r$ the Toeplitz kernel has spectral density $(1-\lambda^2)/(1 - 2\lambda \cos\omega + \lambda^2) \ge (1-r)/(1+r)$, so $v^\top T_h v \ge m \|v\|_2^2$ and $\|v\|_2^2 \le 2B/m$. The transfer bound gives mean-difference squared norm at most $K_*^2 \delta^4 \|v\|_2^2$, hence the KL bound; Pinsker gives (15). For one pulse, cash becomes $B_A - A(P_* + \kappa A/2) - cA$ and inventory $A$; at later marks $E_t = \mu A P_* + A^2 h_{t-1}$ and $\mu |I_t| P_t = \mu A P_* + \mu A^2 h_{t-1}$, so margin holds for $\mu \le 1$.

**Proposition S.2.1.** Parts (i) and (ii) follow from (29) with $\Delta h = \Theta(\delta)$ for a rate shift and $\Theta(\delta^2)$ for symmetric splitting, and from additivity of KL over independent pulse cycles. Part (iii) solves the session constraint for $\delta$; substituting $d_T(\delta) \approx \delta^2 \tilde{d}_T$ gives the fourth-root approximation.

**Theorems 5.1 and 5.2.** For a flat round trip, telescoping the cash account gives $L_i(v) = \frac{1}{2} v^\top T_{h_i} v = \frac{1}{2\pi} \int g_i(\omega) |V(e^{i\omega})|^2 d\omega$, with $g_i > 0$ by Lemma 2.3. Parseval gives $\|(H_1 - H_0)v\|_2^2 = \frac{1}{2\pi} \int |\Delta \hat{H}|^2 |V|^2 \le \sup_\omega (|\Delta \hat{H}|^2 / g_i) L_i(v)$, and the Neyman–Pearson

requirement $\|(H_1 - H_0)v\|_2 \geq 2\sigma z_{1-\alpha}$ gives (18). For adaptive policies, Doob's optional stopping theorem applied to the log-likelihood-ratio martingale and the data-processing inequality give $\mathbb{E}_0[\Lambda_\tau] \geq \mathrm{kl}(\alpha, 1-\beta)$, while $\mathbb{E}_0[\Lambda_\tau] = \mathbb{E}_0 \sum_t (\Delta\mu_t)^2/(2\sigma^2) \leq \sup(|\Delta\hat{H}|^2/g_0)\mathbb{E}_0[L_0]/(2\sigma^2)$, which gives (20).

*Part (iv).* Let $G_\theta$ be the $T \times T$ lower-triangular Toeplitz section of $H_\theta/\kappa$, so the observation covariance is $\Sigma_\theta = \sigma_w^2 G_\theta G_\theta^\top + \sigma_\nu^2 I$. A finite section of a bounded analytic symbol has norm at most its supremum on the circle, so $\|G_\theta\| \leq \sup_\omega |H_\theta|/\kappa \leq \bar{M}$ because $|H_\theta(e^{i\omega})| \leq \sum_j h_j = \kappa/(1-S)$. The inverse of a lower-triangular Toeplitz section is the section of the reciprocal series; $\kappa/H_\theta$ is analytic on and outside the unit circle because $H_\theta$ is minimum phase (Theorem 6.8), and Lemma 2.3 gives $\mathrm{Re}\, H_\theta(e^{i\omega}) = \sum_\ell \omega_\ell(1 - \lambda_\ell \cos\omega)/(1 - 2\lambda_\ell \cos\omega + \lambda_\ell^2) \geq \sum_\ell \omega_\ell/(1+\lambda_\ell) \geq \kappa/2$, so $\|G_\theta^{-1}\| \leq 2$ and $\Sigma_\theta \succeq \sigma_{\mathrm{eff}}^2 I$ (the smallest singular value of $G_\theta$ is at least 0.522 in 762 random positive economies and sections; `hardening/p1_checks.py`). The Gaussian divergence splits into a mean and a covariance part. The mean part is $\frac{1}{2}\Delta\mu^\top \Sigma_1^{-1}\Delta\mu \leq \|(H_1 - H_0)v\|_2^2/(2\sigma_{\mathrm{eff}}^2)$. For the covariance part, the eigenvalues $\lambda_k$ of $A = \Sigma_1^{-1/2}\Sigma_0\Sigma_1^{-1/2}$ are at least $c = \sigma_\nu^2/(\sigma_\nu^2 + \sigma_w^2\bar{M}^2)$, and $\lambda - 1 - \log\lambda \leq (\lambda-1)^2/\lambda$, so it is at most $\|A - I\|_F^2/(2c) \leq \|\Sigma_0 - \Sigma_1\|_F^2/(2c\,\sigma_{\mathrm{eff}}^4)$. Writing $\Sigma_0 - \Sigma_1 = \sigma_w^2(\Delta G\, G_0^\top + G_1 \Delta G^\top)$ and $\|\Delta G\|_F^2 \leq T\|h^\delta - h^0\|_2^2/\kappa^2 \leq TK_*^2\delta^4/\kappa^2$ (proof of Theorem 4.1) gives a covariance part of at most $C_w T\delta^4$. The data-processing inequality turns errors $(\alpha, \beta)$ into $D_{\mathrm{KL}} \geq \mathrm{kl}(\alpha, 1-\beta)$, and part (ii) bounds $\|(H_1 - H_0)v\|_2^2$ by $\sup_\omega(|\Delta\hat{H}|^2/g_i)L_i(v)$, which gives (19).

**Theorem 5.3.** Write a feasible flat probe as $v = Eu$ and expand $u = \sum_j c_j e_j$ in the $G_i$-orthonormal eigenbasis, so $L_i(v) = \sum_j c_j^2$ and the separation constraint reads $\sum_j \lambda_j c_j^2 \geq c_{\mathrm{req}}$. Write the top-two component as $(c_{L-2}, c_{L-1}) = r(\cos\theta, \sin\theta)$ with $r \geq 0$; replacing $\theta$ by $\theta + \pi$ flips $d$ and $s_\theta$ and leaves (21) unchanged, so $\theta \in [0, \pi)$ suffices. Put $L_0(\theta) = c_{\mathrm{req}}/\lambda(\theta)$ and $D = L_i(v) - L_0(\theta)$. The constraint gives $r^2\lambda(\theta) \geq c_{\mathrm{req}} - \sum_{j \leq L-3} \lambda_j c_j^2$, hence

$$D = r^2 + \sum_{j \leq L-3} c_j^2 - L_0(\theta) \;\geq\; \sum_{j \leq L-3} \Big(1 - \frac{\lambda_j}{\lambda(\theta)}\Big) c_j^2 \;\geq\; \Big(1 - \frac{\lambda_{L-3}}{\lambda(\theta)}\Big) \sum_{j \leq L-3} c_j^2 \;\geq 0,$$

and $r^2 \geq L_0(\theta) - \frac{\lambda_{L-3}}{\lambda(\theta)} \sum_{j \leq L-3} c_j^2 \geq L_0(\theta)\big(1 - \gamma(\theta)D\big)$. By Hölder's inequality with the sign vector $s_\theta$, $c_{\mathrm{fee}}\|Eu\|_1 \geq c_{\mathrm{fee}} r \|Ed(\theta)\|_1 + c_{\mathrm{fee}} \sum_{j \leq L-3} c_j\, s_\theta^\top E e_j$, and Cauchy–Schwarz with weights $1 - \lambda_j/\lambda(\theta)$ bounds the second term below by $-\sqrt{D\, W_\perp(\theta)}$. Using $r \geq \sqrt{L_0(1-\gamma D)_+} \geq \sqrt{L_0}\,(1 - \gamma D)_+$ and the fact that the fee is nonnegative, with $x = \sqrt{D}$,

$$L_i(v) + c_{\mathrm{fee}}\|v\|_1 \;\geq\; L_0(\theta) + x^2 + \Big[F_{\mathrm{dom}}(\theta)(1 - \gamma(\theta)x^2)_+ - \sqrt{W_\perp(\theta)}\, x\Big]_+ \;\geq\; L_0(\theta) + \Psi(\theta).$$

Minimizing over the unknown $\theta$ gives (22), and $\Psi \geq 0$, $L_0(\theta) \geq c_{\text{req}}/\lambda_{L-1}$ give the last inequality. On the piece $\gamma x^2 \leq 1$ with $\gamma F_{\text{dom}} < 1$ the bracket is a convex quadratic in $x$ with minimum $F_{\text{dom}} - W_\perp/(4(1-\gamma F_{\text{dom}}))$, which gives the closed form.

**Corollary S.3.1.** Since $\Delta h_t = O(\delta^2)$, the variance difference $\sigma_{1,t}^2 - \sigma_{0,t}^2 = \sigma^2 \xi[(h_1 + h_0) * v]_t [\Delta h * v]_t = O(\delta^2)$, and the variance term of the Gaussian KL, $\frac{1}{2}\sum(x - 1 - \log x)$ with $x = \sigma_{0,t}^2/\sigma_{1,t}^2$, is quadratic in that difference. Convex impact only increases dissipation.

**Proposition 6.1.** Standard Cramér–Rao for a Gaussian mean model with Jacobian $J$. For $K = 2$ the columns of $J$ for the two rates become collinear as $\Delta\eta \to 0$; expanding to third order, the smallest singular value of $J$ is of order $(\Delta\eta)^3$, so $[F^{-1}]_{kk} = \Theta((\Delta\eta)^{-6})$.

**Theorem 6.2.** Expanding $M_m(\theta_1) - M_m(\theta_0)$ under (24) gives zero for $m \leq 2$ and $4\tau p_- p_+ (\Delta\eta)^3$ for $m = 3$. The map from moments to impulse responses is analytic with nonsingular derivative at the common limit, so $\|\Delta h\|_2 = \Theta(\tau(\Delta\eta)^3)$. Le Cam's two-point lemma with $N$ independent Gaussian experiments gives (26). For the permutation-invariant loss, the identity matching of $\theta_0$ and $\theta_1$ costs $2\tau + (p_+ - p_-)\Delta\eta = 2\tau(1 + \Delta\eta/q)$ per atom and the swapped matching costs $\max(2p_+, 2p_-)\Delta\eta = (1 + 2\tau/q)\Delta\eta$; when both are at least $2\tau$, any estimate within $\tau$ of one economy in $d_\infty$ is farther than $\tau$ from the other by the triangle inequality, and the two-point argument is unchanged. For (iii), the projection estimator of $M_3$ is unbiased with variance $\sigma_h^2/(N\|\nabla M_3 h\|^2)$, and sub-Gaussian concentration gives the sample size; the Prony–Hankel determinant identity follows from $g_m = w_1\eta_1 r_1^m + w_2\eta_2 r_2^m$.

**Theorem 6.4.** Let $(s^A, \omega^A)$ and $(s^B, \omega^B)$ be the $K$-point Gauss and Gauss–Radau rules for the uniform weight on $[-1, 1]$, normalized to unit mass. Both are positive and exact for polynomials of degree $2K - 2$, and their moments of degree $2K - 1$ differ by $g_K \neq 0$. With $r = r_c + \Delta\eta\, s$ and weights $W\omega$, the raw moments $\sum w_i r_i^m$ are polynomials in $\Delta\eta$ that agree below degree $2K-1$, so the kernels differ by $O((\Delta\eta)^{2K-1})$ with nonzero leading coefficient; analyticity of the kernel-to-response map gives $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$, and Le Cam's lemma gives the bound. For the $W_1$ version, the two merged measures are the images of the fixed normalized rules under $s \mapsto r_c + \Delta\eta\, s$ scaled to mass $W$, so their distance is $W\Delta\eta$ times the distance $2c_K > 0$ between the rules, and the same two-point argument applies with the metric loss.

**Lemma 6.6.** Let $L = 2n-2$ and let $p^*$ be the best uniform approximation of $|x|$ on $[-1,1]$ by polynomials of degree $L$, with error $E = E_L(|x|)$. By Chebyshev's equioscillation theorem (Rivlin, 1981) there are points $t_0 < \cdots < t_{L+1}$ at which $|x| - p^*(x)$ equals $\pm E$ with alternating signs. The divided-difference weights $w_i = 1/\prod_{j\neq i}(t_i - t_j)$ alternate in sign and satisfy $\sum_i w_i p(t_i) = 0$ for every polynomial $p$ of degree at most $L$; multiplying by $-1$ if necessary, $w_i$ has the sign of $|t_i| - p^*(t_i)$ for every $i$. Exactly $n$ weights are positive and $n$ negative, and the positive and negative parts have equal mass because constants are annihilated. Normalizing each part to unit mass and mapping $x \mapsto \rho(x+1)/2$ gives $n$-atom probability measures $\pi_A$ (positive part) and $\pi_B$ (negative part) on $[0,\rho]$ with equal moments of orders $0,\dots,L$. For the 1-Lipschitz function $f(r) = |r-\rho/2| = (\rho/2)|x|$,

$$\int f\, d(\pi_A - \pi_B) = \frac{\rho}{2}\int \big(|x| - p^*(x)\big)\, d(\pi_A - \pi_B) = \frac{\rho}{2}(E+E) = \rho E,$$

because $p^*$ integrates to the same value under both measures, $|x| - p^* = E$ on the support of $\pi_A$ and $-E$ on that of $\pi_B$. By Kantorovich–Rubinstein duality $W_1(\pi_A,\pi_B) \geq \rho E \geq c_*\rho/L$.

**Theorem 6.7.** Write $K(z) = \sum_m k_m z^m$, $D = 1 - az - \kappa z K$ and $H = \kappa/D$, which is the generating function of the recursion $h_t = a h_{t-1} + \kappa \sum_m k_m h_{t-1-m}$ with $h_0 = \kappa$. Since $k_m \geq 0$ and $\sum_m k_m = \int (1-r)^{-1} d\mu \leq W/(1-\rho)$, the $\ell_1$ norm of $az + \kappa z K$ is at most $s < 1$, so $\|H\|_1 \leq \kappa/(1-s)$ and $\|1/H\|_1 = \|D\|_1/\kappa \leq (1+s)/\kappa$.

(i) Let $\mu_A = W\pi_A$ and $\mu_B = W\pi_B$ with $\pi_A, \pi_B$ from Lemma 6.6; both lie in $\mathcal{M}(\rho, W)$. With $\nu = \mu_A - \mu_B$, $\sum_m (k_m^A - k_m^B) z^m = \int (1-rz)^{-1} d\nu(r) = z^{2n-1} \int r^{2n-1}(1-rz)^{-1} d\nu(r)$, because the first $2n-1$ terms of the geometric series integrate to zero. On $|z| = 1$ the modulus is at most $2W\rho^{2n-1}/(1-\rho)$, so Parseval gives $\|k^A - k^B\|_2 \leq 2W\rho^{2n-1}/(1-\rho)$. Next, $H_A - H_B = z(K_A - K_B) H_A H_B$, so by Young's inequality $\|h^A - h^B\|_2 \leq \|H_A\|_1 \|H_B\|_1 \|k^A - k^B\|_2 \leq C_h \|k^A - k^B\|_2$. The Kullback–Leibler divergence between the $N$-sample Gaussian experiments is $N\|h^A - h^B\|_2^2/(2\sigma^2) \leq (N/\sigma^2)\, 2C_h^2 W^2 \rho^{2(2n-1)}/(1-\rho)^2$, which is at most one as soon as $2(2n-1)\log(1/\rho) \geq \Lambda_N - 2\log(1/\rho)$. Take $n$ the smallest integer with this property and $n \geq 2$; then $2n-2 < \Lambda_N/(2\log(1/\rho))$ when $\Lambda_N \geq 4\log(1/\rho)$. Since $d$ is a norm on signed measures and $d(\mu_A,\mu_B) = W\, W_1(\pi_A,\pi_B) \geq W c_*\rho/(2n-2) > 2c_*\rho W \log(1/\rho)/\Lambda_N$, the Bretagnolle–Huber form of Le Cam's lemma (Tsybakov, 2009, Lemma 2.6 and Theorem 2.2) gives $\max_{j\in\{A,B\}} P_j(d(\hat\mu,\mu_j) \geq c_*\rho W \log(1/\rho)/\Lambda_N) \geq \frac{1}{4}e^{-1}$.

(ii) *Approximation.* Let $f$ be 1-Lipschitz on $[0,\rho]$ with $f(0) = 0$, so $|f| \leq \rho$. The function $g(x) = f(\rho(x+1)/2)$ is $(\rho/2)$-Lipschitz on $[-1,1]$, and Jackson's theorem (Rivlin, 1981) gives a polynomial $p$ of degree $L$ with $\sup|g - p| \leq (\pi/2)(\rho/2)/(L+1) \leq \rho$, hence

$\sup|p| \le 2\rho$. Its Chebyshev coefficients satisfy $|b_j| \le 2\sup|p| \le 4\rho$. The polynomial $q(r) = p(2r/\rho - 1)$ approximates $f$ with the same error, and its monomial coefficients satisfy $\sum_j |q_j| \le \sum_{j\le L} |b_j|\,\|T_j(2r/\rho-1)\|_{\mathrm{coef}} \le 4\rho\sum_{j\le L} B(\rho)^j \le \rho B(\rho)^{L+1}$, where $\|\cdot\|_{\mathrm{coef}}$ is the sum of absolute monomial coefficients, $\|T_j\|_{\mathrm{coef}} \le (1+\sqrt{2})^j$, substituting $y = \alpha r - 1$ multiplies the coefficient sum of $y^i$ by at most $(1+\alpha)^i$, and $B(\rho) - 1 \ge 4$. *Moment comparison.* If $\max_{j\le L} |\int r^j d\hat{\mu} - k_j| \le 2\delta$, then $\hat{\mu}$ has mass at most $W + 2\delta$ and

$$\int f\, d(\hat{\mu} - \mu) = \int (f - q)\, d(\hat{\mu} - \mu) + \sum_{j\le L} q_j \Big( \int r^j d\hat{\mu} - k_j \Big) \le \frac{\pi\rho(W+\delta)}{2(L+1)} + 2\delta\rho B(\rho)^{L+1}.$$

*Kernel recovery.* From $\kappa z K = 1 - az - \kappa/H$, the kernel values $k_0, \dots, k_L$ are the coefficients of $(1 - az - \kappa/H)/(\kappa z)$, which depend on $h_0, \dots, h_{L+1}$ only. Work in the algebra of power series modulo $z^{L+2}$ with the $\ell_1$ norm, which is submultiplicative. If the averaged response is $\hat{H} = H + E$ and $\|1/H\|_1 \|E\|_1 \le \frac{1}{2}$, then $1/\hat{H} = (1/H)(1 + E/H)^{-1}$ and $\|1/\hat{H} - 1/H\|_1 \le 2\|1/H\|_1^2 \|E\|_1$, so $\max_j |\hat{k}_j - k_j| \le \delta \equiv 2((1+s)/\kappa)^2 (L+2) \max_{t\le L+1} |E_t|$. Each $E_t$ is Gaussian with variance $\sigma^2/N$, so with probability at least $1 - \beta$, $\max_t |E_t| \le \sigma\sqrt{2\log(2(L+2)/\beta)/N}$. The true $\mu$ is feasible for the linear program with value at most $\delta$, so the minimizer satisfies the moment condition above. *Choice of $L$.* With $L + 1 = \lfloor \log(N/\sigma^2)/(4\log B(\rho)) \rfloor$, $B(\rho)^{L+1} \le (N/\sigma^2)^{1/4}$, so $\delta B(\rho)^{L+1} = O((\sigma^2/N)^{1/4} L\sqrt{\log(L/\beta)})$ and $\delta/(L+1)$ are $o(1/\log(N/\sigma^2))$, while $\pi\rho W/(2(L+1)) \le 2\pi\rho W \log B(\rho)/(\log(N/\sigma^2) - 4\log B(\rho))$. For $N/\sigma^2$ larger than a threshold depending on $(\rho, W, a, \kappa, \beta)$ the side condition $\|1/H\|_1 \|E\|_1 \le \frac{1}{2}$ holds on the event and the total is below $3\pi\rho W \log B(\rho)/\log(N/\sigma^2)$, uniformly over $\mathcal{M}(\rho, W)$. The final statement combines (i) and (ii).

**Theorem 6.9.** For stationary Gaussian processes with spectral densities $f_j \propto |H_j(e^{i\omega})|^2$, the divergence rate is

$$D = \frac{1}{4\pi} \int_{-\pi}^{\pi} \big(\varrho - 1 - \log\varrho\big)\, d\omega, \qquad \varrho = f_1/f_0 = |H_1/H_0|^2.$$

By the resolvent identity, $H_1/H_0 - 1 = H_1 \Delta G$ with $\Delta G = G_1 - G_0$, and uniform stability bounds $|H_1|$ on the circle, so $u \equiv \varrho - 1 = 2\,\mathrm{Re}(H_1 \Delta G) + O(|\Delta G|^2)$. Write $G(z) = \sum_k w_k \phi(z, \eta_k)$, with $w_k$ the cohort weights (any factor of $\eta_k$ is absorbed in $\phi$), where $\phi$ is analytic in $\eta$ on a neighbourhood of $[\eta_{\min}, \eta_{\max}]$ uniformly for $|z| = 1$ (its pole $z = 1 - \eta$ stays inside the disk). Expanding $\phi$ around $\bar{\eta}$, $\Delta G = \sum_{j\ge 0} \frac{\partial_\eta^j \phi(z,\bar{\eta})}{j!} \Delta\tilde{M}_j$, where $\tilde{M}_j = \sum_k w_k (\eta_k - \bar{\eta})^j$. For the pair, $\Delta\tilde{M}_0 = \Delta\tilde{M}_1 = 0$ and $\Delta\tilde{M}_2 = q\delta^2$; for the moment-

matched economies, $\Delta\tilde{M}_j = 0$ for $j \le 2$ and $\Delta\tilde{M}_3 = 4\tau p_- p_+ (\Delta\eta)^3$ by (25); for the $K$-cohort Gauss and Gauss–Radau economies with rates $\bar{\eta}+\Delta\eta\, s$ and weights $W\omega$, both rules integrate polynomials of degree $2K-2$ exactly, so $\Delta\tilde{M}_j = W(\Delta\eta)^j \sum_k (\omega_k^A (s_k^A)^j - \omega_k^B (s_k^B)^j)$ vanishes for $j \le 2K-2$ and equals $\pm W g_K (\Delta\eta)^{2K-1} \neq 0$ at $j = 2K-1$. Higher terms are smaller by a factor of the spacing: $|\Delta\tilde{M}_j| \le 2W(\Delta\eta)^j$, and analyticity of $\phi$ in $\eta$ on a disk of radius $\varrho > 0$ around $\bar{\eta}$, uniformly on the circle, gives $|\partial_\eta^j \phi|/j! \le C\varrho^{-j}$, so for $\Delta\eta < \varrho/2$ the tail beyond the first nonzero term is $O((\Delta\eta)^{j_0+1})$. Hence $\sup_\omega |u| = O(\delta^2)$, $O(\tau(\Delta\eta)^3)$, respectively $O((\Delta\eta)^{2K-1})$, and $u = c(\omega)\epsilon + o(\epsilon)$ with $\epsilon = \delta^2$, $\tau(\Delta\eta)^3$ or $(\Delta\eta)^{2K-1}$ and $c(\omega)$ proportional to $2\,\mathrm{Re}\big(H_0\, \partial_\eta^{j_0}\phi(\cdot,\bar{\eta})\big)/j_0!$, $j_0 = 2$, 3 or $2K-1$. The function $c$ does not vanish identically: $H_0 \partial_\eta^j \phi$ is analytic outside the disk and non-constant, and an analytic function whose real part vanishes on the circle is an imaginary constant. For $|u| \le \frac{1}{2}$, $u^2/4 \le u - \log(1+u) \le u^2$, so $D \asymp \int c(\omega)^2 d\omega\, \epsilon^2$, which gives (i)–(iii). The sample requirements follow from $D(P_1^T \| P_0^T) = TD + O(1)$ for stationary Gaussian processes with smooth spectra and the Bretagnolle–Huber inequality (Tsybakov, 2009, Lemma 2.6); attainment follows from Stein's lemma for the likelihood-ratio test.

**Theorem 6.8.** In the delay variable $\lambda = z^{-1}$, $H = \kappa \prod_i (1 - r_i\lambda) / \big[(1-a\lambda)\prod_i (1 - r_i\lambda) - \kappa\lambda \sum_i w_i \prod_{j\neq i}(1 - r_j\lambda)\big]$, which is (4) divided through by $z^{K+1}$; its zeros are the memory rates, which do not cancel by the argument for Theorem 3.1, and stability puts the poles inside the disk. A rational spectral density $\sigma^2 |H(e^{i\omega})|^2$ has a unique minimum-phase factor up to a unimodular constant, which is $H$; Theorem 3.1 then gives the merged measure.

**Proposition 8.1.** With tagged flow, $u_{i,t} = r_i u_{i,t-1} + w_i x_t$ holds exactly and involves only cohort $i$, so $\tilde{u}_{i,t} = r_i \tilde{u}_{i,t-1} + w_i x_t + e_{i,t}$ with the MA(1) error $e_{i,t}$. Prices are independent of the measurement noise, so $\mathbb{E}[Z_t e_{i,t}] = 0$ and the IV estimator is consistent when $\Sigma_i = \mathbb{E}[Z_t(\tilde{u}_{i,t-1}, x_t)]$ is nonsingular. Since $u_{i,t-1} = w_i \sum_{s\ge 0} r_i^s x_{t-1-s}$, $\mathbb{E}[x_{t-1} u_{i,t-1}] = w_i A_0$ and $\mathbb{E}[x_t u_{i,t-1}] = w_i A_1$, which gives $\Sigma_i$. The price is a nonnegative mixture of geometric responses to its innovation (Lemma 2.3), so $\gamma_x(s) = \sum_\ell c_\ell \lambda_\ell^s$ with $c_\ell = \sigma_\xi^2 \kappa^{-2} \omega_\ell \sum_m \omega_m/(1-\lambda_\ell\lambda_m) \ge 0$; substituting, $\det \Sigma_i = w_i(\gamma_x(0)A_0 - \gamma_x(1)A_1) = w_i \sum_{\ell,m} c_\ell c_m (1-\lambda_\ell\lambda_m)/(1-r_i\lambda_m)$, each term positive. The scores $Z_t e_{i,t}$ are one-dependent with $\mathbb{E}[Z_t Z_t^\top e_{i,t}^2] = \sigma_f^2(1+r_i^2)\Gamma_0$ and $\mathbb{E}[Z_t Z_{t-1}^\top e_{i,t} e_{i,t-1}] = -r_i \sigma_f^2 \Gamma_1$, so their long-run variance is $\Omega_i$, and the central limit theorem for $m$-dependent sequences gives (27). On a uniformly stable family the modes and weights lie in a compact set with $\lambda_\ell \le \bar{\lambda} < 1$, so $\det \Sigma_i \ge w_{\min}(1-\bar{\lambda}^2)\gamma_x(0)^2 > 0$ and $\|\Omega_i\|$ is bounded, uniformly in the spacing.

**Theorem S.4.1.** With $J_\theta$ upper triangular the eigenvalues are fixed. The stationary covariance solves $P = J_\theta P J_\theta^\top + \mathrm{diag}(s_1, s_2)$, and $P_{11}$ is affine in $s_2$ with positive slope, so for each $\theta$ a unique $s_2(\theta) \geq 0$ gives $P_{11} = V$ when $V$ exceeds the value at $s_2 = 0$. The one-step price variance is $s_1$ throughout, and $e_1^\top J_\theta e_2 = \theta$ gives $\sup_k \|e_1^\top J_\theta^k\| \geq |\theta|$.

# S.7 Notation

| Symbol | Meaning |
|---|---|
| $x_t$ | pricing error (price minus fundamental) |
| $a$ | price persistence (anchor), $0 \leq a < 1$ |
| $\kappa$ | price impact per unit of executed quantity |
| $q_i,\ \eta_i,\ r_i$ | cohort weight, adaptation rate, memory rate $r_i = 1 - \eta_i$ |
| $w_i = q_i \eta_i$ | merged weight |
| $M = \sum_i w_i \delta_{r_i}$ | merged rate–weight measure |
| $k_m,\ G(z)$ | feedback kernel and its transform |
| $H(z),\ h_j$ | closed-loop transfer function and impulse response, $h_0 = \kappa$ |
| $J,\ S$ | state transition matrix and $S = a + \kappa \sum q_i$ |
| $v_t,\ V$ | executed probe quantity and total quantity $\sum \lvert v_t \rvert$ |
| $d_T$ | impulse-response difference between hypotheses, first $T$ taps |
| $\sigma$ | observation-noise standard deviation |
| $\delta,\ \Delta\eta$ | half-separation of a known pair; spacing of unknown cohorts |
| $K_*$ | transfer-difference constant $2\kappa^2 q / [\epsilon^3 (1 - S)^2]$ |
| $T_h$ | Toeplitz matrix $(h_{\lvert i-j \rvert})$ of execution cost |
| $g_i(\omega)$ | dissipation kernel $\mathrm{Re}\, \hat{H}_i(e^{i\omega}) - \kappa/2$ |
| $B,\ \mu,\ c,\ \phi$ | capital, maintenance-margin fraction, proportional cost, taker fee |
| $V_{\mathrm{eff}}$ | per-trade execution capacity $\min(V_{\max}, D_{\max}/\kappa)$ |
| $\hat{Q},\ \hat{S}$ | reduced-form loop gain and persistence indices |

# S.8 Internet appendix

## S.8.1 A finite-window price-surveillance bound

**Theorem S.8.1** (Finite-window abstention)**.** *Observe* $x_t = sx_{t-1} + w_t$, $w_t \sim N(0, \sigma^2)$, $x_0 = 0$, $t \le T$, *with* $s \in (0, 2)$. *Let a possibly randomized rule* $\Psi$ *output Stable, Unstable or Abstain, with* $\sup_{s<1} P_s(\Psi = \text{Unstable}) \le \alpha$ *and* $\sup_{s\ge1} P_s(\Psi = \text{Stable}) \le \alpha$, $\alpha < 1/2$. *Choose* $0 < c_0 < 1$ *with* $c_0 e^{c_0}/(2\sqrt{2}) < 1 - 2\alpha$. *On* $B_T = (1 - c_0/T, 1 + c_0/T)$,

$$\inf_{s\in B_T} P_s(\Psi = \text{Abstain}) \;\ge\; 1 - 2\alpha - \frac{c_0 e^{c_0}}{2\sqrt{2}} > 0,$$

*and for every finite* $T$ *the minimax worst-case error of binary rules over* $s \in (0, 2)$ *is at least* $1/2$.

*Proof.* Pair $s_0 \in (1 - c_0/T, 1)$ with $s_1 = s_0 + c_0/T$. Then $D_{\mathrm{KL}}(P_{s_0} \| P_{s_1}) = \frac{(s_1-s_0)^2}{2\sigma^2} \sum_t \mathbb{E}x_{t-1}^2 \le c_0^2 e^{2c_0}/4$, so Pinsker bounds total variation by $c_0 e^{c_0}/(2\sqrt{2})$. Error control gives $P_{s_0}(\text{Stable}) \le \alpha + \text{TV}$ and $P_{s_0}(\text{Unstable}) \le \alpha$, which yields the abstention bound. For binary rules, continuity in total variation and Le Cam's two-point bound give worst-case error tending to $1/2$. □

The interval $B_T$ is an abstention layer that any error-controlled monitor must respect in the scalar Gaussian case; richer disclosure can remove it.

## S.8.2 Passive distinction of a known pair in finite samples

**Proposition S.8.2** (Passive distinction with zero controlled quantity)**.** *Keep the known pair, but add independent uniform innovations* $w_t \in [-b, b]$ *of variance* $s_w^2 = b^2/3$ *at each price update, observe* $\tilde{x}_t = x_t + \nu_t$ *with* $\nu_t \sim N(0, \sigma^2)$, *use independent three-step reset blocks and no probes, with* $P_* > 3b$. *Let* $b_1 = a + \kappa q\eta$, $b_{2,0} = b_1^2 + \kappa q\eta(1 - \eta)$ *and* $b_{2,\delta} = b_{2,0} - \kappa q\delta^2$. *In each block the covariance of the first and third observations differs by* $\Delta = \kappa q s_w^2 \delta^2$. *With*

$$R = \max_{r\in\{0,\delta\}} \Big\{ (s_w^2 + \sigma^2)\big[s_w^2(1 + b_1^2 + b_{2,r}^2) + \sigma^2\big] + s_w^4 b_{2,r}^2 \Big\},$$

*the midpoint test of the average product has worst-case error at most* $\alpha$ *whenever* $n \ge 4R/(\alpha\Delta^2)$ *blocks are observed: zero controlled quantity and order* $\delta^{-4}$ *blocks.*

The Chebyshev bound is conservative but the order is real. With $b = 0.03$, observation s.d. 0.01 and $\delta = 0.10$, the bound asks for $3.9 \times 10^7$ blocks at $\alpha = 0.1$; in simulation the

midpoint test still errs 32–38% of the time with $2 \times 10^5$ blocks. Independent resets and known innovation laws are extra conditions. The proposition does not show that anonymous public records identify participants; it shows that passive excitation can accumulate memory information while the probe account stays untouched.

*Proof.* In each block $x_1 = w_1$, $x_2 = b_1 w_1 + w_2$ and $x_3 = b_{2,r} w_1 + b_1 w_2 + w_3$, so $|x_t| \le 3b$. Observation noise changes variances but not the covariance $s_w^2 b_{2,r}$. The product variance is the product of marginal variances plus the squared covariance plus a fourth-cumulant term $b_{2,r}^2(\mathbb{E}w^4 - 3s_w^4) \le 0$ for uniform innovations, so it is at most $R$. Chebyshev bounds each midpoint error by $4R/(n\Delta^2)$. □

### S.8.3 Blind inversion of the closed loop

1. *Impulse decoupling:* $g_t = h_{t+1} - a h_t$.
2. *Minimal realization:* form the Hankel matrix $H_{ij} = g_{i+j-1}$ and run Kung's eigensystem realization with a truncated SVD.
3. *Transmission zeros:* compute the zeros of the realization, which under known $(\kappa, a)$ isolate the roots of the denominator of $G$.
4. *Weights:* with recovered rates, project $u_t = (x_t - a x_{t-1})/\kappa$ on the reconstructed filters $y_{k,t-1}$.

Without noise each step is exact: the realization has the order of the true system, its transmission zeros are the memory rates (Theorem 3.1), and the projection returns the weights, so the procedure removes the pole-shift bias of fitting poles directly to the price response. With noise, per-cohort recovery degrades as rates move together at the rate given by Theorems 6.4 and 6.5, which is why we do not report it as an estimator of individual cohorts.

### S.8.4 Probing protocols

### S.8.5 Coordinate-weighted output

For $J = \left(\begin{smallmatrix} a & \kappa q \\ \eta a & 1-\eta+\eta\kappa q \end{smallmatrix}\right)$ with $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.3$ ($\rho(J) = 0.8224$, non-normal), the price output has $\sup_t \|e_x^\top J^t\| = 1$. The weighted output $C_w = (1, \lambda/\mathrm{MMR}) = (1, 160)$ with $\lambda = 0.8$ and $\mathrm{MMR} = 0.005$ has $\sup_t \|C_w J^t\| = \|C_w\| = 160.003$, attained at $t = 0$, then 122.32, 98.32, 80.79 [RECOMPUTED IN CLOSED FORM]. The factor 160 is a static

Table 28: Probing protocols at observation horizon $T = 4$ per cycle

| Protocol | $\delta$ | Executed vol. | Max trade | Error (SE) | Capital |
|---|---|---|---|---|---|
| A: single pulse (unconstrained) | 0.03 | 131.5 | 131.5 | 0.047 (0.003) | 6,985.91 |
| | 0.05 | 47.3 | 47.3 | 0.054 (0.004) | 921.73 |
| | 0.08 | 18.5 | 18.5 | 0.053 (0.004) | 146.73 |
| | 0.12 | 8.2 | 8.2 | 0.044 (0.003) | 31.45 |
| B: repeated capped pulses | 0.03 | 2,768.8 | 6.2 | 0.050 (0.003) | 7,035.75 |
| | 0.05 | 362.5 | 6.2 | 0.054 (0.004) | 923.88 |
| | 0.08 | 56.2 | 6.2 | 0.047 (0.003) | 146.00 |
| | 0.12 | 12.5 | 6.2 | 0.038 (0.003) | 34.88 |
| C: sequential test, capped | 0.03 | 1,410.2 | 6.2 | 0.045 (0.003) | 3,585.14 |
| | 0.05 | 195.6 | 6.2 | 0.038 (0.003) | 500.00 |
| | 0.08 | 35.9 | 6.2 | 0.032 (0.003) | 94.30 |
| | 0.12 | 10.5 | 6.2 | 0.011 (0.002) | 29.85 |
| D: generalized eigenvector | 0.03 | 2,359.7 | 3.7 | 0.050 (0.003) | 2,632.77 |
| | 0.05 | 314.8 | 3.7 | 0.048 (0.003) | 353.67 |
| | 0.08 | 49.2 | 3.7 | 0.046 (0.003) | 57.53 |
| | 0.12 | 9.9 | 3.8 | 0.045 (0.003) | 13.56 |

2,000 replications per hypothesis and cell. Log–log capital slopes: −3.90, −3.84, −3.47, −3.81. Protocols A and B reproduce the earlier version exactly. C and D are re-implemented because the earlier code was not retained: C is a Wald test with capped pulses; D repeats the top generalized eigenvector of a flat $(T + 1)$-step probe, scaled to a maximum trade of 3.75, and is cheaper than the earlier D. Kind: [RECOMPUTED] (`recompute_p1_protocols.py`).

coordinate weight; $C_w z$ is not a margin quantity because $z$ contains no inventory. Margin must be computed on a cash-and-inventory ledger.

## S.9 Reproducibility: source of every number

The replication package contains numbered Colab notebooks and stand-alone scripts; its README lists them in run order. "Part $N$" below and in table notes refers to notebook $N$ (Part 1: theorem checks and spot tests; Part 2: BTCUSDT futures depth; Parts 4–13: the cross-section, replications, fresh-day tests, second venues, the CFTC test, the certified guard and post-hoc checks). Result identifiers such as `X6` or `U6` name the JSON file that the notebook writes. "Kind" distinguishes closed-form or simulation results (recomputed), results from public market data (real data), and audits.

| Result | Source | Kind |
|---|---|---|
| Complete monotonicity, 892 economies | `L26_complete_monotonicity` | recomputed |
| $d_2 = -\kappa^2 q \delta^2$ | `P1.7-T1_third_tap` | recomputed |

| Result | Source | Kind |
|---|---|---|
| Ledger identity; $T_h$ positive definite | `R2-T1`, `R2-T3` | recomputed |
| Figure 2 | `make_figs2.py` | recomputed |
| $\delta_{\min}(3)$, Table 4 recomputed columns, 61.14, 29.94 | `recompute_theory.py` → `R_P1_theory_constants.json` | recomputed |
| Pulse benchmarks (271, 58, 12, 11 pulses), Figures 1–7 | `make_figs3.py` → `R_P1_pulse_benchmarks.json` | recomputed |
| Table 2, Table 10, worked example, passive simulation | `recompute_extra.py` → `R_P1_extra.json` | recomputed |
| Daily noise (Table 7) | `P1.3-T1_sigma_obs` | real data |
| Table 9 recomputed columns | same | recomputed |
| $K$-cohort slopes (Table 11) | `T1_K_cohort_rate` | recomputed |
| ESPRIT slope $-6.15$ | `P1.2-T1_esprit_rate` | recomputed |
| Minimum phase, 1,733 economies; rank 6 of 6 | `T2a`, `T2b` | recomputed |
| Passive slopes $-4.21$, $-5.93$ | `T3_passive_rate` | recomputed |
| Tagged-flow slope 0.011 | `T4_disclosure` | recomputed |
| Spectral blindness gains | `T5_spectral_blindness` | recomputed |
| Noise and capital grid (Table 6) | `P1.3-T1_sigma_obs` | real data |
| Loop gain and placebo (Table 24) | `P1.6-T1_ema_placebo` | real data |
| Estimator table (Table 25) | `P1.5-T1`, `T2`, `T3` | recomputed |
| Data integrity (111 files) | `R2-T9_data_integrity` | audit |
| Tables 3, 4 (certified, feasible, adaptive columns) | `recompute_p1_capital.py` → `R_P1_capital_ledger.json` | recomputed |
| Table 15; resolution floors; Table 28 | `recompute_p1_protocols.py` → `R_P1_protocols.json` | recomputed |
| SPRT capital figures | `recompute_p1_sprt.py` → `R_P1_sprt.json` | recomputed |
| Table 8 ($N^*$, $\sigma_{\text{crit}}$) | `recompute_crb.py` → `R_P1_crb.json` | recomputed |
| Table 8 (ML columns), Table 9 (estimators) | `recompute_p1_estimators.py` | recomputed |
| Symmetric multi-asset network (Section S.4) | `recompute_oct20.py` | recomputed |

| Result | Source | Kind |
|---|---|---|
| Proof audit: $K_*$ bound and $d_2$ identity (Theorem 4.1), moment gaps (Theorem 6.2), spectral-blindness example (Theorem S.4.1) | `recompute_audit.py` → `R_audit_oct2026.json` | recomputed |
| Theorem 6.5, Table 12 | `kcohort_upper.py` → `R_P1_kcohort_upper.json` | recomputed |
| Lem. 6.6, Thm 6.7, Tab. 13 | `general_minimax.py` | recomputed |
| Theorem 8.2 numbers | `disclosure_value.py` → `R_P1_disclosure_value.json` | recomputed |
| Tables 22, 23, 26 | Part 8: `V2_spot_panel`, `V3_fee_events` | real data |
| Table 17, top panel | Part 4: `X9_disclosure_metrics` | real data |
| Table 17, bottom panel; class-position forecasts | Part 6: `Z6_C_identification`, `Z6_secondary`; `P6_fresh_days.json` (65acda36…) | real data |
| Table 19 | Part 11: `C3_cot_data`, `C4_cot_disclosure`; `P11_cot.json` (6f1d6f35…) | real data |
| Basis-trade signature; real-data Cochrane–Orcutt | Part 13: `R6`, `R7` | real data (post hoc) |
| Feedback across 30 perpetuals | Part 4: `X3_T6_cross_section` | real data |
| Theorem 5.3 full bound (Table 5), direct minimization | `hardening/rh_bound.py`, `rh_check.py` → `R_P1_rh_explicit.json` | recomputed |
| Minimal dimension at $a = 0$; Toeplitz sections; passive slopes for $K \leq 4$; IV determinant | `hardening/p1_checks.py` → `R_P1_hardening_checks.json` | recomputed |
| Stylized nonlinear and trained online rules (Table 14, Figure 6) | `recompute_p1_learners.py` → `R_P1_learners.json`; `P1_trained_learners.json` (3e8bf30c…) | recomputed |

| Result | Source | Kind |
|---|---|---|
| Disclosure in dollars (Table 16) | `recompute_p1_cat_costs.py` → `R_P1_cat_disclosure_dollars.json` | recomputed |
| Out-of-sample post-shock return test (Section 8.3) | `recompute_prereg_oct21_22nd.py` → `R_P1_disclosure_oos.json`; `P1_disclosure_oos.json` (f0182ce5…) | real data |
| Out-of-sample cohort-state trajectory test (Table 18) | `recompute_prereg_oct21_followup.py` → `R_P1_disclosure_state_oos.json`; `P1_disclosure_state_oos.json` (5136cc6b…) | real data |